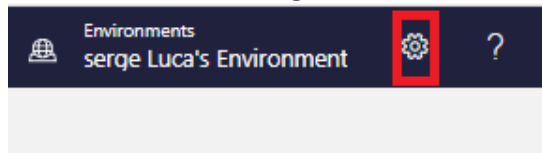


Lab 0. Setting up the experimental designer & activate your Power Apps developer plan

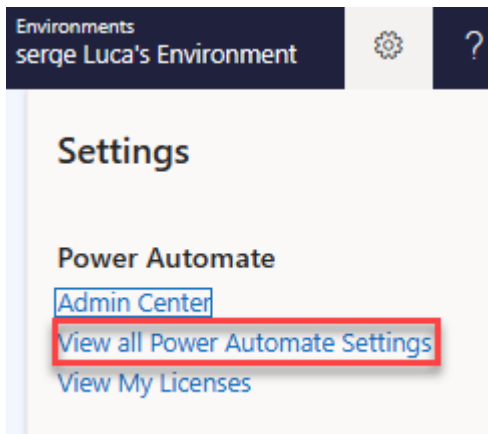
Task: Select the new Power Automate Designer

The new Power Automate designer is still in the experimental stage at the time of writing. To select the new designer,

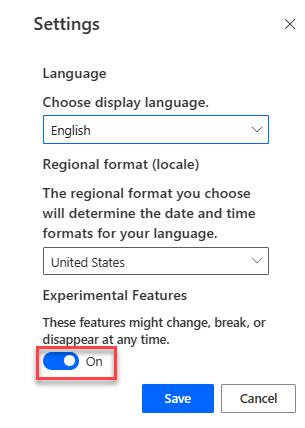
1. go to <https://powerautomate.com>, and **sign-in**.
2. This will redirect you to <https://make.powerautomate.com>.
3. Select the **settings** button:



4. Select **View all Power Automate settings**:



5. Check the option **Experimental Features**:



6. Click **Save**.

Task: Activate your developer license

The Microsoft365 license provides a Standard Power Apps and Power Automate license which is good enough for most labs. However, some labs require a Premium connector; the Power Apps Developer Plan gives you access to **Premium features** of the product for free. The main restriction of the Developer license is that flows and applications created with this license cannot be used in production.

Follow the instruction described here [Sign up for Power Apps Developer Plan - Power Apps | Microsoft Learn](#) to activate this license.

A dedicated Power Platform environment will be generated as part of this activation, and this is where you will do the exercises.



Lab 1. Building a time-tracking flow

Lab 1. Building a time-tracking flow

Author: Serge Luca aka "Doctor Flow"

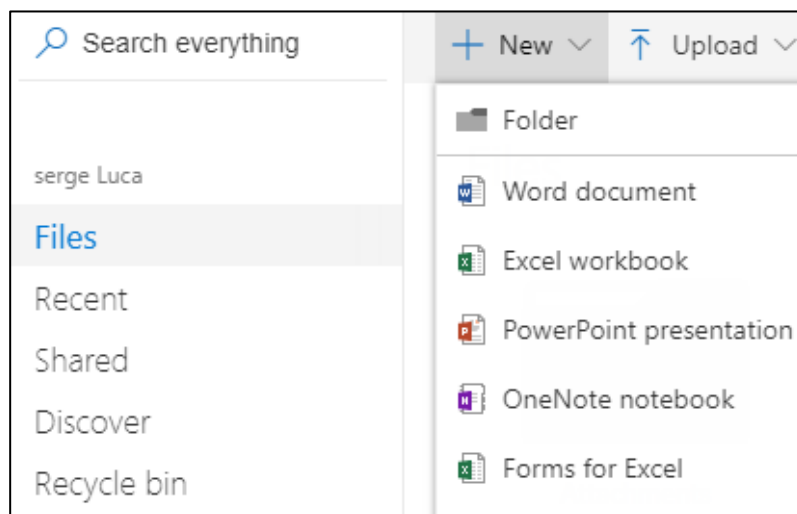
Learning objectives: Building a Flow, hello world, button, time.

Duration: 30 minutes

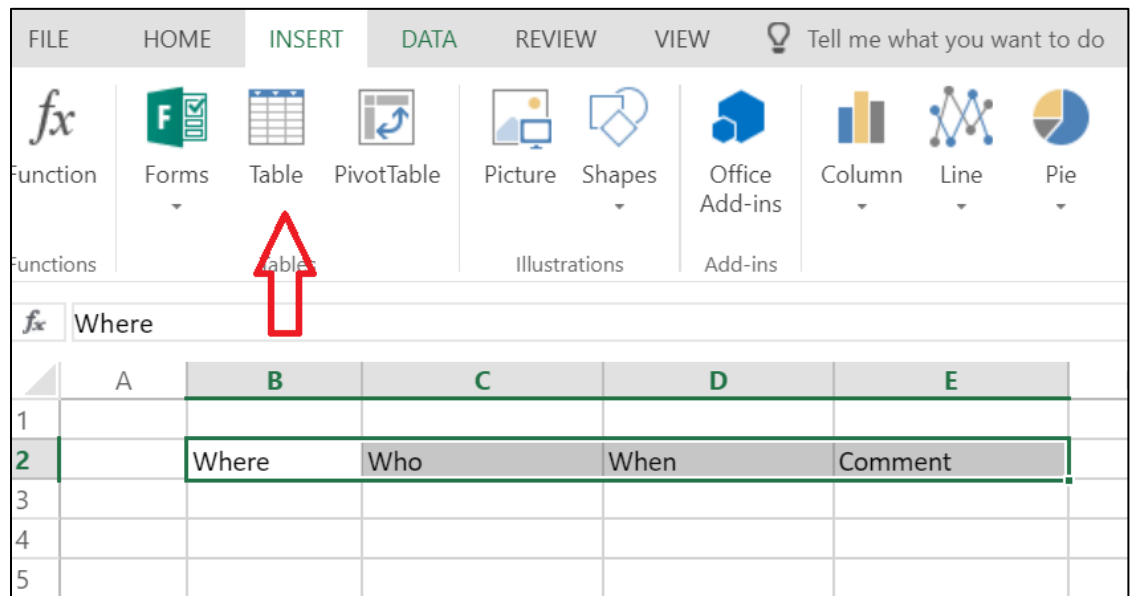
Scenario: When a user pushes a button, the current time and the current location will be saved in an Excel document stored in OneDrive for Business.

Task 1.1: Build an instant flow from blank

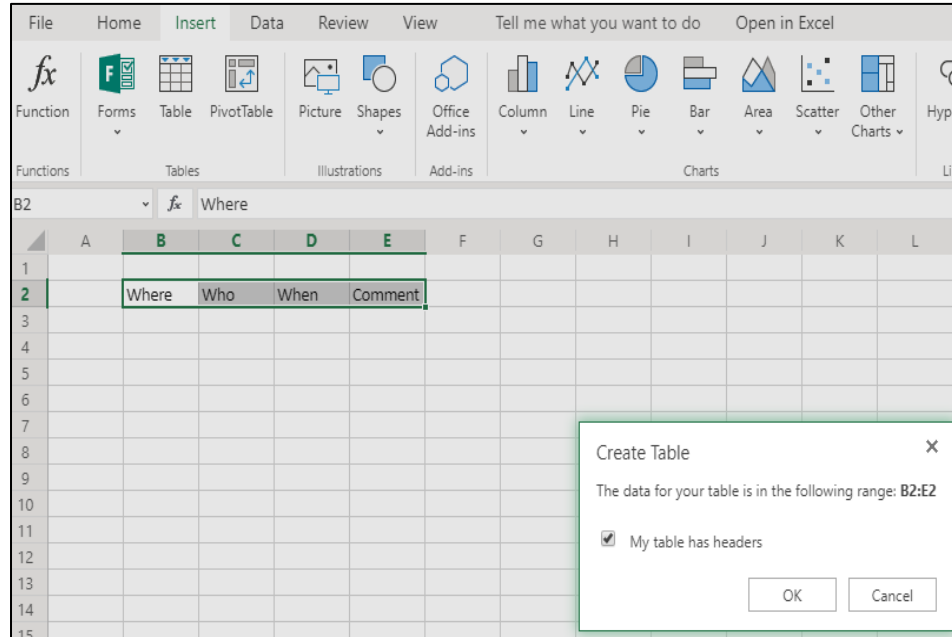
1. Navigate to **OneDrive** (for Business), and create a new Excel workbook:
 - a. Click **New** and select **Excel workbook**.



- b. Create **four** columns, **where, who, when, comment**. Then, click the "Insert" Menu and select "Table".

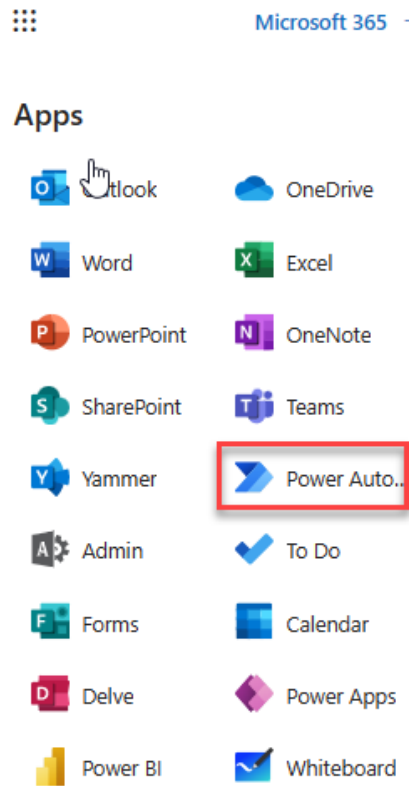


- c. Format as a table (check the box, has headers)

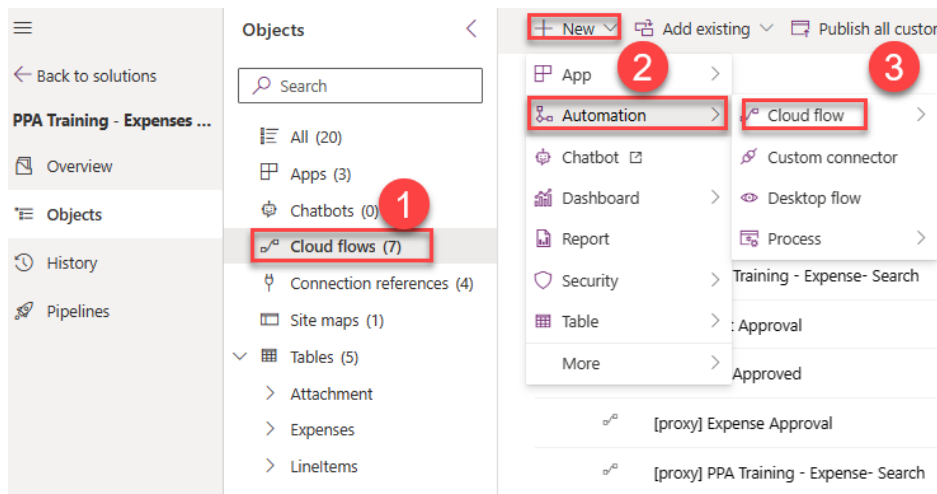


- d. Save the file (**File** > **Save as - Rename**) with the name **Timesheet.xlsx**.

2. Create a flow named **Track Time** using the following steps:
 - a. Navigate to <https://make.powerautomate.com> or click the Power Automate icon in the Microsoft 365 app launcher (and if requested, sign-in) :



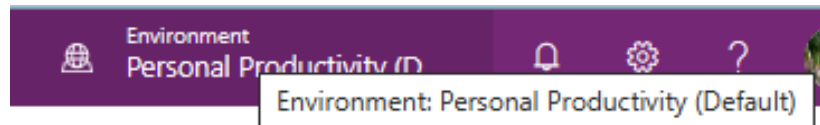
- b. If you are already familiar with the concept of "Solution", creating the flows of all labs in a dedicated solution is a better practice. Create a Solution, or use an existing Solution, and select the folder cloud flows as illustrated below:



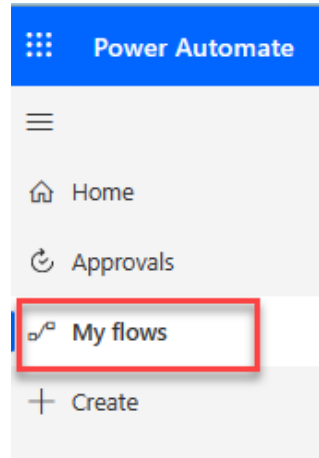
If you are not familiar with the concept of Solution, move to the next step.

- c. Now that you are in Power Automate Studio, you have to select an environment to create your flows; for these labs, you can create your flows in the **default environment**: on the top-right side of the Power Automate site, select the environment with a name containing

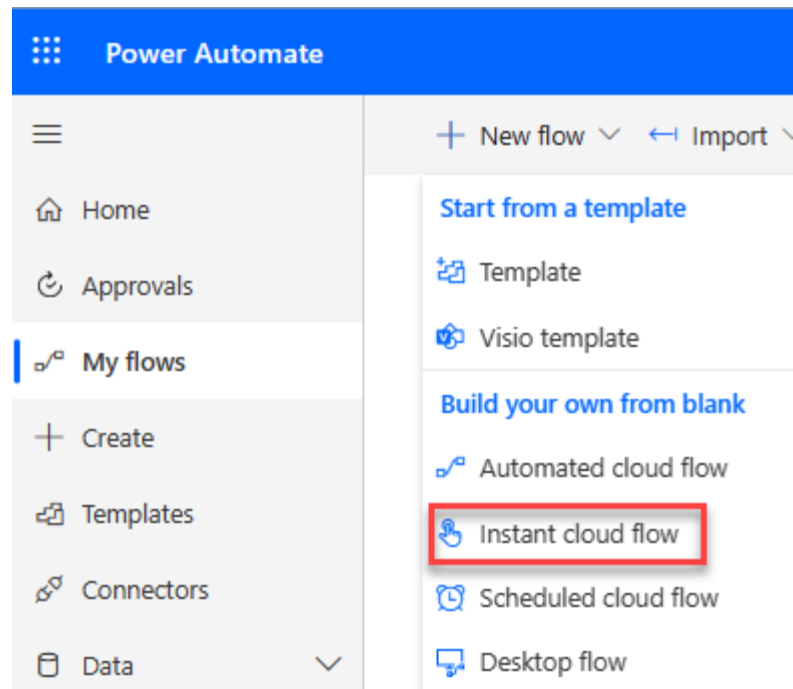
(default) . The environment name varies from tenant to tenant because most companies rename it, but **(default)** must be part of the name. This is your private space :



- d. Select **My flows** :



- e. On the left side of the screen (unless you work in a Solution), select **New flow > Instant cloud flow**.



- f. On the **Build an instant cloud flow** dialog, in the **Flow name** text box, type **Track time**, and select the trigger **Manually trigger a flow**.

Build an instant cloud flow

Flow name
Track time

Choose how to trigger this flow *

- ☒ Manually trigger a flow
Flow button for mobile
- ☐ PowerApps
PowerApps
- ☐ When Power Virtual Agents calls a fl...
Power Virtual Agents
- ☐ When a flow step is executed
Common Data Service (current environment)
- ☐ For a selected message
Microsoft Teams
- ☐ For a selected file
OneDrive for Business
- ☐ For a selected file

Triggers manually from any device, easy-to-share instant flows automate tasks so you don't have to repeat yourself.

Examples:

- Get an automatic mobile alert whenever a VIP client emails you
- Save all your email attachments to a folder automatically

Skip Create Cancel

- g. Click **"Create"** button.
A flow is generated with the trigger "Manually trigger a flow".

Manually trigger a flow

Note: This trigger will start the flow when the user presses a button. The button is also visible in the Flow mobile app, including on the home screen of a smartphone. A form can be displayed when the user selects the button if needed. In any case, default information such as the current user's coordinates (Latitude & Longitude), current time, and much more are captured automatically from the device by flow when the button is pressed.

3. In the flow, add an action to create a row in the Excel table by completing the following steps:
- Click on **"New step"** button:

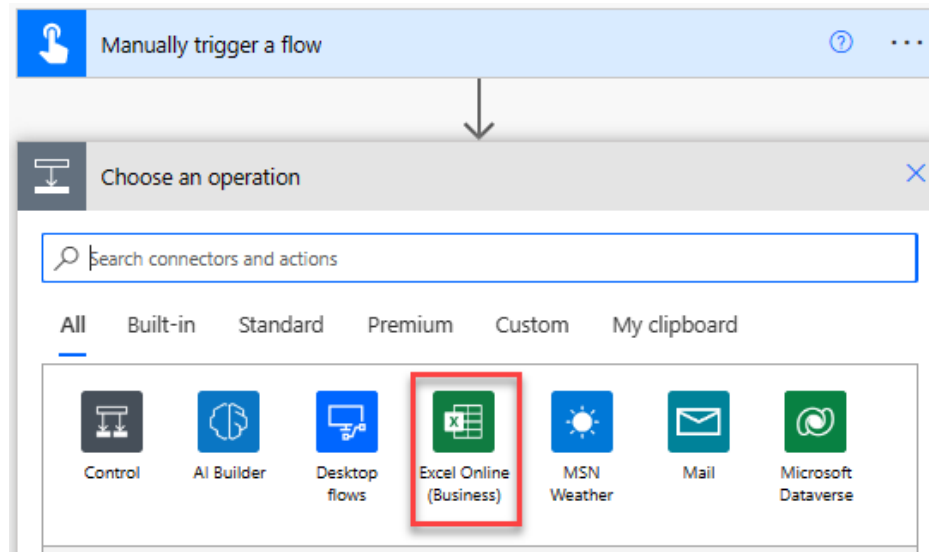
Manually trigger a flow

+ Add an input

+ New step Save

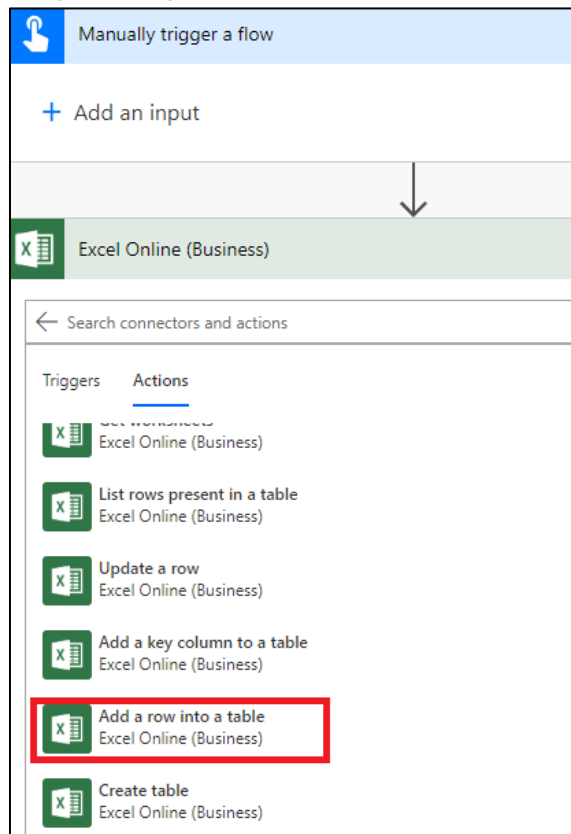
- In the **Choose an action** text box, type **Excel** and select **Excel Online (Business)** connector.

DO NOT select Excel Online OneDrive; be careful here, as many users choose the wrong connector.



Once you select the Option, this will prompt you to choose an associated action.

- c. Select the **Excel Online (Business) Add a row into a table** action.



Your flow should look similar to the following screenshot:

Manually trigger a flow

↓

Add a row into a table

* Location

* Document Library

* File

* Table

Show advanced options

4. Configure the action, add a row into a table, input properties to add the current user's location and time stamp to the spreadsheet, using the following steps:
 - a. From the **Location** drop down select **OneDrive for Business**, and in the **Document Library** drop down choose **OneDrive**.
 - b. To the far right of the **File** text box, select the folder icon, browse and select the spreadsheet you created earlier, **Timesheet.xlsx**.
5. When the spreadsheet has been selected, the fields will automatically be visible:

Manually trigger a flow

↓

Add a row into a table

* Location

* Document Library

* File

* Table

Who

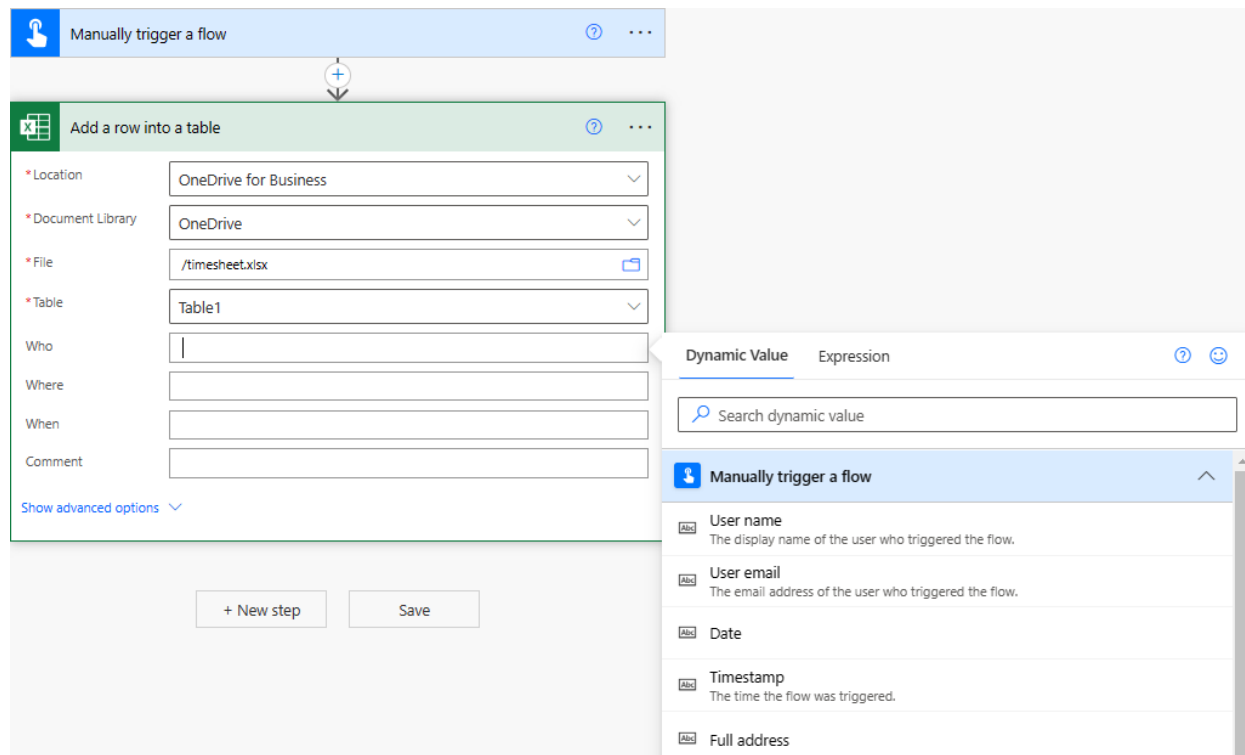
Where

When

Comment

Show advanced options

6. In this step, we are going to map the fields to the actual values
- a. Move the cursor to the Where field and a panel with Dynamic Value and Expression will be visible:



- b. Select and map the properties as follows :

Excel Column Name	Dynamic Value Selection
Where	Full address
Who	User name
When	Timestamp

Important Note : To retrieve your location service, your computer must have the location service activated, and this is not the case in most organizations. So do not use the Full address

dynamic value but write your address directly in the field.

Manually trigger a flow

Add a row into a table

* Location: OneDrive for Business

* Document Library: OneDrive

* File: /Timesheet.xlsx

* Table: Table1

Where: Full address

Who: User name

When: Timestamp

Comment:

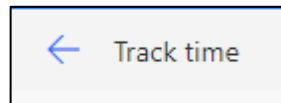
Show advanced options

+ New step Save

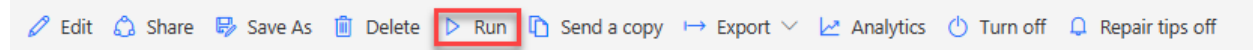
7. Click **Save**.

8. Run the Flow using the following steps:

a. Click the back arrow to return to the flow page



b. then click Run now (you could also use the Test button from the designer).



Flows > Track time

Note: If needed, click **Allow** to allow the flow to access your device to obtain your location, and if required, "Sign In" right side slider panel.

c. Ensure the flow runs successfully by checking your Excel sheet. You should see a new row with the requested information

where	who	when
Jean en Pierre Carsoellaan 243, 1180 Ukkel, Belgium	Paul Pieter	2018-06-28T07:17:51.35413Z

Note: You might have to refresh the spreadsheet to see the updated values.

9. If you see the message "Location services must be enabled for this flow", that means it is not activated, and many organizations deactivate it on their corporate computers:

Run flow



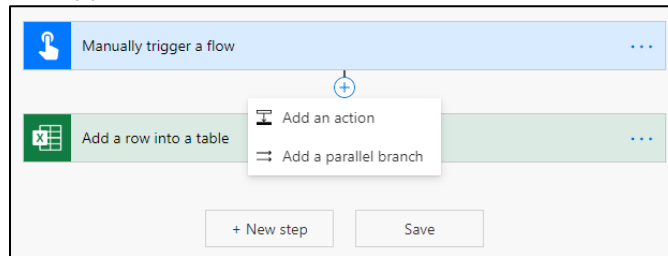
Location services must be enabled to run this flow.

So, write your address directly in the field (do not use the dynamic value).

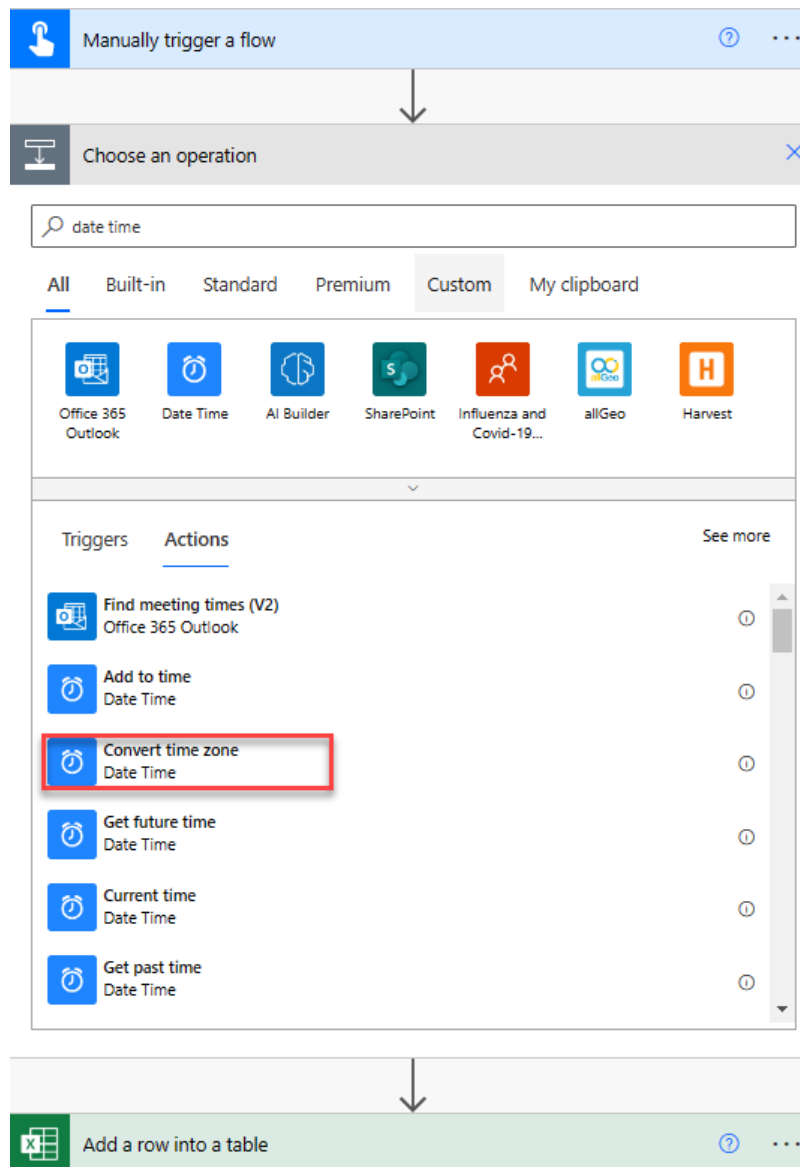
If you still get the same message even though you don't use the dynamic value, **remove** the trigger **Manually trigger a flow** and add a new one.

Task 1.2: Convert the timestamp to a more user-friendly date/time format

1. By default, Power Automate uses UTC as its default time zone. Use the following steps to display the timestamp that matches your time zone. Click on the edit button and follow the below steps :
 - a. Hover over the arrow joining the Manual trigger and the Excel action and click the **+** icon that appears and click **Add an action**.



- b. In the **Search** text box, type **Date Time** and select **Convert time zone**.



- c. In the **Base time** property, select the trigger's output property, **timestamp**.
- d. Use the drop-down list to select an appropriate time zone for the **Source time zone** and **Destination time zone** properties.
- e. In the **Format string** property, scroll to the bottom of the drop-down list and select **add a custom item** and then type **dd/MM/yyyy-T-hh:mm:ss**.
Notice there are several predefined Date & Time formats. Your Convert time zone action should look similar to the following screenshot.

Manually trigger a flow

Convert time zone

- * Base time: Timestamp
- * Source time zone: (UTC) Coordinated Universal Time
- * Destination time zone: (UTC+01:00) Brussels, Copenhagen, Madrid, Paris
- Format string: dd/MM/yyyy-T-hh:mm:ss

Add a row into a table

- Update the **Add a row into a table** action with the **Converted time** value:

Manually trigger a flow

Convert time zone

- * Base time: Timestamp
- * Source time zone: (UTC) Coordinated Universal Time
- * Destination time zone: (UTC+01:00) Brussels, Copenhagen, Madrid, Paris
- Format string: dd/MM/yyyy-T-hh:mm:ss

Add a row into a table

- * Location: OneDrive for Business
- * Document Library: OneDrive
- * File: /timesheet.xlsx
- * Table: Table1
- Who: User name
- Where: Full address
- When:

Dynamic Value Expression

Search dynamic value

Converted time

User name
The display name of the user who triggered the flow.

User email
The email address of the user who triggered the flow.

Date

Timestamp
The time the flow was triggered.

Full address

Country/Region

City

State

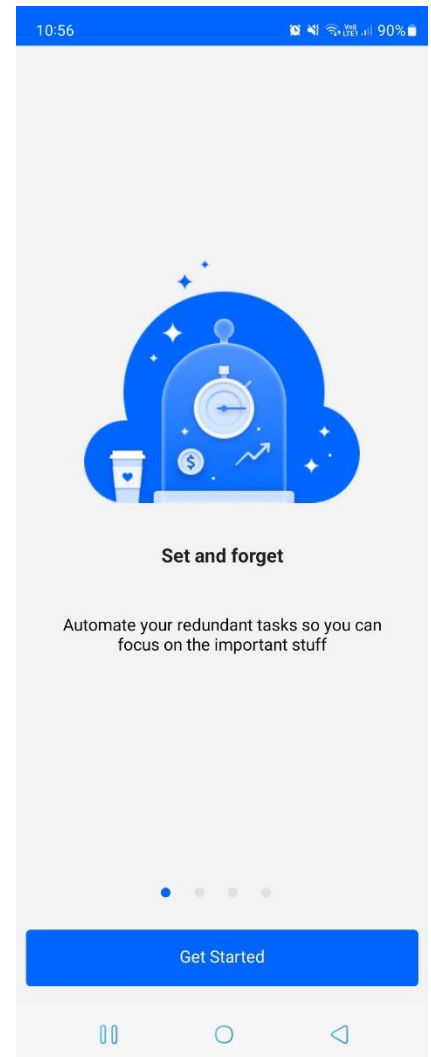
- Save the Flow, rerun it, and check your Excel sheet:

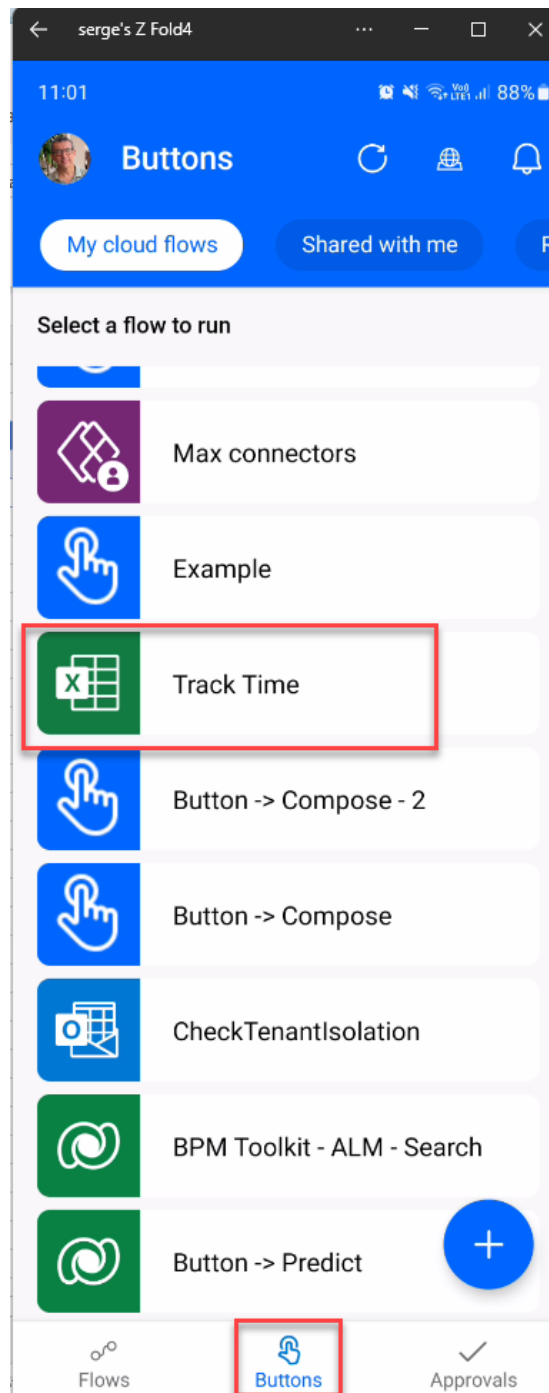
where	who	when
Jean en Pierre Carsoellaan 243, 1180 Ukkel, Belgium	Paul Pieter	2018-06-28T07:17:51.35413Z
Jean en Pierre Carsoellaan 243, 1180 Ukkel, Belgium	Paul Pieter	28/06/2018-T-09:42:51

Task 1.3: Optional, interact with your flow on your mobile

Note: This task has been tested using an Android phone and iPhone; Screenshots are taken from the Android phone. On iPhone, the look and feel might be slightly different than on Android phones.

1. Prepare the mobile app on your phone.
 - a. If you do not already have the mobile app on your phone, download the **Power Automate** mobile application from the store (Android/iOS).
2. Sign-in
3. Start the flow from your phone by completing the following steps:
 - a. Click **Buttons** (to filter the flows that can be triggered by a button), and run the Track Time flow:





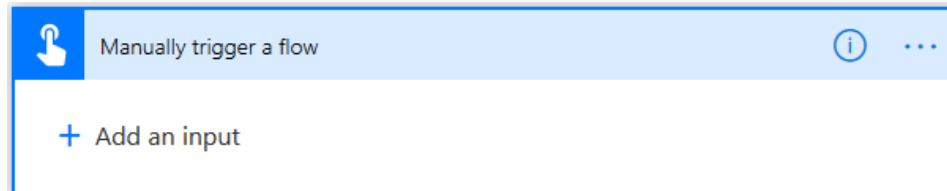
- b. Ensure that the flow ran successfully by checking that a new line was added to the Excel spreadsheet.

Task 1.4: Amend the flow to update the comment field in the spreadsheet

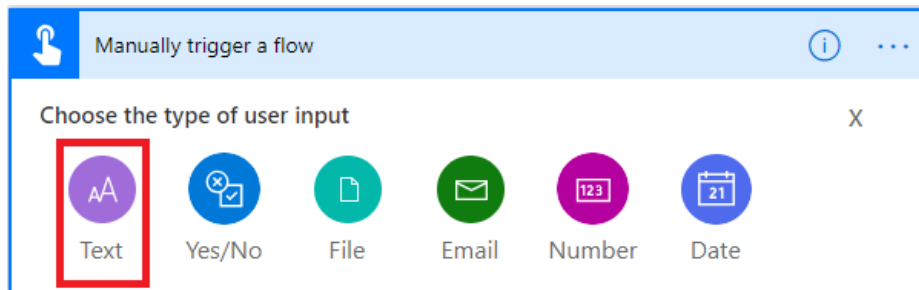
Now you are going to get the flow to update the **comment** field within your spreadsheet:

when	comment
2018-06-28T07:17:51.35413Z	

1. Add an input parameter to the flow button trigger using the following steps:
 - a. Edit the Flow, click on trigger heading (**Manually trigger a Flow**) and then click **Add an input**:

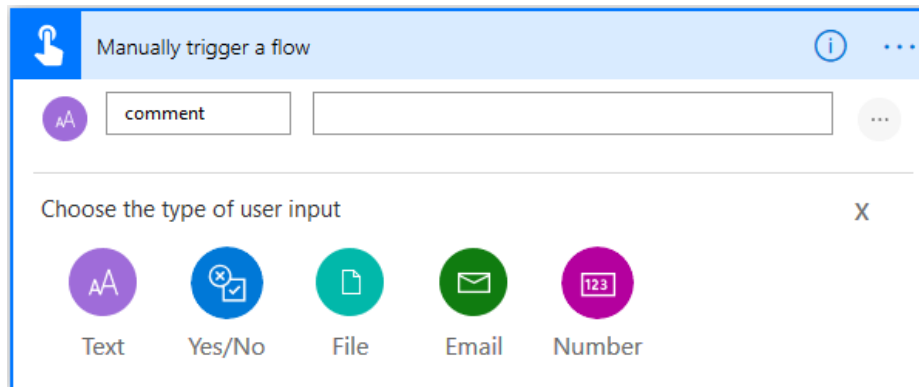


- b. Click **Text** to add a text parameter, named input.



Notice: There are many parameter types to assist the interaction with users when they click on a Flow button. Use these data input types to improve the data integrity of the information provided by the users.

- c. Delete **Input** and type **comment** and in the text box to the right of comment, press the space bar to use the space character as the default value:



2. Modify the Excel action to update the comment value
 - a. Click the action **Add row into a table**.
 - b. Add a **dynamic value** to update the Excel action **comment** property with the trigger **comment** output property:

Add a row into a table

Location*
OneDrive for Business

Document Library*
OneDrive

* File
/Flow Workshop/Timesheet.xlsx

Table*
Table1

Where
Full address

Who
User name

When
Timestamp

Comment
comment

3. Save, run and check.
 - a. Save the Flow and return to the flow details page.
 - b. Then click Run now (you could also use the Test button).

Note: If needed, click **Allow** to allow the flow to access your device to obtain your location, and if required, "Sign In" right side slider panel.

Run flow

Track time
Owner: serge Luca

comment *

Please enter your input

This flow uses Excel Online (Business).
[Review connections and actions](#)

Run flow Cancel

- c. Type some text in the comment parameter in the form.
- d. If you run it from your mobile, you will see something similar to the following screenshot after the button is pushed:

11:23

← Buttons

Track Time

This flow uses one or more connections
[Review connections and actions](#)

My comment

just picking up my laptop

Run flow

- 4. Check the spreadsheet, and you should see a new row with your comment:

where	who	when	comment
Jean en Pierre Carsoellaan 243, 1180 Ukkel, Belgium	Paul Pieter	2018-06-28T07:17:51.35413Z	
Jean en Pierre Carsoellaan 133, 1180 Ukkel, Belgium	Paul Pieter	29/06/2018-T-08:52:42	just picking up my laptop

We need your feedback

Do you want to report an issue or suggest something? We need your feedback:

<https://github.com/Power-Automate-in-a-day/Training-by-the-community/issues>

Lab 2. Approvals (Part 1) - Travel Approval

Author: Serge Luca aka "Doctor Flow."

Learning objectives: Approvals and conditions.

Duration: 20 minutes.

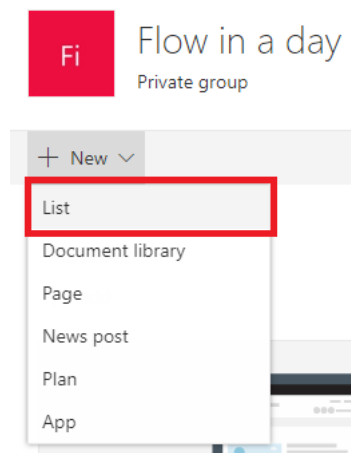
Scenario: A user stores travel information in a SharePoint list named **Travels**. When a new travel request is created, a flow is triggered and asks a manager to Approve/Reject.

Prerequisites: Each student must have a dedicated custom SharePoint list named **Travels_<name>**. The list must have three fields: **Title**, **Amount** (currency), and **Status** (single line of text). The student will create this list in the first task of this lab.

Remarks: the first time approval is created in an environment, the system provisions the infrastructure (approval system tables are made in the internal database, which is Dataverse) needed for the approvals. Therefore, the first approval may take some time to appear.

Task 2.1: Create a SharePoint list

1. Create your own custom SharePoint list using the following steps:
 - a. Go to your SharePoint site. The site URL is: <tenant URL>sites/Flowinaday.
 - b. Click **New > List**:



- c. In the **Name** text box type something similar to, **Travels_<YourFirstnameYourLastname>**. Ensure the list name is unique.

- d. Click **Create**.

Create list

Name *

Travels_AyrtonSenna

Description

☒ Show in site navigation

Create

Cancel

2. Add two columns, **Amount**, and **Status** using the following steps:

- a. Click **Add column** > **Currency**.

Travels_AyrtonSenna

Title ▾

+ Add column

- b. Provide the following information, and then click **Save**.

Create a column

Learn more about column creation.

Name *

Amount

Description

Type

Currency

Number of decimal places

Automatic

Currency format

\$123,456.00 (United States)

Default value

Enter a number

☐ Use calculated value ⓘ

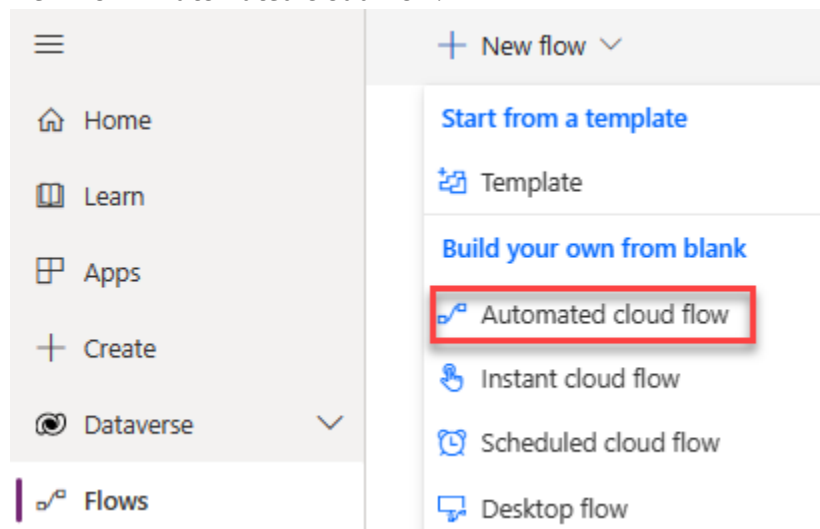
More options

Save Cancel

3. Add a **Single line of text** column named **Status**.

Task 2.2: Create an approval flow

1. Create a new Automated flow triggered when an item is created in a SharePoint list, using the following steps:
 - a. **New flow> Automated cloud flow.**



- b. In the **Flow name** text box, type **Expense Approval** and select the SharePoint trigger **When an item is created**.

Build an automated cloud flow



Free yourself from repetitive work just by connecting the apps you already use—automate alerts, reports, and other tasks.

Examples:

- Automatically collect and store data in business solutions
- Generate reports via custom queries on your SQL database

Flow name
Expense Approval

Choose your flow's trigger * ⓘ
Search or select a trigger from the list below to create a flow. (Required)

Search all triggers

When a new response is submitted Microsoft Forms	ⓘ
When an item is created SharePoint	ⓘ
When an item is created or modified SharePoint	ⓘ
When a file is created in a folder (de... SharePoint	ⓘ
When a file is created OneDrive for Business	ⓘ

Skip Create Cancel

- c. Click **Create**.

Note: If you have not connected to Office 365, SharePoint, or Approvals before, you will need to provide your credentials to create connections to these services.

- d. From the **Site Address** drop-down list, select your SharePoint site, and then from the **List Name** drop-down list select your list.

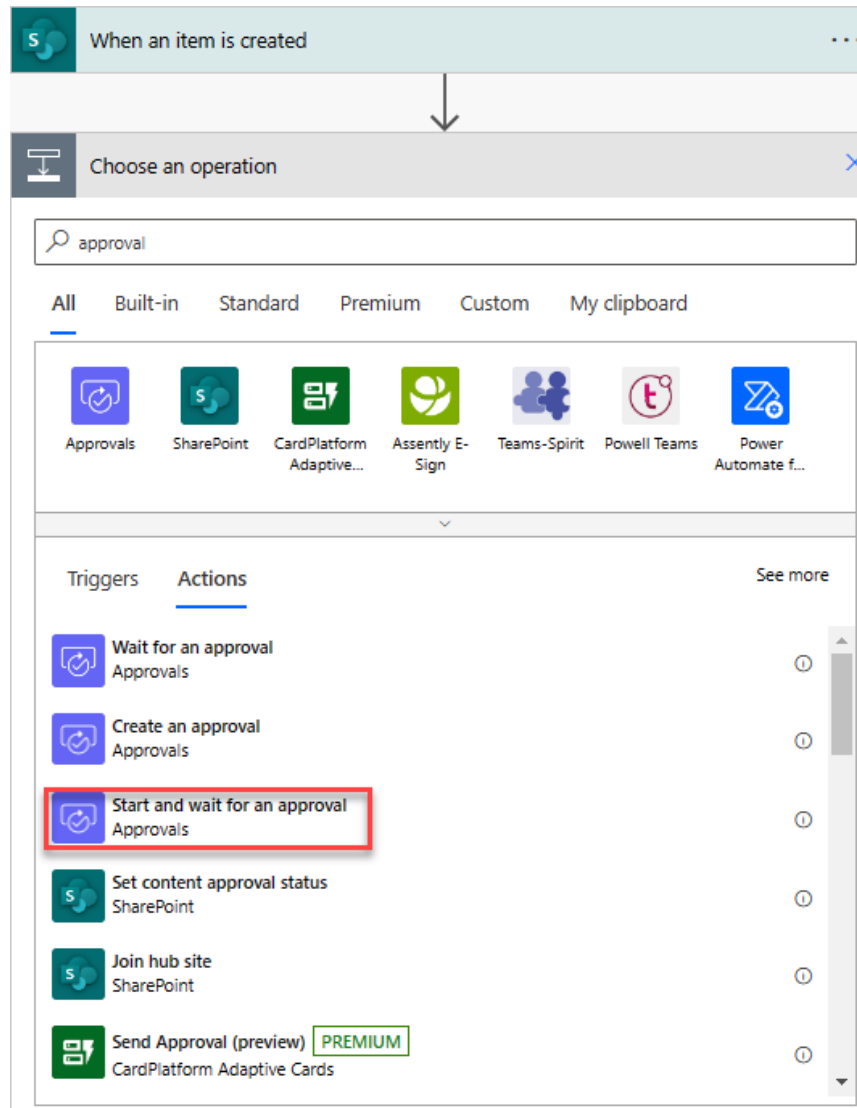
When an item is created ⓘ ...

Site Address *
[Blurred dropdown menu]

List Name *
Travels

Show advanced options ▾

2. Add an approvals action
- a. **New step > Start and wait for an approval.**
Hint: search for approval.



- b. **Approval type** > **Approve/Reject -First to respond**
- c. **Title** > type **Please approve the following expense**
- d. **Assigned to** > type the e-mail address (the one provided by the trainer in the training tenant)
- e. **Details** > type the text as per the following screenshot:

Start and wait for an approval ...

* Approval type: Approve/Reject - First to respond

* Title: Please approve the following expenses

* Assigned to:

Details: Destination: Amount: has been submitted by

Item link: Add a link to the item to approve

Item link description: Describe the link to the item

Show advanced options

- f. To define the destination, place the cursor to the right of **Destination;** in the action panel, click the **Add a dynamic value** button.
- g. Scroll down to dialog and select **Title** from the trigger **When an item is created**:

Start and wait for an approval ...

* Approval type: Approve/Reject - First to respond

* Title: Please approve the following expenses

* Assigned to: ;

Details: Destination; Amount: has been submitted by:

Item link: Add a link to the item to approve

Item link description: Describe the link to the item

Show advanced options

+ New step Save

Dynamic Value Expression

Search dynamic value

When an item is created

ID
List item id. Use this value for specifying the item to act on in other list related actions.

Title

- h. Repeat the step above to select **Amount** and for the submitter: **Created By Display Name** properties.


Start and wait for an approval
...

* Approval type

Approve/Reject - First to respond

* Title

Please approve the following expenses

* Assigned to

Details

Destination:  Title  Amount:  Amount  has been submitted by:  Created By ... 

Item link

Add a link to the item to approve

Item link description

Describe the link to the item

[Show advanced options](#)

- i. Use the Dynamic value button to set the **Item link** > **Link to item**


Start and wait for an approval
?
...

* Approval type

Approve/Reject - First to respond

* Title

Please approve the following expense

* Assigned to

Details

Destination:  Title  Amount:  Amount  has been submitted by:  Created By ... 

Item link

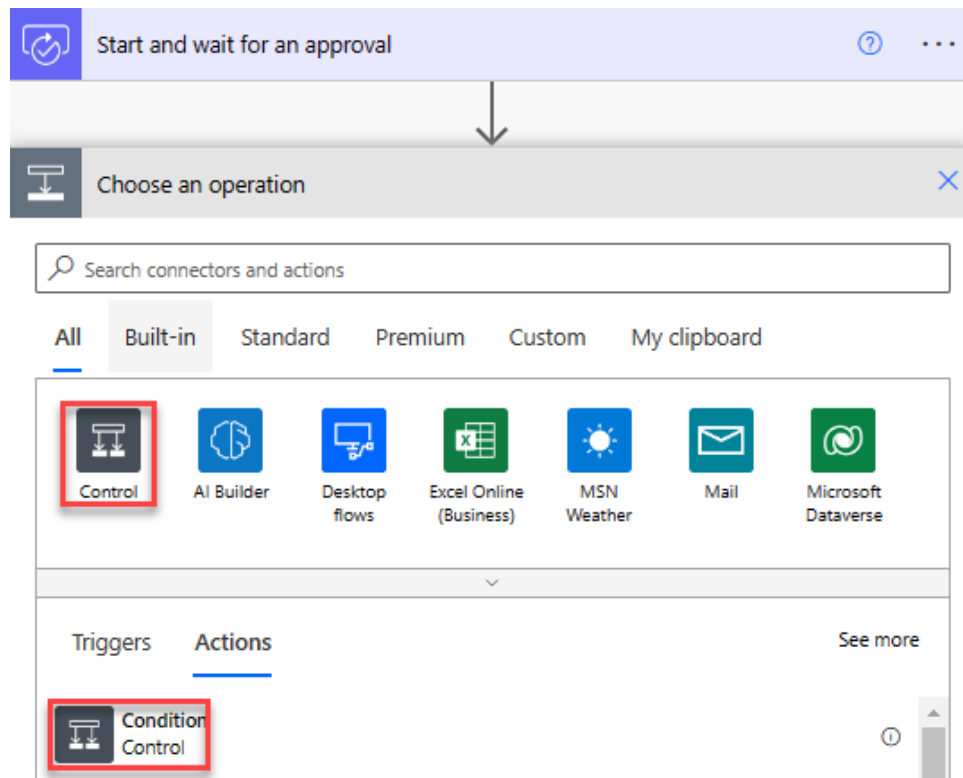
 Link to item 

Item link description

Describe the link to the item

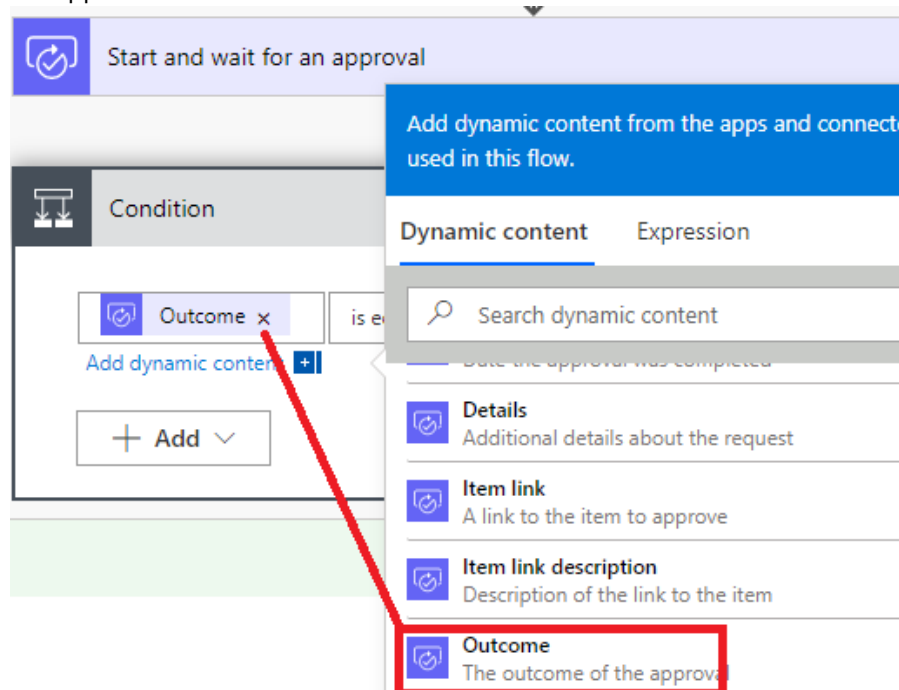
[Show advanced options](#)

3. Add a condition action using the following steps:
 - a. After the **Start and wait for approval** action, select **New step** > **Condition control**



A condition is added to the flow containing three boxes, **Choose a value**, **is equal to** and **Choose a value**.

- b. In the left **Choose a value** text box, use the Dynamic content tab to select **Outcome** from the approval action:



- c. In the right **Choose a value** text box, type **Approve:** (don't type Approved)

Condition

Outcome × is equal to Approve

Add dynamic content +

+ Add ▾

4. Add a SharePoint Update item action to the *If yes* branch of the condition using the following steps:
 - a. In the **if yes** branch, click **Add an action**, and select the SharePoint **Update item** action:

Start and wait for an approval

Condition

Outcome × is equal to Approve

+ Add ▾

✓ If yes

✗ If no

Update item

* Site Address

* List Name Travels

* Id Unique identifier of item to be updated
'Id' is required.

* Title
'Title' is required.

Amount

Status

Show advanced options ▾

- b. Use the **dynamic value** tab to set **Id** > **ID** from the trigger:

Dynamic Value

Learn more X

Search content from previous steps

Update item

ID
Sample data not available

1

When an item is created

ID
Sample data not available

2

ID X

Insert a dynamic value

Title*

Amount

Status

Approved

Show advanced options

- c. Use the **Dynamic value** button to set **Title** > **Title** from the trigger.

- d. **Status** > type **Approved**:

✓ If yes

Update item

* Site Address

* List Name: Travels

* Id: ID

* Title: Title

Amount

Status: Approved

Show advanced options

5. Complete the *If no* branch of the condition using as a starting point the SharePoint action from the *If Yes* branch.

- a. On the **Update item** heading, click the three dots and select **Copy to my clipboard**.

✓ If yes

✗ If no

Update item

Site Address*

List Name*: Travels

Id*

Copy to my clipboard (Preview)

Rename

Add a comment

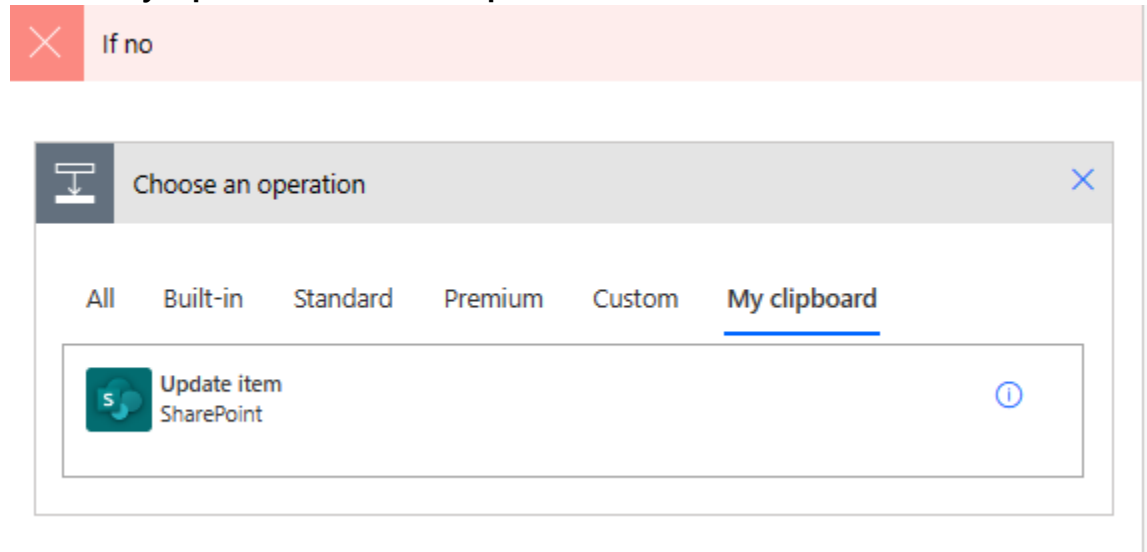
Static result (Preview)

Settings

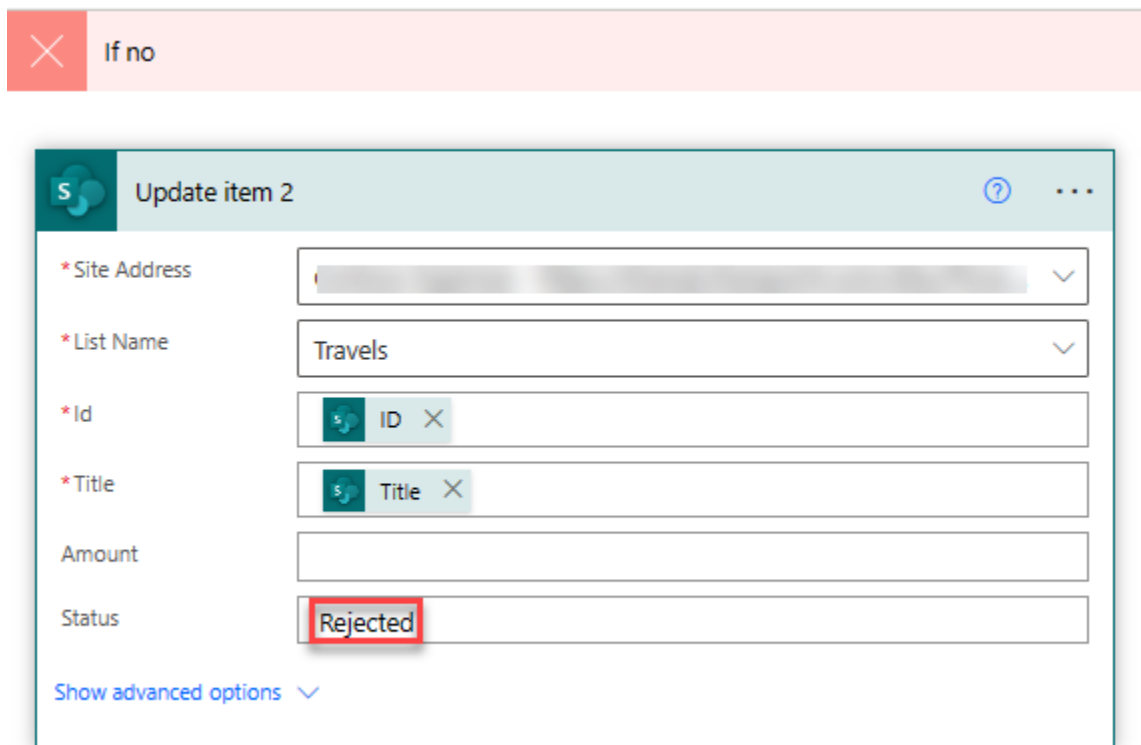
Configure run after

- b. In the **If no** branch, click **Add an action**.

- c. Click the **My clipboard** tab, and select **Update item**.



- d. Change **Status** to **Rejected**:



Rename the condition and the two update item actions as shown in the following screenshot: It is recommended to keep the default action name and to append additional details; if the original action name is **Update item**, then rename it "**Update item – set status to approved**" or "**Update item – set status to rejected**:"

e.

The image shows two parallel workflow branches. The left branch is labeled 'If yes' with a green checkmark icon. It contains an action 'Update item - set status to Approved' with a red box around the title. Below it, the 'Site Address' is set to 'Contoso Expenses - https://shareql.sharepoint.com/sites/Flows...' and the 'List Name' is 'Travels'. The right branch is labeled 'If no' with a red X icon. It contains an action 'Update item - set status to Rejected' with a red box around the title. Below it, the 'Site Address' is the same as the left branch, and the 'List Name' is 'Travels'.

6. Send an e-mail if the expense is approved or rejected by using the following steps:
 - a. In the **If Yes** branch, **Add an action** > **Send an e-mail (V2)** action (Office 365 Outlook).
 - b. Complete like the following screenshot (include the Title field from the trigger)

The image shows the configuration for the 'Send an email (V2) 2' action. The title bar says 'Send an email (V2) 2'. The 'To' field contains a contact with initials 'SL'. The 'Subject' field is 'Expense Approved'. The 'Body' field has a rich text editor with the text 'Your expense [SharePoint icon] Title X has been approved by'. The 'Body' field also shows a font size of 14 and bold, italic, and underline formatting options. At the bottom, there is a link 'Show advanced options' with a dropdown arrow.

Rename the action name.

- c. Place the cursor to the right of **approved by**; in the dialog pane, click the **Add an expression** button,

Send an email (V2)

To*
Created By Email X

Subject*
Expense Approved

Body*
Font 12 B I U [icons] [fx]
Your expense [dynamic value: Title] X has been approved by [dynamic value:]

Show advanced options v

- d. You will now add the name of the approver, and you will have to rely on code (expression) that we will fully explain in a subsequent lab; 1° move the cursor next to approved by, 2 °click **Expression**, 3° **Dynamic values** , 4° select **Responses** (click see more if you don't see Responses) . This will generate an expression as illustrated here:

Dynamic Value Expression **2** Format data by examples fx

`outputs("Start_and_wait_for_an_approval")["body/responses"]`

Save Cancel

Formulas **Dynamic values** **3**

Search dynamic value

- body
- Responses responder
- Responses Item
- Responses** **4**
All of the responses
- When an item is created
- Title

Show advanced options v

Condition

Outcome X is equal to

+ Add v

If yes

Update item - set status to Approved

Send an email (V2)

* To sergeluca@ShareQL.com X

* Subject Expense Approved

* Body Font 14 B I U [icons] [fx]
Your expense [dynamic value: Title] X has been approved by **1**

Show advanced options v

But you still need to add some code to this expression:

Dynamic Value Expression [Format data by examples](#) *fx* 😊

```
outputs('Start_and_wait_for_an_approval')?['body/responses']  
['responder']['displayname']
```

...and click **Save**:

Dynamic Value Expression [Format data by examples](#) *fx* 😊

```
outputs('Start_and_wait_for_an_approval')?['body/responses']  
['responder']['displayname']
```

Save Cancel

Your **Send an e-mail** action should look like the following screenshot.

 Send an email (V2) ? ...

*To  sergeluca@ShareQL.com X

*Subject Expense Approved

*Body

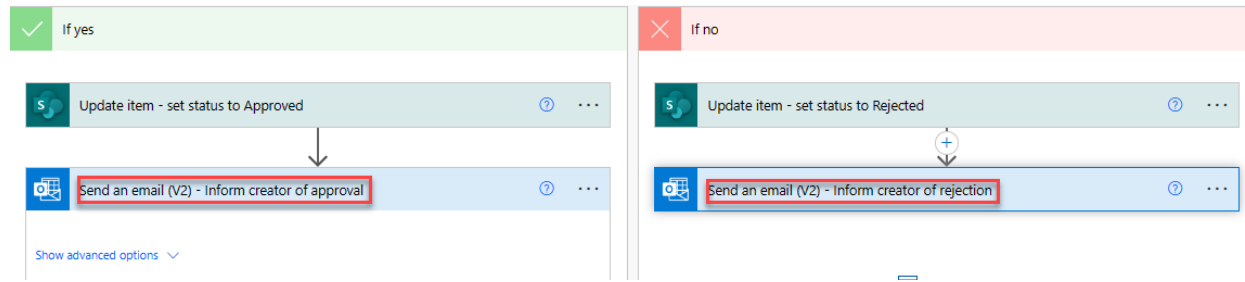
Font ▾ 14 **B** *I* U        

Your expense  Title X has been approved by  outputs(...) X

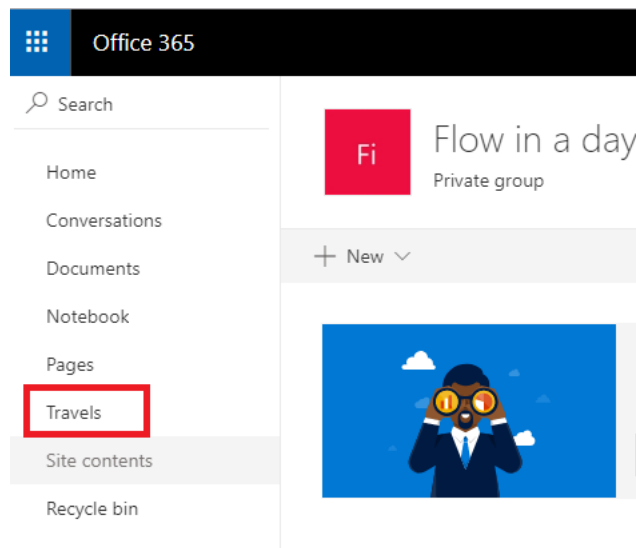
[Show advanced options](#) ▾

- e. Update the "Send an e-mail action in the other branch with this expression.

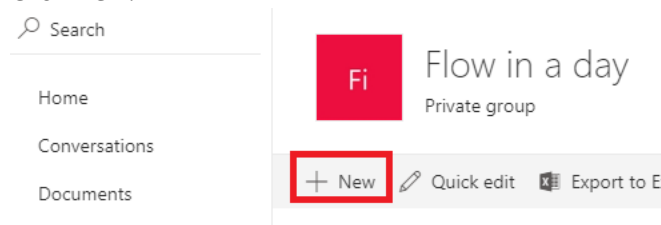
- f. Rename the **Send an e-mail** actions in both branches as illustrated here:



7. Save the flow.
8. Test the flow by adding an expense item to the SharePoint list using the following steps:
- a. Navigate to your SharePoint list, for example, by clicking on the list name on the Quick Launch bar.



- b. Click **New**:



- c. Complete the form similar to the following screenshot (keep the **Status** field empty):

 Save  Cancel  Copy link

New item

Title *
Tokyo

Amount
800

Status
Enter value here

Attachments
[Add attachments](#)

Save Cancel

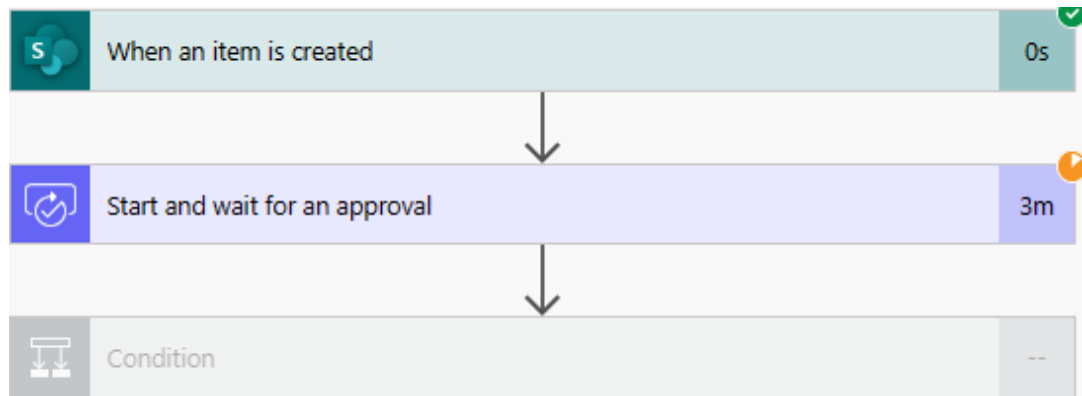
- d. Save the new record, which will automatically trigger the flow.
e. Check the flow status. It should be **running**:

Flows > Expense Approval

Details		
Flow	Expense Approval	
Owner		
Status	On	
Created	Feb 3, 02:51 PM	
Modified	Feb 3, 03:49 PM	
Type	Automated	
Plan	This flow runs on owner's plan	

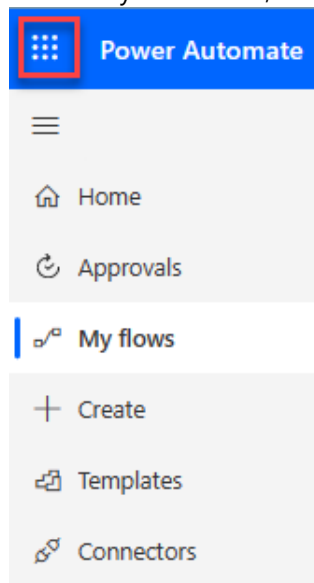
28-day run history ⓘ			Edit columns
Start	Duration	Status	
Feb 3, 03:52 PM (15 sec ago)	00:00:15	Running	

- f. Click the running flow to see the progress of this instance of the flow:



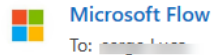
In this screenshot, the flow is waiting for approval. Approvals send an e-mail to the person(s) in charge of the approval.

- g. Check your e-mail to display your approval e-mail:
- i. To check your e-mails, click the app launcher (the waffle).



- ii. In the list of Apps, right-click **Outlook**, and select **Open in new tab**.
- iii. In the Focused tab or the Other tab you should see your Flow approval e-mail:

Please approve the following expense



 **Approvals | Power Automate**

Please approve the following expense

Requested by [redacted] <[redacted]@ShareQL.com>

Date Friday, February 3, 2023 3:57 PM

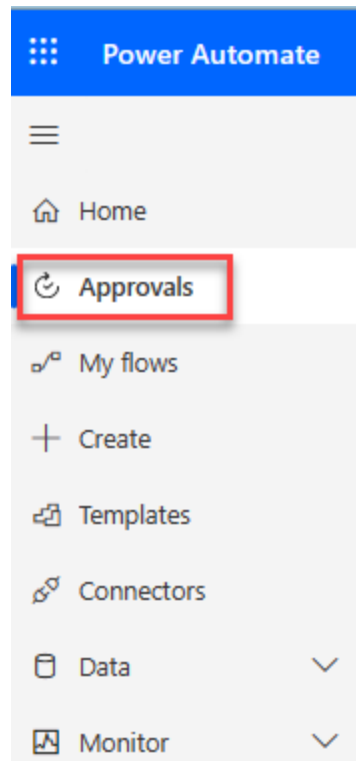
Created [redacted]

Link https://shareql.sharepoint.com/_layouts/15/listform.aspx?PageType=4&ListId=518e02b0%2D0c43%2D4dc7%2Da691%2D85e1a378bc6c&ID=3&Cor

Destination:Hawai Amount:800 has been submitted by [redacted]

Get the Power Automate app to receive push notifications and grant approvals from anywhere. Learn more. This message was created by a flow in Power Automate. Do not reply. Microsoft Corporation 2020.

- h. Approvers can also find the pending approvals in the **Approvals center**:
 - i. Expand **Action items**, and then click **Approvals**.



The Approvals center is displayed.



 Home

 Action items

▼

 My flows

 Create

 Templates

 Connectors

Approvals

Received

Sent

History

Request	Received	Details
Please approve the following expense	Oct 11, 01:19 PM (3 min ago)	The following exper

- ii. On the **Received** tab, click the item to be approved/rejected to display an information pane.

Respond: Approve ✕

Overview
Approval
Please approve the following expense
Requester

Received
Feb 3, 03:57 PM (25 min ago)
Link
<https://sharepoint...PageType=4&ListId=5...>

Details

Destination:Hawai
Amount:800 has been submitted by:serge

Choose your response *

Approve ▼

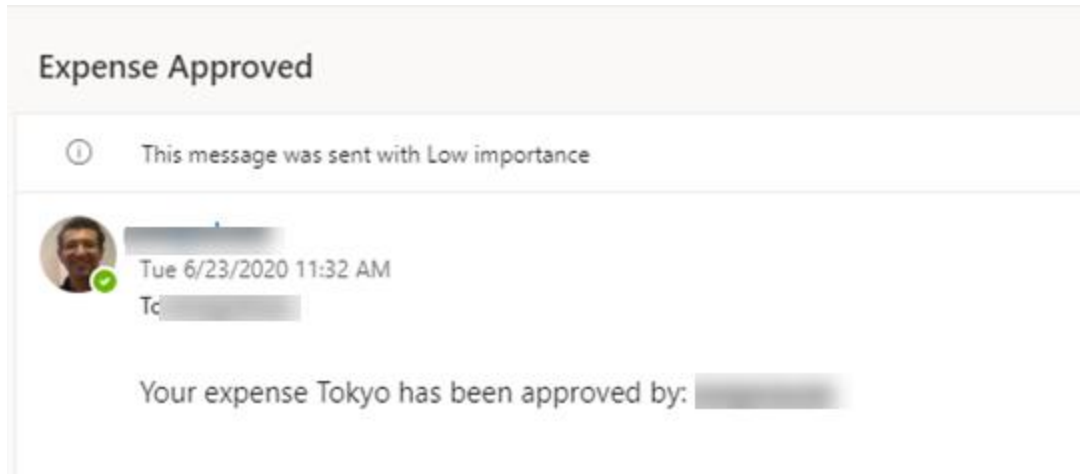
Add a comment
(optional)

Confirm

Cancel

- iii. Select **Approve**, and then click **Confirm**.

The flow will send you a notification by e-mail:



Task 2.3: Extending the flow by adding a new status when the flow is waiting

1. Click **Edit** on your flow's details page.
2. Add a SharePoint **Update item** action between the trigger and the condition, and complete it as shown in the following screenshot.

A screenshot of the "Update item" action configuration in a flow. The action is preceded by a trigger "When an item is created". The configuration fields are: "Site Address" (a dropdown menu), "List Name" (a dropdown menu with "Travels" selected), "Id" (a text box containing a SharePoint ID with a delete icon), "Title" (a text box containing a SharePoint title with a delete icon), "Amount" (an empty text box), and "Status" (a dropdown menu with "Being approved" selected and highlighted by a red rectangle). At the bottom, there is a link "Show advanced options" with a downward arrow.

Hint:

- The **ID** Dynamic value comes from the trigger and identifies the corresponding SharePoint list item.

- **Title** also comes from the trigger,
- Type the value **Being approved** in the **Status** field:

Update item

Site Address*

List Name*

Travels

Id*

1

fx

Insert a dynamic value

Dynamic Value

Learn more X

Search content from previous steps

When an item is created

ID

Sample data not available 2

3. Apply the same logic to display Approved or Rejected in the SharePoint list
4. Add a new travel expense item to the SharePoint list and check the value in its status column (it might take a few seconds before the flow starts):

Fi Flow in a day
Private group

+ New Quick edit Export to Excel Flow PowerApps ...

Travels

Title	Amount	Status
Las vegas	\$900.00	
Brussels	\$750.00	Being approved

2. Approve, and once the approval is completed, check the status value in the list.



Flow in a day

Private group

+ New Quick edit Export to Excel Flow ▾ PowerApps ▾ ...

Travels

Title ▾	Amount ▾	Status ▾
Las vegas	\$900.00	
Brussels	\$750.00	Approved

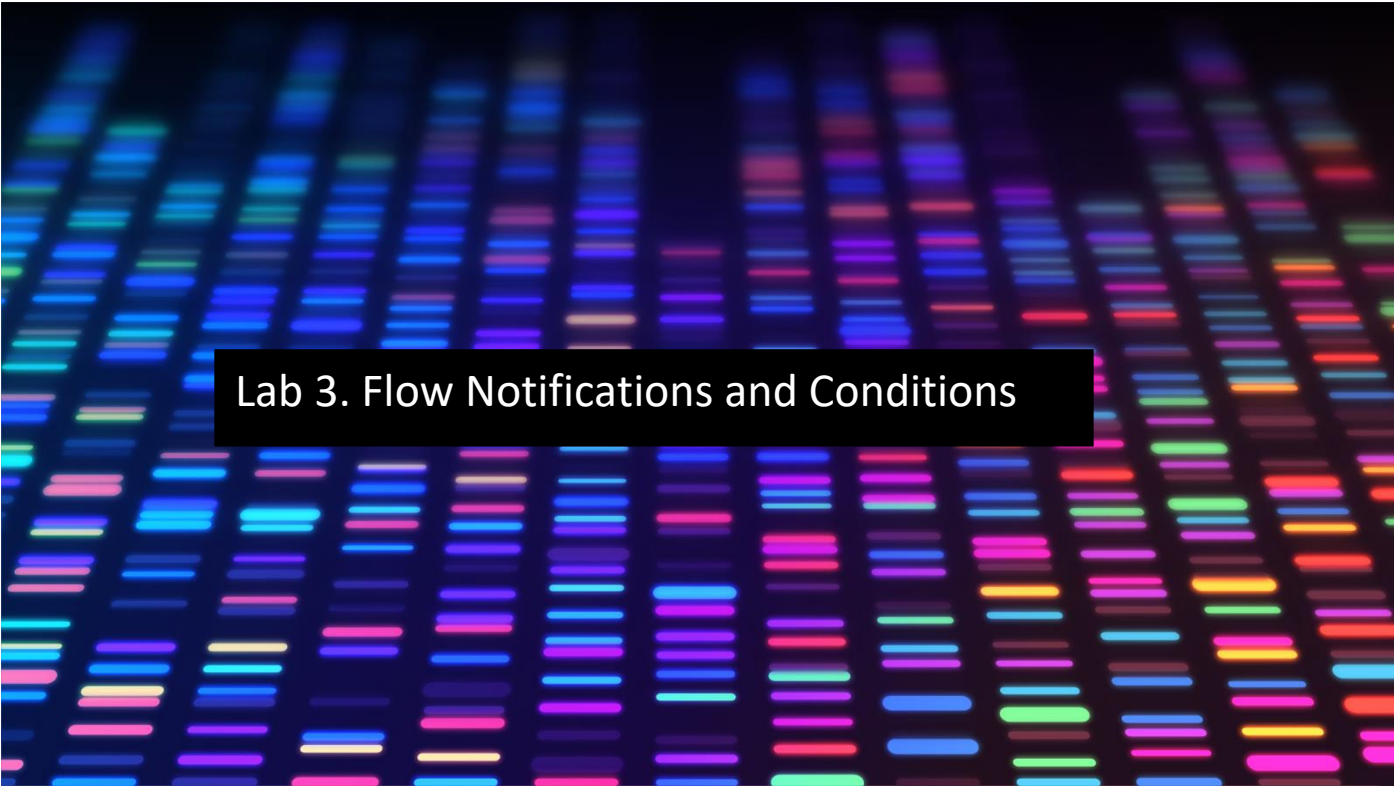
Optional exercise if time permits:

Amend the flow to automatically approve the expense If the amount is smaller than \$500; otherwise, the expense item will progress through the approval process.

We need your feedback

Do you want to report an issue or suggest something? We need your feedback:

<https://github.com/Power-Automate-in-a-day/Training-by-the-community/issues>



Lab 3. Flow Notifications and Conditions

Lab 3. Flow Notifications and Conditions

Author: Serge Luca, aka "Doctor Flow"

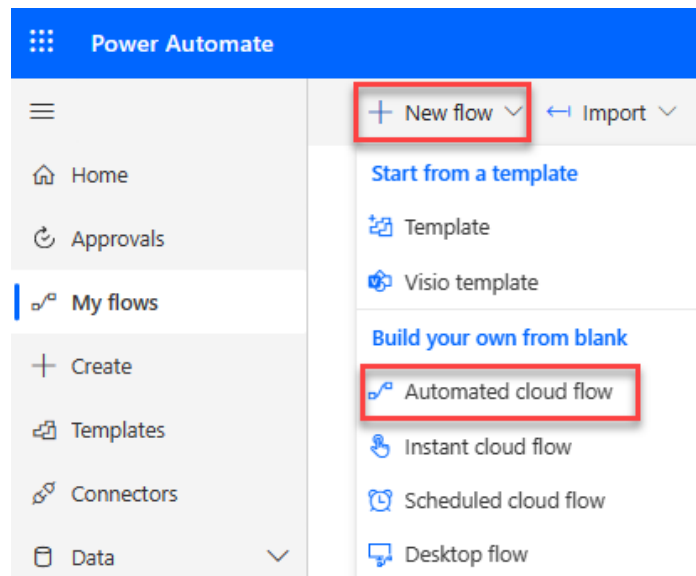
Learning objectives: Notifications and conditions

Duration: 20 minutes

Scenario: You have set an Out of Office notification in your e-mail box. In the Out of Office message, you will mention that if the message is urgent, people can send an e-mail containing just one exclamation point '!'. You also want to get a mobile notification when such a message shows up.

Task 3.1: Create and test a notification flow

1. If you haven't done so, install the **Microsoft Power Automate** application on your phone and enable notifications.
2. Create a new Flow from scratch; the flow should start when your mailbox receives a new e-mail, so it must be an automated cloud flow:



3. Name the flow **important e-mail** and select the trigger **When a new e-mail arrives (v3) Office 365 Outlook**. At the time of writing, the latest version of the trigger is v3; use the most recent version if there is one. Your flow should look like this.

 When a new email arrives (V3)  

Folder

Inbox 

Show advanced options 

+ New step

Save

4. Add a condition to check if the mail subject contains !
 - a. Between the **trigger** and the **Send** action, add a **Condition control**.
 - b. On the left, **Choose a value** text box, and use the **Dynamic content** tab to select **subject** from the trigger.
 - c. In the middle text box, select the **contains** operation.
 - d. On the right, **Choose a value** text box, and type **!**

When a new email arrives (V3)

Condition

Subject X contains !

+ Add v

- e. Update the **If yes** condition by moving the **send me a mobile notification** action into it:

When a new email arrives (V3)

Condition

Subject X contains !

+ Add v

If yes

Send me a mobile notification

*Text You have received an important e-mail!

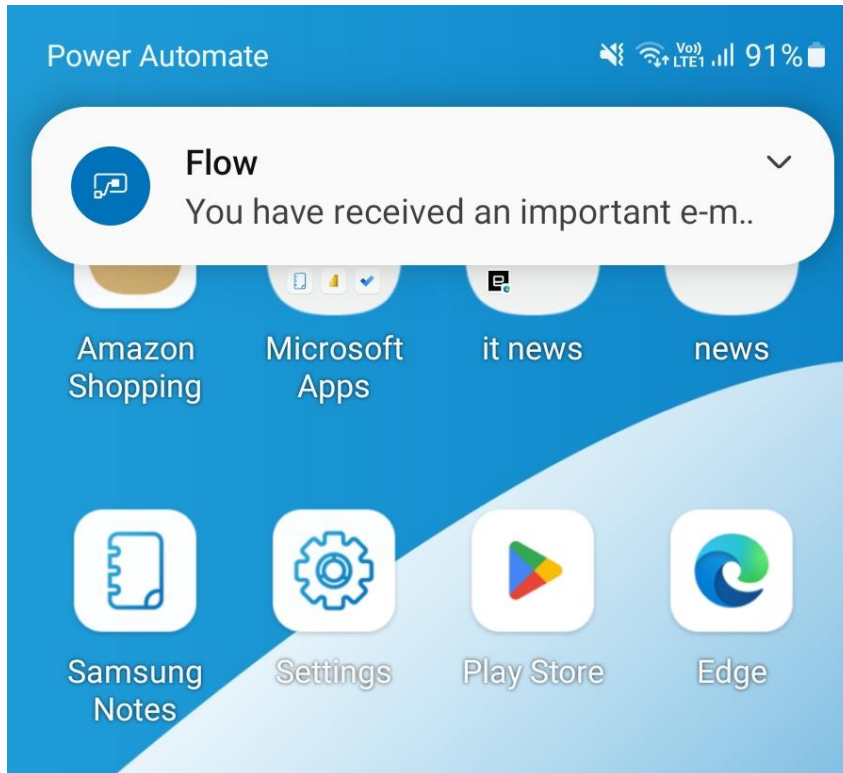
Link Include a link in the notification

Link label The display name for the link

If no

5. Test the flow by sending an e-mail to yourself with the subject **You won the lottery!**
You get a phone notification that looks like this (in Android)

Note: When using your Flow mobile application, you may be prompted to enable notifications.



Task 3.2: Filter the trigger

1. The condition can be avoided by filtering the e-mails directly in the trigger. Click the trigger and **add !** in the **Subject Filter**

 When a new email arrives (V3) ? ...

Folder	Inbox 
To	Recipient email addresses separated by semicolons (If any match, the
CC	CC recipient email addresses separated by semicolons (If any match,
To or CC	To or CC recipient email addresses separated by semicolons (If any m
From	Sender email addresses separated by semicolons (If any match, the ti
Include Attachments	No 
Subject Filter	!
Importance	Any 
Only with Attachments	No 

[Hide advanced options](#) 

2. You can remove the condition and test again. Your flow will look like this:

When a new email arrives (V3)

Folder
Inbox

To
Recipient email addresses separated by semicolons (If any match, the

CC
CC recipient email addresses separated by semicolons (If any match,

To or CC
To or CC recipient email addresses separated by semicolons (If any m

From
Sender email addresses separated by semicolons (If any match, the ti

Include Attachments
No

Subject Filter
!

Importance
Any

Only with Attachments
No

[Hide advanced options](#)

+

Send me a mobile notification

*Text
You have received an important e-mail!

Link
Include a link in the notification

Link label
The display name for the link

This simplifies your flow; it is also essential to know that the number of flow steps your account can perform daily is limited. Preventing the flow from starting when it is unnecessary to optimize this as well.

- You should know that more elaborate expressions can be defined in the trigger. In the trigger, click the Settings option:

When a new email arrives

Folder
Inbox

To
Recipient email addresses separated by semicolons (If any match, the trigger will run).

Copy to my clipboard (Preview)
Rename
Add a comment
Static result (Preview)
Settings

Settings for 'When a new email arrives'

Split On
Enable split-on to start an instance of the workflow per item in the selected array. Each instance can also have a distinct tracking id.
Split On ☒ On

Array

Split-On Tracking Id

Custom Tracking Id
Set the tracking id for the run. For split-on this tracking id is for the initiating request.
Tracking Id

Secure Inputs (Preview)
Secure inputs of the operation.
Secure Inputs ☐ Off

Secure Outputs (Preview)
Secure outputs of the operation and references of output properties.
Secure Outputs ☐ Off

Retry Policy
A retry policy applies to intermittent failures, characterized as HTTP status codes 408, 429, and 5xx, in addition to any connectivity exceptions. The default is an exponential interval policy set to retry 4 times.
Type

Concurrency Control
Limit number of concurrent runs of the flow, or leave it off to run as many as possible at the same time. Concurrency control changes the way new runs are queued. It cannot be undone once enabled.
Limit ☐ Off

Trigger Conditions
Specify one or more expressions which must be true for the trigger to fire.
[+ Add](#)

4. Click **Add**
5. Paste the following expression in the trigger condition textbox
`@contains(triggerOutputs()?['body/subject'], '!')`

Settings for 'When a new email arrives'

Split On

Enable split-on to start an instance of the workflow per item in the selected array. Each instance can also have a distinct tracking id.

Split On ☒ On

Array

Array

@triggerOutputs()?['body/value']

Split-On Tracking Id

Custom Tracking Id

Set the tracking id for the run. For split-on this tracking id is for the initiating request.

Tracking Id

Secure Inputs (Preview)

Secure inputs of the operation.

Secure Inputs ☐ Off

Secure Outputs (Preview)

Secure outputs of the operation and references of output properties.

Secure Outputs ☐ Off

Retry Policy

A retry policy applies to intermittent failures, characterized as HTTP status codes 408, 429, and 5xx, in addition to any connectivity exceptions. The default is an exponential interval policy set to retry 4 times.

Type

Concurrency Control

Limit number of concurrent runs of the flow, or leave it off to run as many as possible at the same time. Concurrency control changes the way new runs are queued. It cannot be undone once enabled.

Limit ☐ Off

Trigger Conditions

Specify one or more expressions which must be true for the trigger to fire.

@contains(triggerOutputs()?['body/subject'], '!')

+ Add

Done

Cancel

(You will become more familiar with expressions in the following labs).

6. **Remove the Subject Filter** you defined before
7. **Save your flow** and test it again.
8. **If your test succeeds, don't forget to turn this flow off :**
 - a. Click the arrow:

Important e-mails

When a new email arrives (V3)

Folder
Inbox

Show advanced options

Send me a mobile notification

*Text
You have received an important e-mail!

Link
Include a link in the notification

Link label
The display name for the link

b. On the flow pane, click the **Turn off** button:

Edit
 Share
 Save As
 Delete
 Send a copy
 Export
 Analytics
 Turn off
 Repair tips off

Flows > Important e-mails

Details
Edit

Flow	Important e-mails	Status	On
Owner	serge Luca	Created	Feb 14, 02:28 PM
		Modified	Feb 28, 10:19 AM
		Type	Automated
		Plan	This flow runs on owner's plan

We need your feedback

Do you want to report an issue or suggest something? We need your feedback:

<https://github.com/Power-Automate-in-a-day/Training-by-the-community/issues>

Lab 6. Task overdue

Author: Serge Luca aka "Doctor Flow"

Updated by: Dattatray-Patil

Learning objective: Controls, manipulation Excel, conditions, Date & time, expressions

Duration: 20 minutes

Scenario: We have an Excel document with a set of tasks, and some of these tasks are overdue. You will create a Flow that will find all overdue tasks and send a report of these tasks.

Tasks:

1. Create a new Excel file that looks like this in your OneDrive for Business (use the same columns). Before adding data in the Deadline column, ensure it is in **Text** format (otherwise, your flows will crash!). Create a table, and replace the existing e-mails with your e-mail.

Task	Status	in charge	Deadline
Feed the cat	Started	User1@dynamicjuly.onmicrosoft.com	1/1/2021
Call jon	Not Stared	User1@dynamicjuly.onmicrosoft.com	2/4/2021
Patch sql server	Started	User1@dynamicjuly.onmicrosoft.com	7/6/2021
Call mum	Started	User1@dynamicjuly.onmicrosoft.com	6/7/2021
Buy fruits	Started	User1@dynamicjuly.onmicrosoft.com	8/8/2021
Call Kent	Not Started	User1@dynamicjuly.onmicrosoft.com	8/8/2021
Buy a Porsche	Done	User1@dynamicjuly.onmicrosoft.com	1/9/2021

Task	Status	in charge	Deadline
Feed the cat	Started	User1@dynamicjuly.onmicrosoft.com	1/1/2021
Call jon	Not Stared	User1@dynamicjuly.onmicrosoft.com	2/4/2021
Patch sql server	Started	User1@dynamicjuly.onmicrosoft.com	7/6/2021
Call mum	Started	User1@dynamicjuly.onmicrosoft.com	6/7/2021
Buy fruits	Started	User1@dynamicjuly.onmicrosoft.com	8/8/2021
Call Kent	Not Started	User1@dynamicjuly.onmicrosoft.com	8/8/2021
Buy a Porsche	Done	User1@dynamicjuly.onmicrosoft.com	1/9/2021

2. If you already have existing data, change the format to Text :

	A	B	C	D	E	F
1						
2	tasks	Status	in charge	Deadline		
3	Feed the cat	Started	sergeluca@doctorflow.onmicrosoft.com	1/1/2019		
4	Call Jon	Not Started	sergeluca@doctorflow.onmicrosoft.com	10/10/2019		
5	patch SQL Server	Started	sergeluca@doctorflow.onmicrosoft.com	7/6/2019		
6	Call Mum	Started	sergeluca@doctorflow.onmicrosoft.com	6/7/2019		
7	Buy fruits	Done	sergeluca@doctorflow.onmicrosoft.com	8/8/2019		
8	Call kent	Not started	sergeluca@doctorflow.onmicrosoft.com	8/8/2019		
9	Buy a Porsche	Done	sergeluca@doctorflow.onmicrosoft.com	1/9/2019		
10						
11						

3. Name it **Tasks.xlsx**
4. Change the value of the **in-charge column** to your e-mail address and adjust some deadline value.
5. We want to write a Flow that will loop through all tasks, and that will check if the task is overdue. To do so, **create a new Instant cloud flow** and use a **Manually trigger a flow** as a trigger. Name it **my Overdue Tasks**.
6. Define a variable of type Array by adding **an Initialize variable** action as illustrated hereafter:

Manually trigger a flow

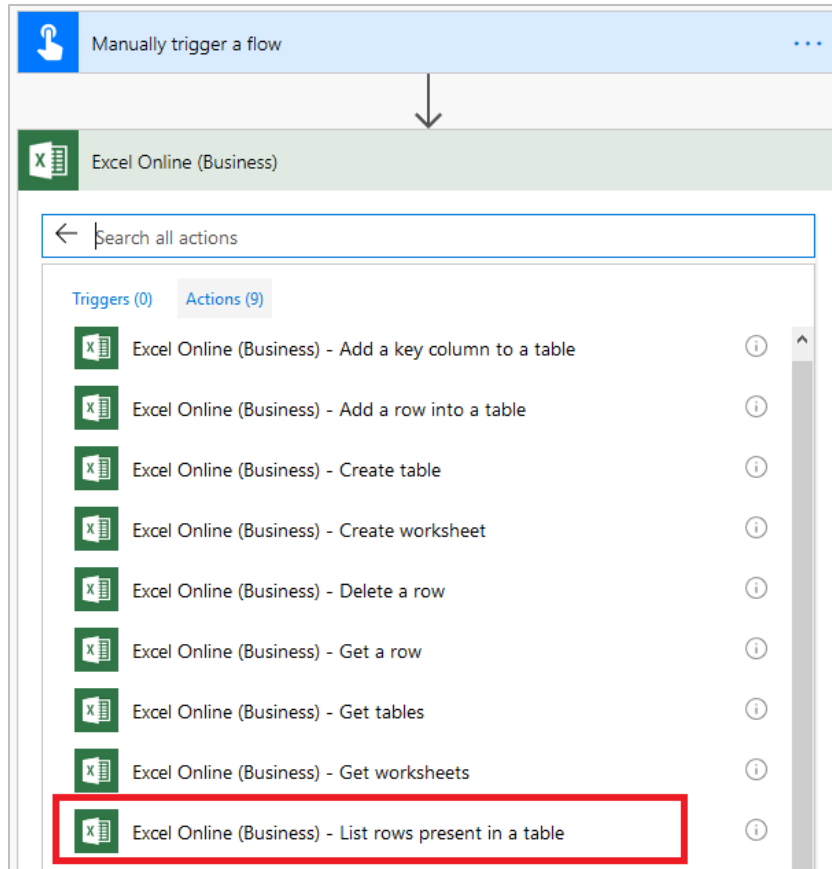
Initialize variable - Tasks Overdue

* Name: Tasks overdue

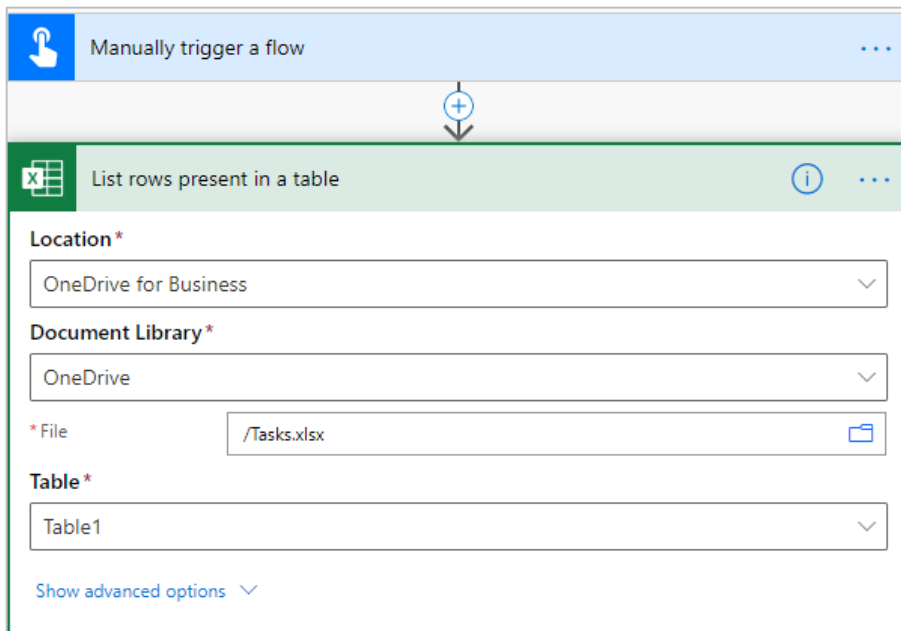
* Type: Array

Value: Enter initial value

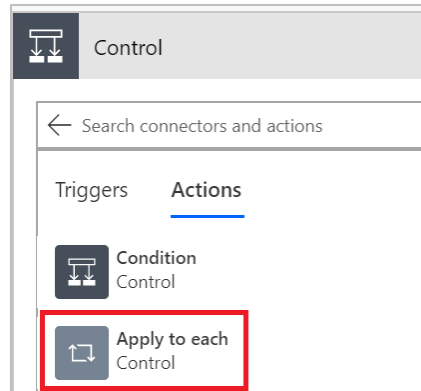
7. The Flow needs to connect to the **Tasks.xlsx** file, so add an **Excel Online (Business) – List rows present in a table** action (Not OneDrive !!!) as illustrated below:



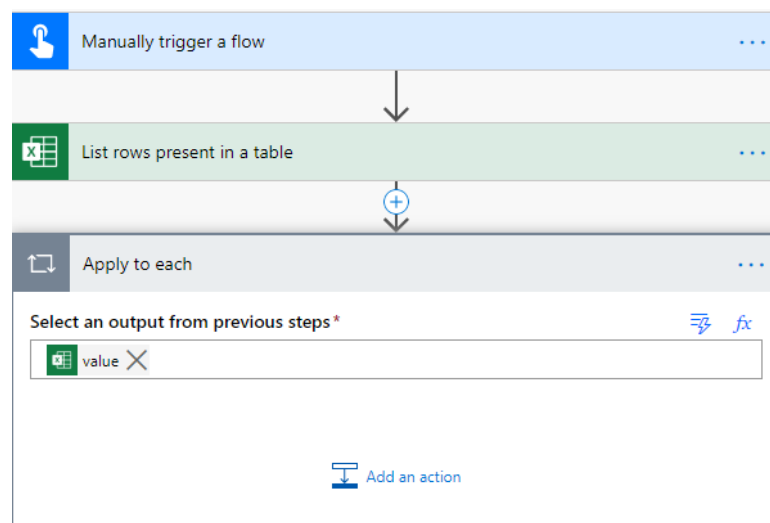
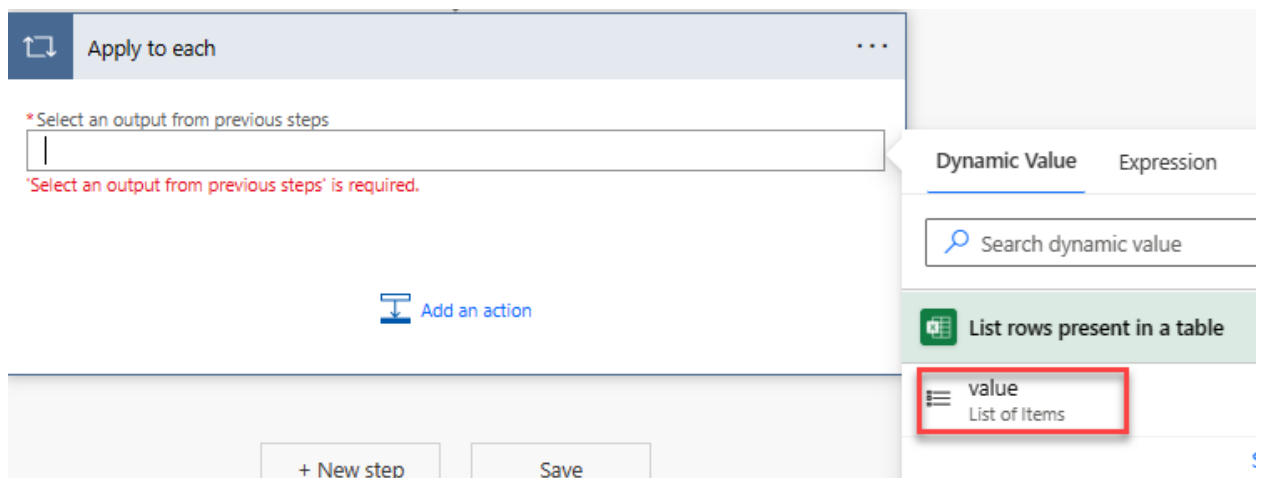
8. Set the action properties, as illustrated in the following picture:



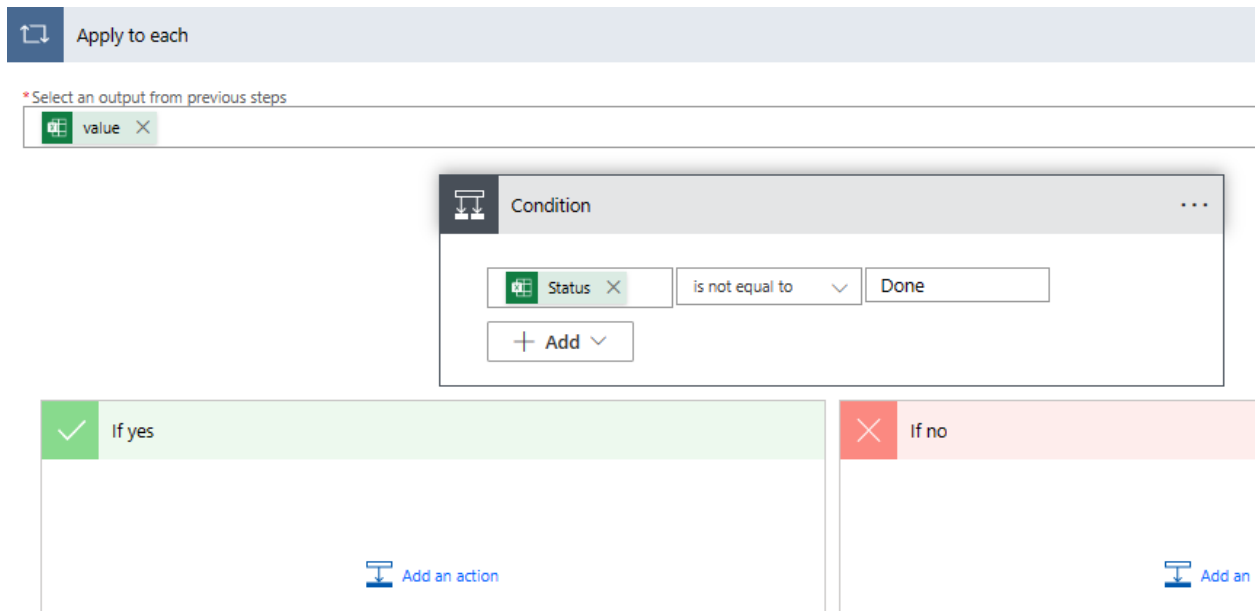
9. Let's loop through all tasks, so we need to add an **Apply to each**:



10. Select the add a **Dynamic value** in the **Apply to each** action:



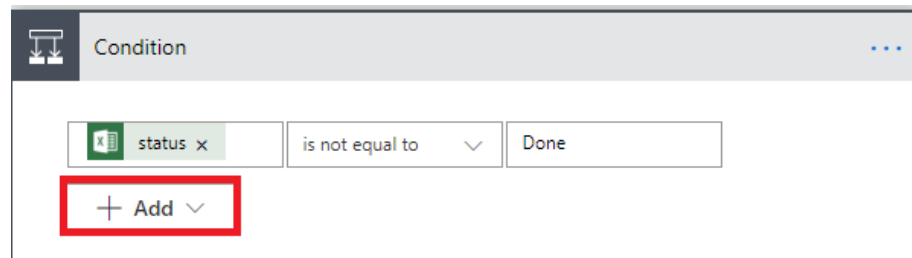
11. Add a **Condition** to filter the task where **Status** is not equal to **Done**:



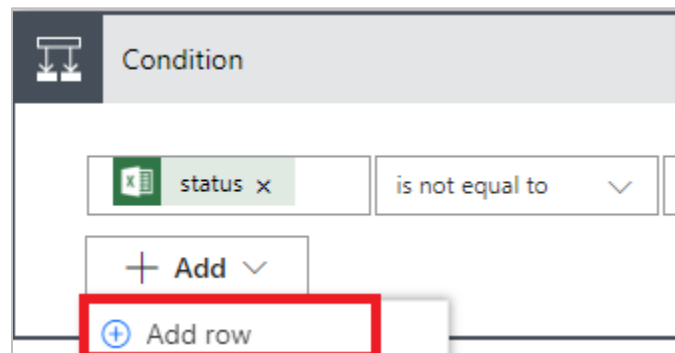
Make sure the operator used is **is not equal**.

12. Add a new sub condition that will check if the task is overdue:

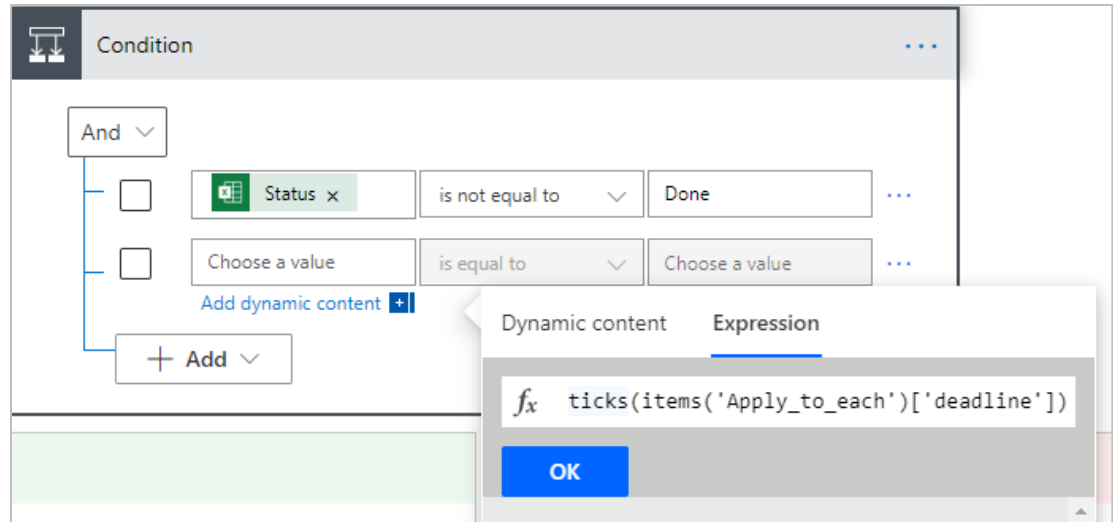
a) Click Add:



b) Select Add row:

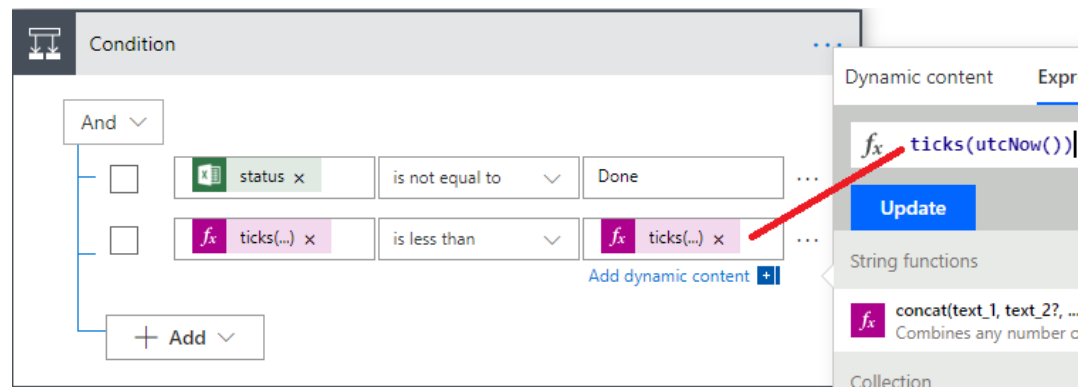


c) In "Choose a value", type the expression **(ticks())**, and In the parentheses, grab the dynamic value for Deadline and click Ok:



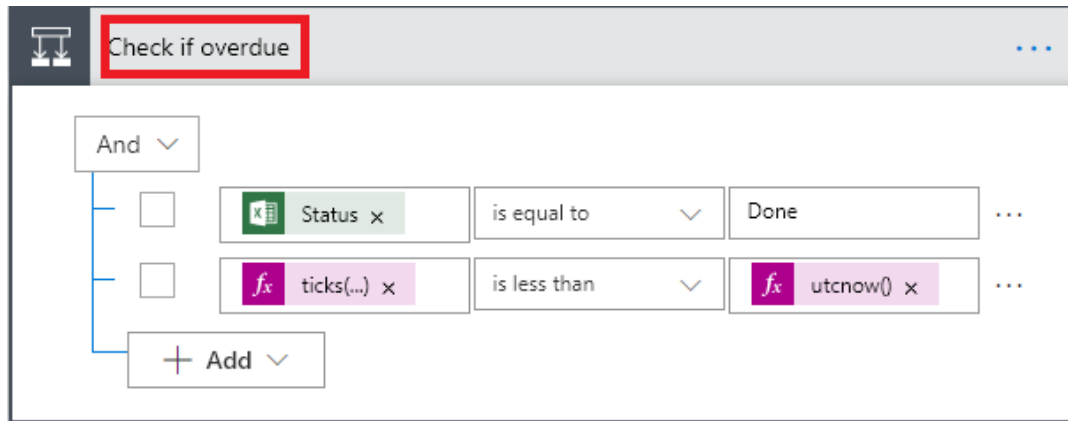
Since Deadline is a timestamp and has a string format, the **ticks** expression returns the number of ticks (100 nanosecond intervals) since 1st January 1601. By using ticks, we can compare two different timestamp values.

- d) Select the operator is less than
- e) Type the following expression (and click Ok):

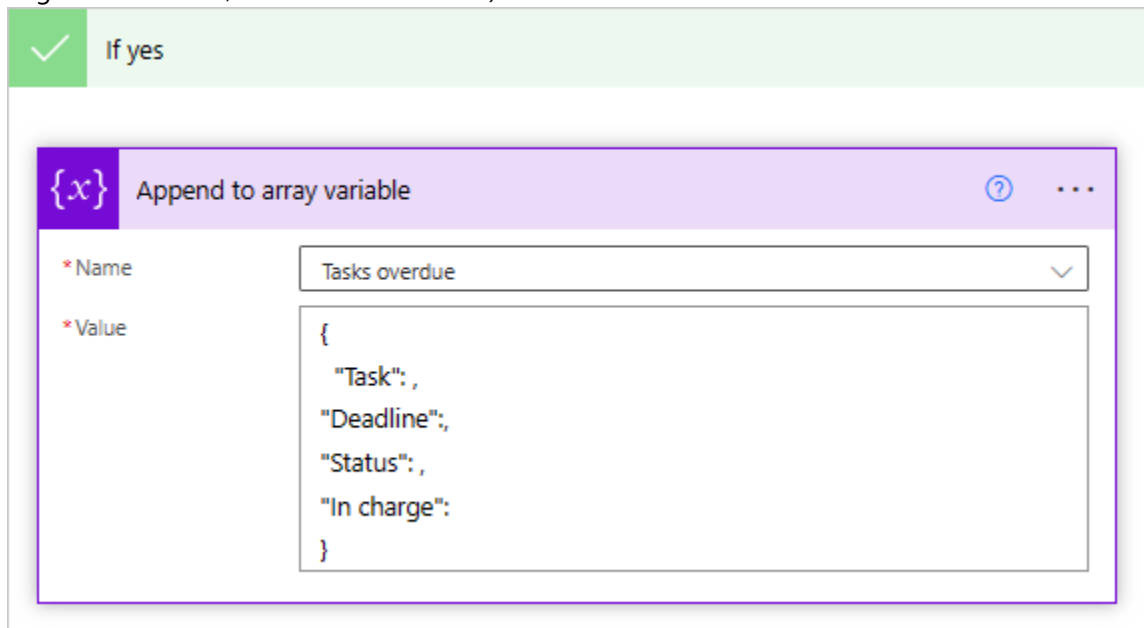


13. In the **If yes branch**, add a new **condition** where we will check if the due date is overdue:

14. Rename the condition to **Check if overdue** to provide your Flow with more clarity:



15. In the If yes branch, add an **Append to array variable** action with the following JSON text (don't forget the commas, where it makes sense):



16. Add the dynamic value of **tasks**, **Deadline**, **Status** and **In charge** into this action:

✓ If yes

{x} Append to array variable

* Name Tasks overdue

* Value

```

{
  "Task": tasks ,
  "Deadline": Deadline ,
  "Status": Status ,
  "In charge": in charge
}

```

17. After the **Apply to each** add a Create HTML table action where you will generate a report:

↩ Apply to each

{ Create HTML table

* From {x} Tasks overdue

* Columns Custom

Header	Value
Task	f_x item(...) ✕
Deadline	f_x item(...) ✕
In charge	f_x item(...) ✕
Status	f_x item(...) ✕

18. Here is how the Task value has been implemented:

The screenshot shows a workflow editor with the following steps:

- Initialize variable - Tasks Overdue
- List rows present in a table
- Apply to each
- Create HTML table

The 'Create HTML table' action is configured with the following settings:

- From:** Tasks overdue
- Columns:** Custom
- Header:** Task, Deadline, In charge, Status
- Value:** item(...)

The 'Dynamic Value' panel on the right shows the expression `item()?['Task']` for the 'Task' column. The 'Formulas' tab is selected, and the 'String functions' section is visible.

Run the flow and check the output of the Create HTML table action:

Create HTML table

0s

INPUTS

Show raw inputs >

Format

Html

From

```
{
  "Task": "Feed the cat",
  "Deadline": "01/01/2019",
  "Status": "Started",
  "In charge": "Started"
},
```

OUTPUTS

Show raw outputs >

Body

Task	Deadline	In charge	Status
Feed the cat	01/01/2019	Started	Started
Call Jon	01/01/2020	Not Started	Not Started
patch SQL Server	01/01/2021	Started	Started
Call Mum	01/01/2022	Started	Started

19. We can drastically simplify this flow : instead of filtering the Tasks with the Status Done in a condition, you can also keep it more straightforward by using an OData Filter (click on Show advanced options) in the List rows present in a table action:

List rows present in a table

Location *

OneDrive for Business

Document Library *

OneDrive

* File

/Tasks.xlsx

Table *

Table1

Filter Query

Status ne 'Done'

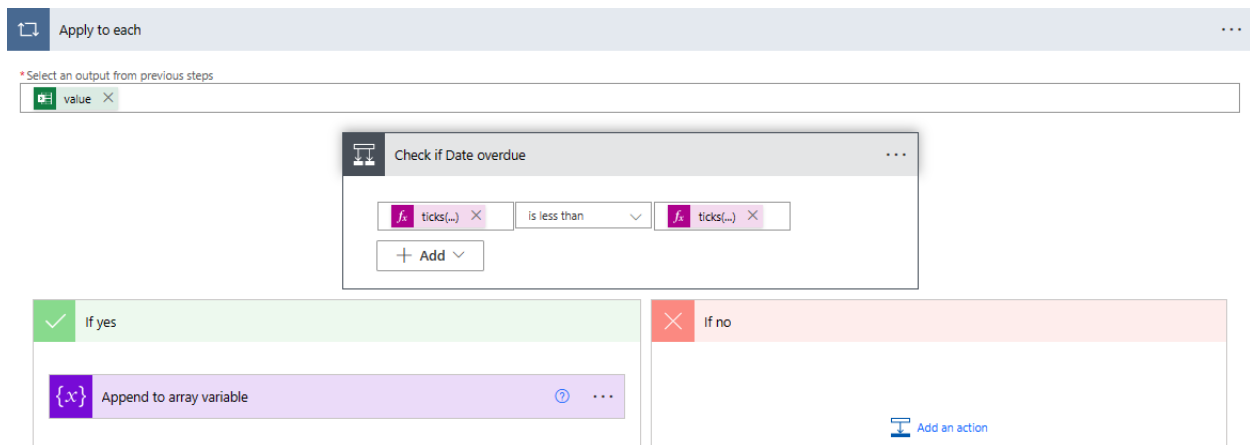
Order By

Using OData Filter (Filter query) is often considered as a good practice because the data are filtered on the server-side, and that can make your Flow running faster by using fewer resources.

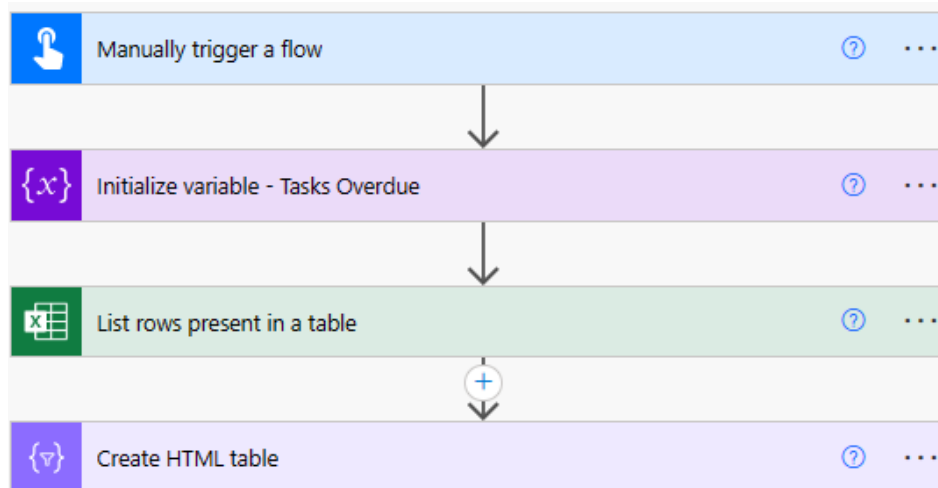
Note: oData expressions are case sensitive!

How can this simplify how flow ?

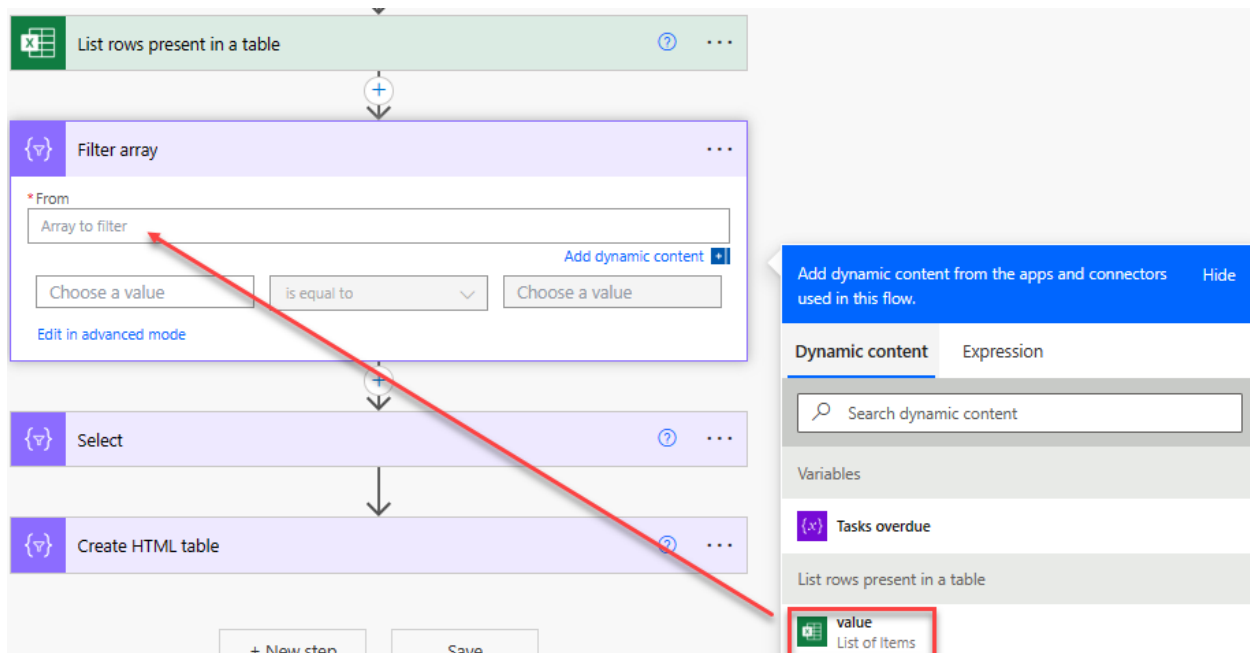
We can remove the **is not equal to Done** condition:



20. But the oData filter does not allow us to compare two dates. Nevertheless, we can **remove** the **Apply to Each**:

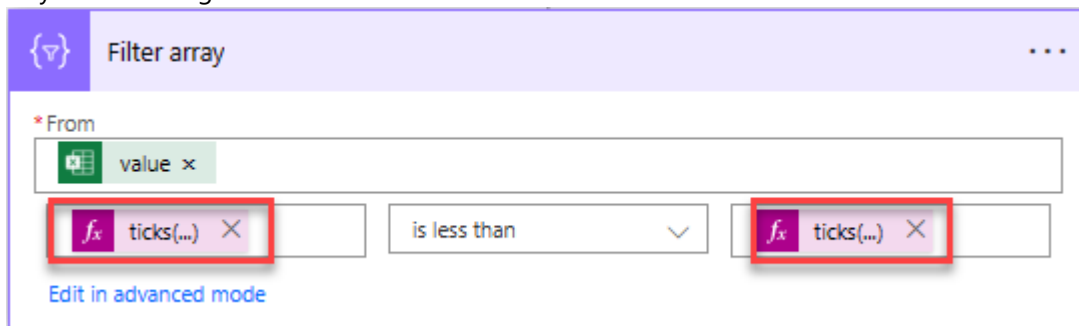


But we still need to feed the Create HTML table action by adding a Filter Array action that we will **connect** to the **List rows present in a table** action:

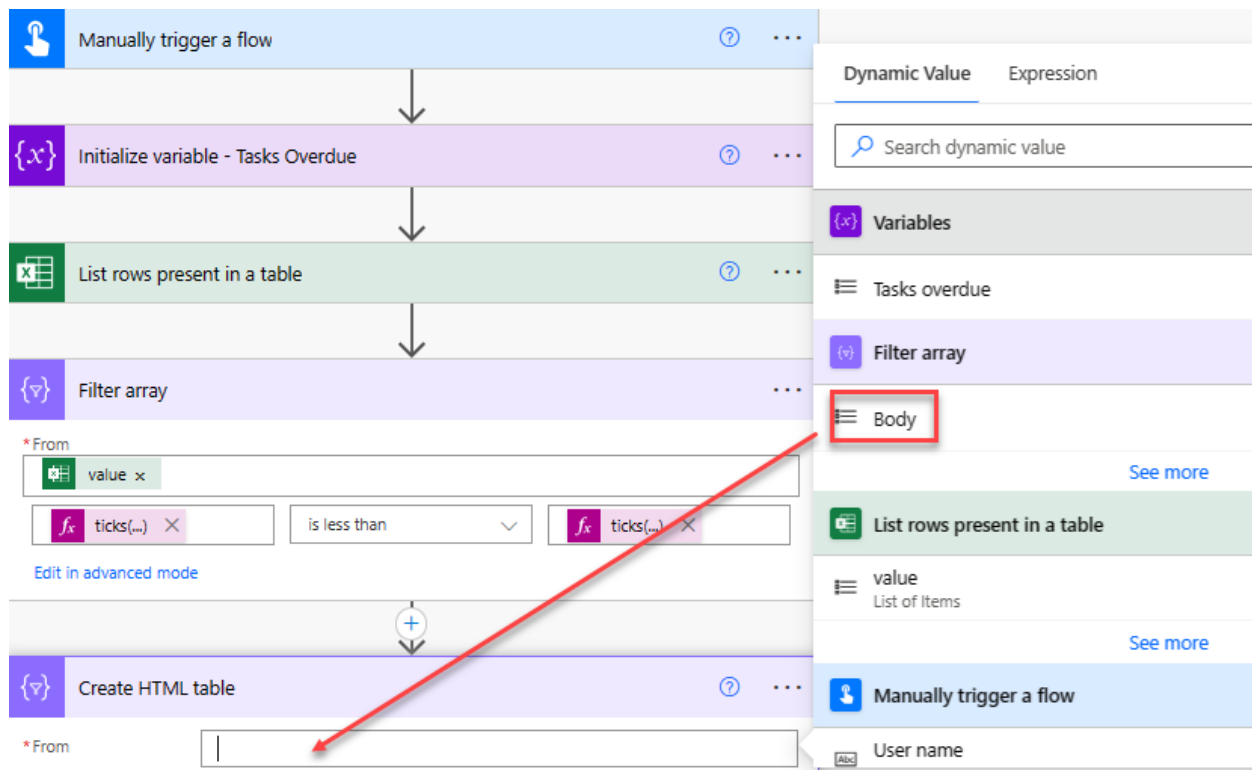


In some cases, filtering data in the Flow itself cannot be avoided. In this case, you can use the Filter Array action instead of adding conditions. The ticks() expression defined before can be reused in the Filter array:

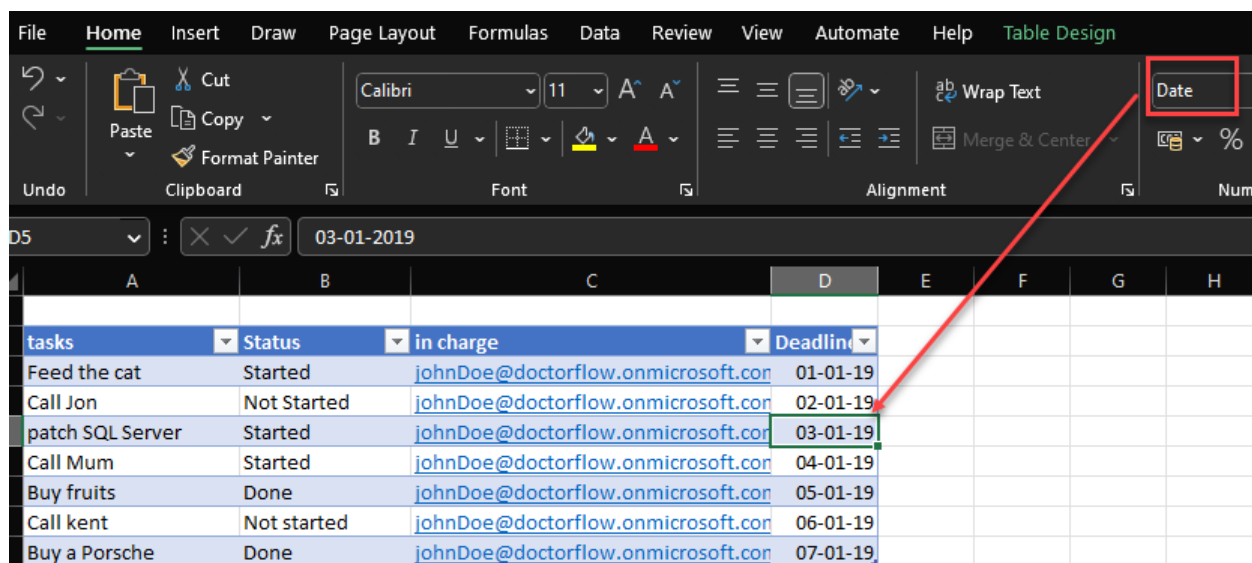
21. Try the following code:



22. We still need to **connect** the **Create HTML table** action and the **Filter array**, by using its **Body** property:



23. You will notice that the Deadline format in the Excel document is in a text format, but if we switch this format to a Date format (and re-enter the dates), our flow will crash:



Indeed, if we switch the Deadline column type to **General** and if re-write (or just update) the data our flow run will generate an error:

Error Details

The execution of template action 'Filter_array' failed: The evaluation of 'query' action 'where' expression '@less(ticks(item()['Deadline']), ticks(utcNow()))' failed: 'In function 'ticks', the value provided for date time string '43466' was not valid. The datetime string must match ISO 8601 format.'

24. Deadline field is now a number that give you the number of days between 1899-12-30 and your specific date. The ticks() function expects a real date, so it fails here. It should be updated with:

```
ticks(addDays('1899-12-30',int(items('Apply_to_each')['deadline'])))
```

25. You can test this using the Excel file **Tasks with Date format.xmls** in the provided in the resources.zip file.

We need your feedback

Do you want to report an issue or suggest something? We need your feedback:

<https://github.com/Power-Automate-in-a-day/Training-by-the-community/issues>

Lab 5. Using the Microsoft Form Connector

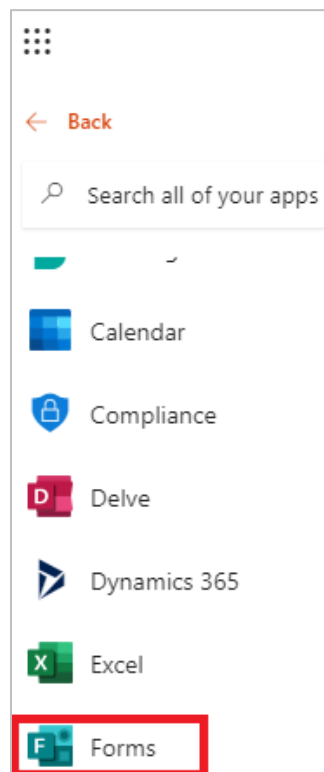
Author: Serge Luca, aka "Doctor Flow"

Duration: 15 minutes.

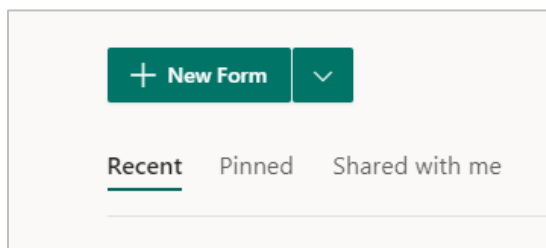
Scenario: In this lab, we will explore how to catch data submitted by a Microsoft Form. The case of a course evaluation will be implemented.

Tasks:

1. Go to your Office 365 Applications and select **Forms**:



2. Click **New Form** :



3. Name the form **Training evaluation**:

Training evaluation - Saved

Questions Responses

Training evaluation

Enter a description

+ Add new

- Click **Add new**, select **Text**, and name the field **Your First Name**:

Training evaluation

1. Your First Name

Enter your answer

☐ Long answer ☒ Required ...

+ Add new

- Proceed like this for the fields **Your Last Name**, and **your e-mail**.
- Add a field How as the course content and Select **rating**

When a new response is submitted

Form Id *

Training evaluation

- After the trigger, add a new action **Get response details** from the Form connector; select the Form id **Training** evaluation and grab the **Response Id** from the trigger:

When a new response is submitted

Get response details

Form Id *

Training evaluation

Response Id *

Response Id

- Add a Compose for each field of the response and bind them to their corresponding dynamic value:

FirstName

Inputs *

Your First Name

LastName

Email

Course evaluation

Trainer evaluation

13. Check if one of the rating is less than 4; if that is the case, send an e-mail to yourself with the evaluation details: in the condition statement, don't forget to use the **int()** function in order to compare the ratings to 4:

The screenshot displays a Power Automate flow configuration. At the top, a 'Condition' step is set to 'Or'. It contains two identical conditions: 'int(...) is less than 4'. Below the condition, there are two paths: 'If yes' (green background) and 'If no' (red background). The 'If yes' path leads to a 'Send an email (V2)' action. The email action is configured with the following details:

- To:** serge Luca
- Subject:** Bad course evaluation
- Body:**
 - from : Your First Name, Your Last Name
 - The course evaluation was: How was the course content ?
 - The trainer evaluation was: How was the trainer

The email body uses dynamic content (represented by icons) to insert values from previous steps. A 'Show advanced options' link is visible at the bottom of the email configuration panel.

14. Submit a bad evaluation and check your e-mail.

We need your feedback

Do you want to report an issue or suggest something? We need your feedback:

<https://github.com/Power-Automate-in-a-day/Training-by-the-community/issues>



Lab 4. Flow control, variables, expressions

Lab 4. Flow control, variables, expressions

Author: Serge Luca, aka "Doctor Flow"

Learning objectives: Flow control, expressions, variables, using Date/Time

Duration: 50 minutes.

Scenario: We have a list of offices in an Excel sheet. Create a Flow that will send a report describing this list of offices, including the biggest office.

Task 4.1: Create an Excel workbook and a scheduled flow

1. Create an Excel workbook to use in this lab.
 - a. In your One Drive (for Business), create an Excel workbook named **Offices.xlsx**.
 - b. Add two columns like the screenshot below, with the cities and capacities data, and then format the data as a table with headers:

1	Contoso Offices	
2		
3	city	capacity
4	London	100
5	Brussels	250
6	Seattle	80
7	Vancouver	200
8	Toronto	400
9	Antwerpen	15
10	Warsaw	300
11	Paris	54
12	Berlin	70
13	Amsterdam	60
14	Montreal	78
15		
16		
17		
18		

Note: This document contains the list of offices of Contoso Corp. Each office has a limited number of seats.

Every month a report describing the list of offices and the total number of seats is sent to the management (in this case the management is...yourself). The e-mail should look like this:

Office Capacity Report



Today, 3:10 PM

This message was sent with low importance.

The biggest office is : Toronto

Its capacity is : 400

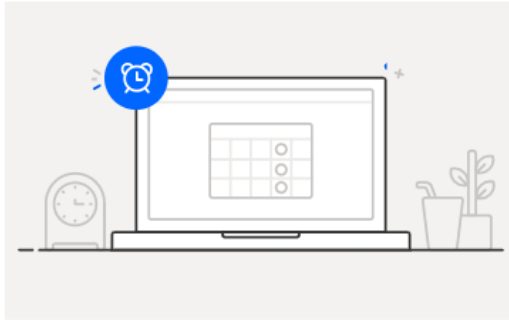
The total capacity is :

List of offices

City	Capacity
London	100
Brussels	250
Seattle	80
Vancouver	200
Toronto	400
Antwerpen	15
Warsaw	300
Paris	54
Berlin	70
Amsterdam	60
Montreal	78

2. Create a flow to generate this e-mail report.
 - a. Create a **New flow** > **Scheduled cloud flow**
 - b. Use the following screenshot to name the flow, and set the flow frequency.

Build a scheduled cloud flow



Stay on top of what's important without the effort—you choose when and how often the flow runs.

Examples:

- Automate team reminders to submit expense reports
- Auto-backup data to designated storage on a regular basis

Flow name

Office capacity

Run this flow *

Starting 2/5/23 at 10:00 AM

Repeat every 1 Month

This flow will run:

Every month

Skip

Create

Cancel

c. Click **Create**.

The following flow will be generated:

Recurrence

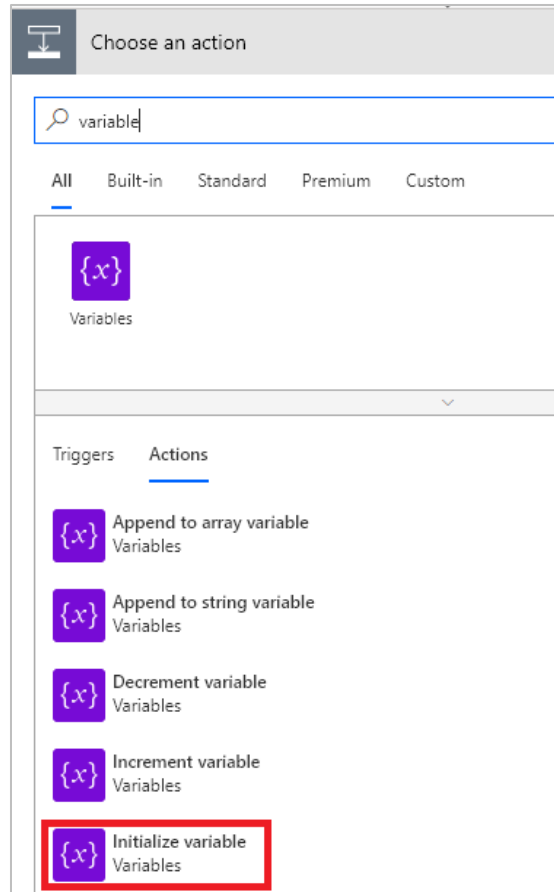
* Interval1

* FrequencyMonth

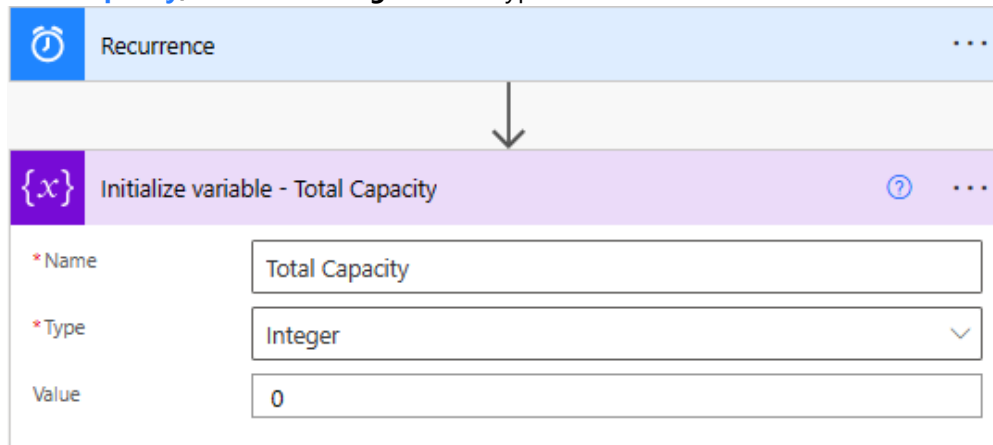
Show advanced options

3. The first challenge will be to define the **Total capacity**. Use the following steps to create a variable that will contain that value.

a. Select **New Step** and select **Initialize variable**:



- b. Rename this action to **Initialize variable Total capacity**, set the variable name **Total Capacity**, and select **Integer** as the type with an initial **Value** of **0**:



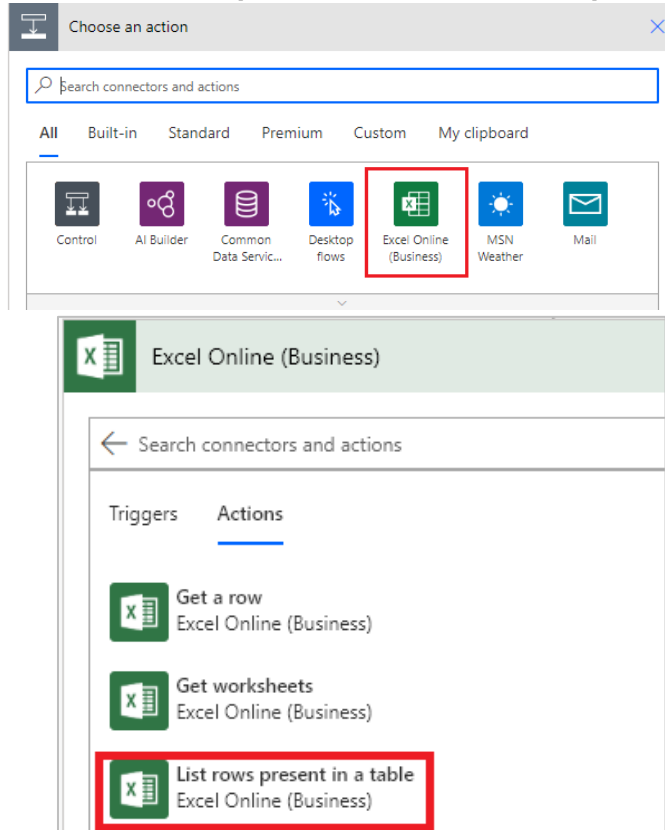
4. Click on **Save** button to **Office Capacity** flow .

Task 4.2: Extend the flow to loop through all offices

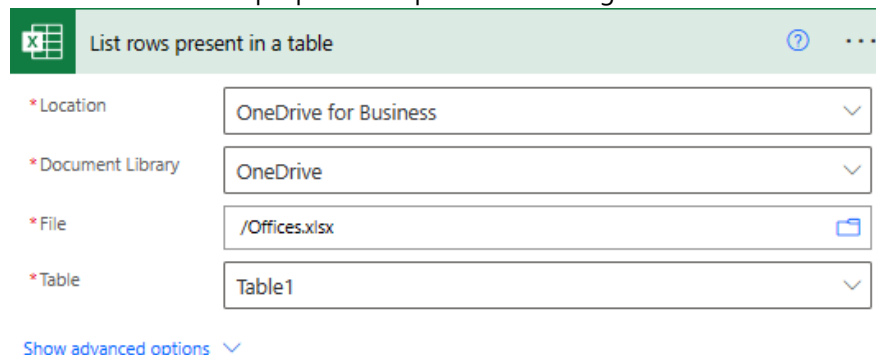
In this task, you will make it loop through all offices, retrieve their capacity, and increment the Global Capacity variable to calculate the total capacity.

1. To retrieve the list of offices.

a. **Select New step > Excel Online > List rows present in a table:**



b. Set the Excel action's properties as per the following screenshot:



c. Click **Show Advanced option** and type **capacity DESC** in the **Order By** field:

List rows present in a table

* Location: OneDrive for Business

* Document Library: OneDrive

* File: /Offices.xlsx

* Table: Table1

Filter Query: An ODATA filter query to restrict the entries returned.

Order By: **capacity DESC**

Top Count: Total number of entries to retrieve (default = all).

Skip Count: The number of entries to skip (default = 0).

Select Query: Comma-separated list of columns to retrieve (first 500 by default).

DateTime Format: Select an item

[Hide advanced options](#)

2. Loop through the cities

a. **Select New step > Control**

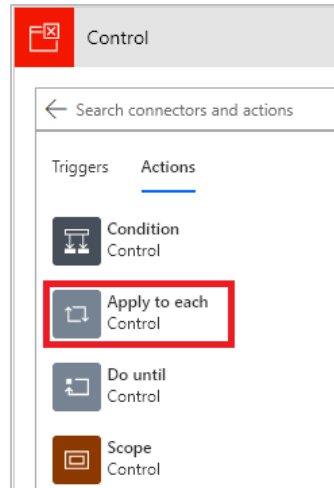
Choose an action

Search connectors and actions

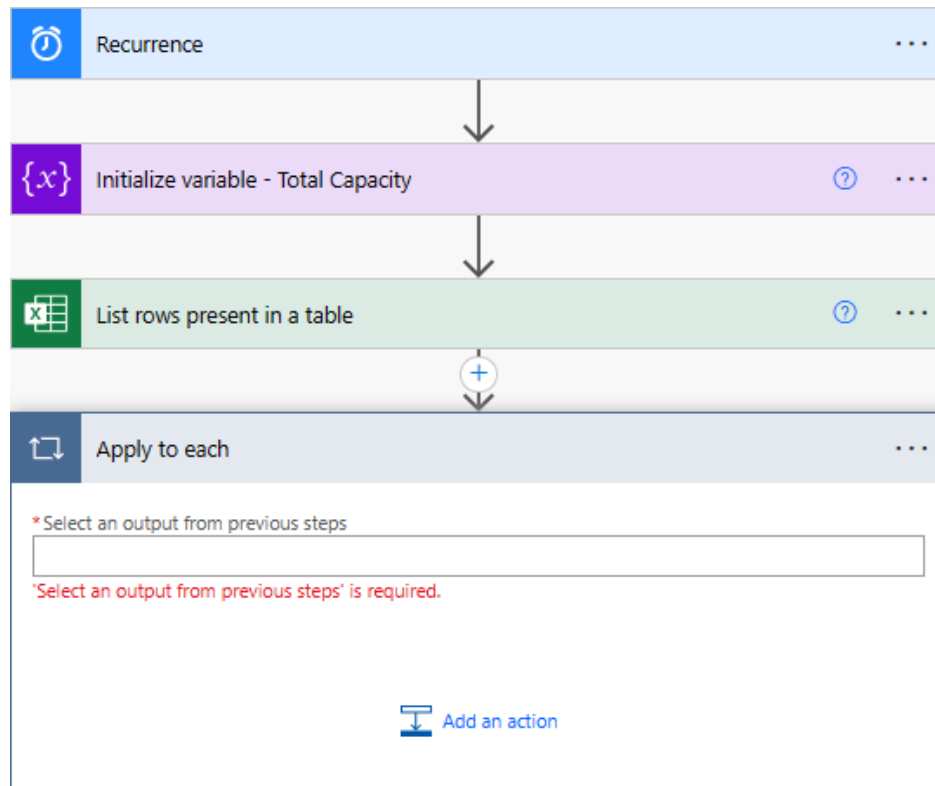
All Built-in Standard Premium

Control Notifications Office 365 Outlook

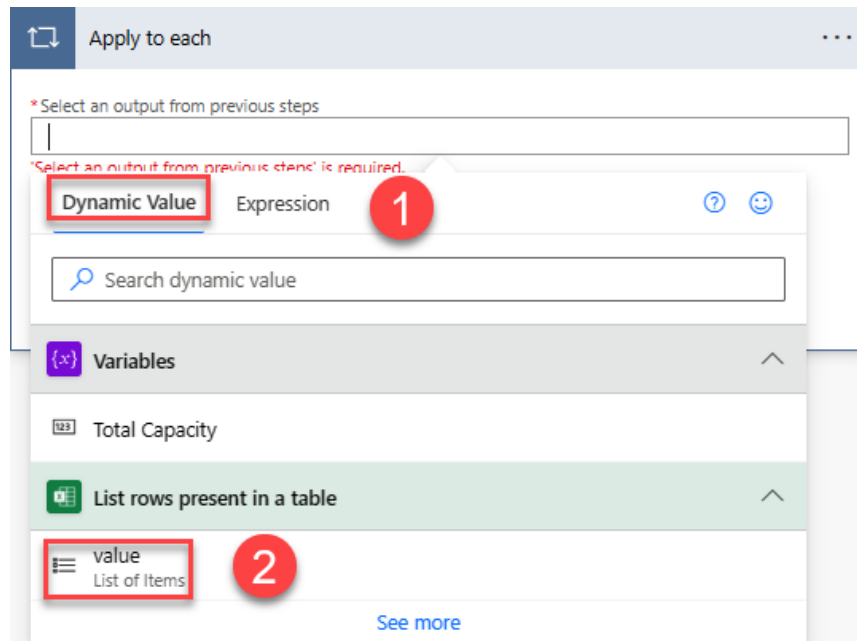
b. Click **Apply to each**:



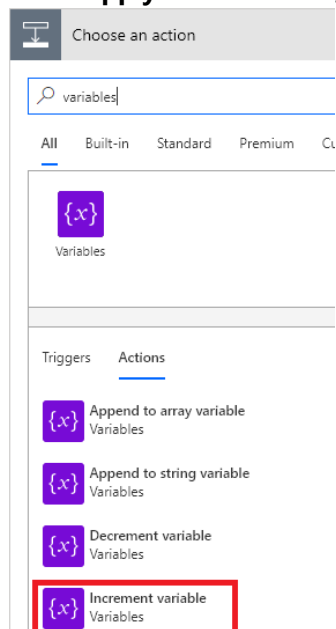
Your flow should now look like this:



3. Configure the **Apply to each** action (it expects a list of values), using the "Add a dynamic value" to select the **value** property from the **List rows present in a table** action.



4. Calculate the current office capacity using a variable and an expression.
 - a. In the **Apply to each** action, click **Add an action > Increment variable**:



- b. In the **Name** drop-down list, select **Total capacity**, and rename the action:

Apply to each

*Select an output from previous steps

value

{x} Increment variable - Total Capacity

* Name Total Capacity

Value Enter a value

- c. Click inside the **Value** text box, and click the **Expression** tab to add an expression

List rows present

Dynamic Value Expression

Search dynamic value

{x} Variables

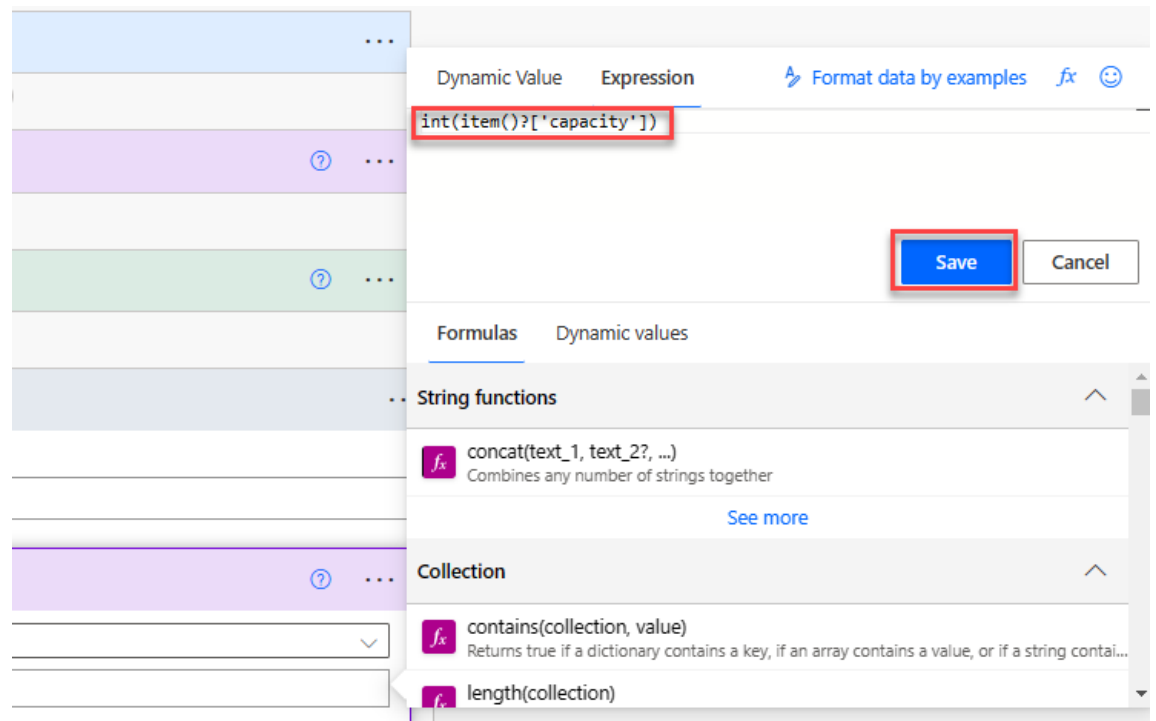
Total Capacity

Apply to each

Current item
Current item

Value Enter a value

- d. Type the following expression: and don't forget the click **Save**:
int(item()?['capacity'])

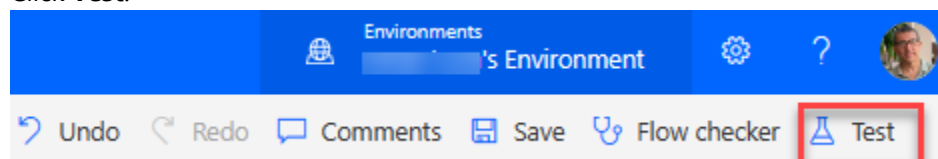


Note:

The **item()** expression retrieves the current record information in the current loop, and **['Capacity']** provides the field name to retrieve. **item()['Capacity']** returns a string. The question mark **?** makes your code more robust by avoiding it crashing if there is no such field (here 'capacity') in the record. The information coming from Excel is a string, and we need to transfer it as a number to increment it to the variable **Total capacity**. To transform a string to an integer (), we use the **int()** function.

There are many other expressions available in Power Automate, and we encourage you to read the documentation related to expressions after doing the labs. You can start from the following web page, <https://powerautomate.microsoft.com/en-us/blog/use-expressions-in-actions/>. The list (reference) of all functions can be found here: <https://learn.microsoft.com/en-us/azure/logic-apps/workflow-definition-language-functions-reference>

5. To test the flow, without waiting one month before it starts, use the **Test** button to manually start the flow on demand (in test mode). This is convenient for testing and debugging purposes.
 - a. Click **Test**.



- b. Select **Manually**:

Test Flow

☒ Manually

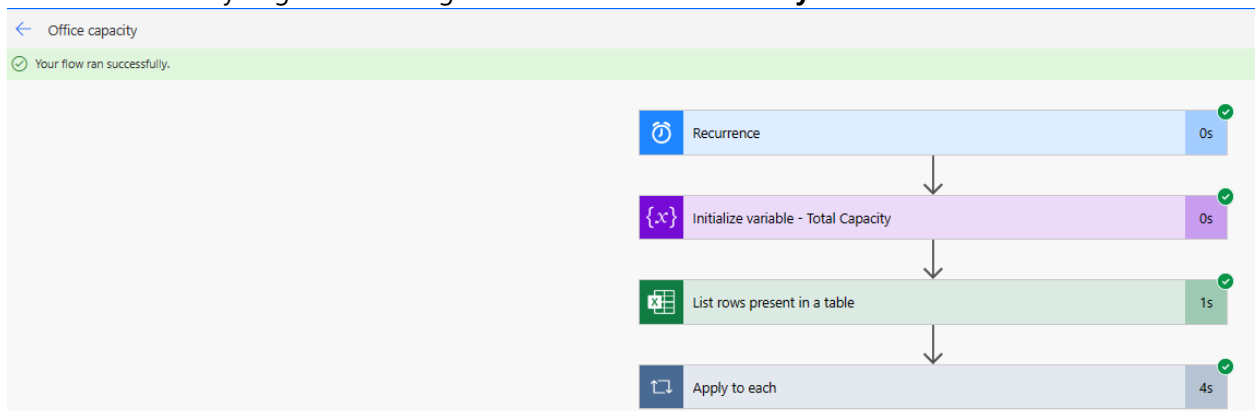
☐ Automatically

Test

Cancel

c. Click on **Run Flow**, and **Done**.

d. Wait until you get the message: **Your Flow ran successfully.**



To check the **Total Capacity** value, you can examine the value of **Total capacity** for each step. For example, in our case, we will check its value once it has completed the loop 11 times: so, type **11** in the **Show** textbox:

Apply to each 4s

< Previous < Previous failed Show 11 of 11 Next failed > Next >

{x}

Increment variable 0s

- e. Click **Increment variable – Total Capacity** to display a value of **1607** (if you use the values in the Excel workbook as defined at the beginning of the lab).

 Apply to each 4s

[< Previous](#) [< Previous failed](#) Show of 11 [Next failed >](#) [Next >](#)

 Increment variable - Total Capacity 0s

INPUTS [Show raw inputs >](#)

Name

Total Capacity

Increment By

78

OUTPUTS [Show raw outputs >](#)

Name

Total Capacity

Value

1607

6. Define 2 new variables

a. Below the variable, **Total capacity** and before the loop, add two new variables named:

- **Biggest office** (type string)
- **Biggest capacity** (type integer)

The screenshot shows a sequence of three 'Initialize variable' actions in a workflow editor. The first action is 'Initialize variable - Total Capacity'. An arrow points down to the second action, 'Initialize variable - Biggest Office', which is highlighted with a red border. This action has fields for Name (Biggest Office), Type (String), and Value (Enter initial value). Below it, another arrow points down to the third action, 'Initialize variable - Biggest Capacity', also highlighted with a red border. This action has fields for Name (Biggest Capacity), Type (Integer), and Value (0).

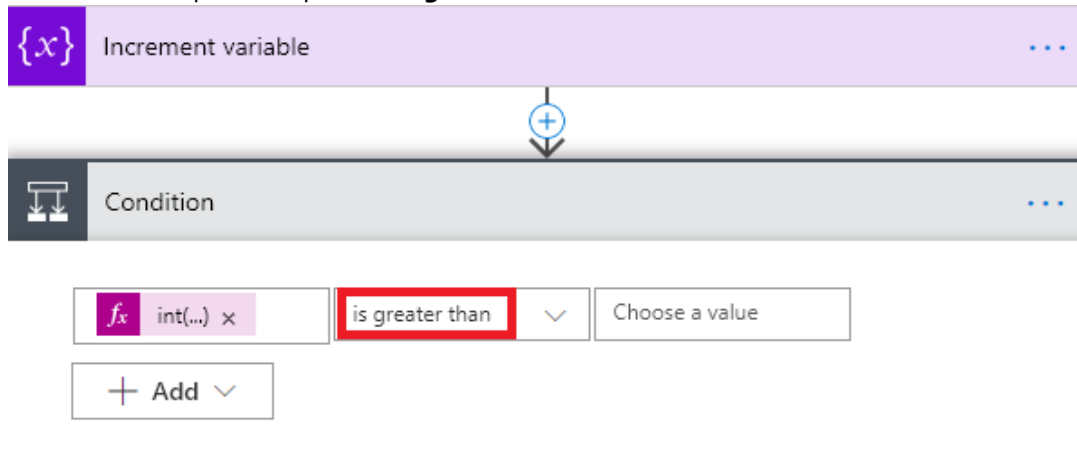
7. **Add a Condition** (from) the Control connector) in the **Apply to each** action:

8. The goal is to compare two numbers and select the larger one. To do so, we need to transform our capacity values into integers. On the left side of the condition, click **Choose a value** and click on **expression**. As we already did it before, type **int(items()?['Capacity'])** as illustrated below:

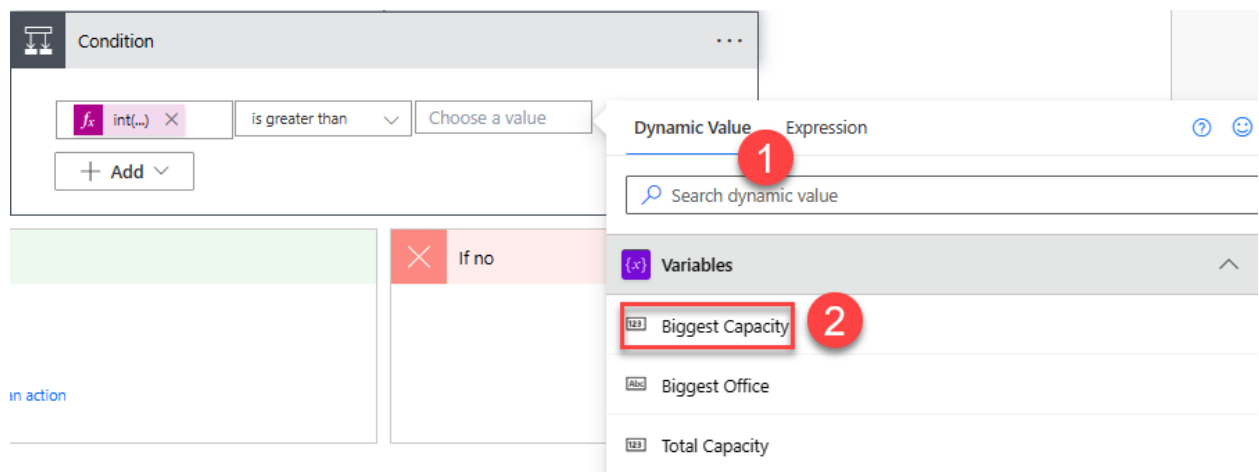
The screenshot shows the 'Condition' configuration window. On the left, there is a 'Choose a value' button and an 'Add' button. The main area has tabs for 'Dynamic Value' and 'Expression'. The 'Expression' tab is selected, and the text 'int(item()?['Capacity'])' is entered in the input field. At the bottom right, there are 'Save' and 'Cancel' buttons.

9. Click **Save**.

10. Select the comparison operator **is greater than**:

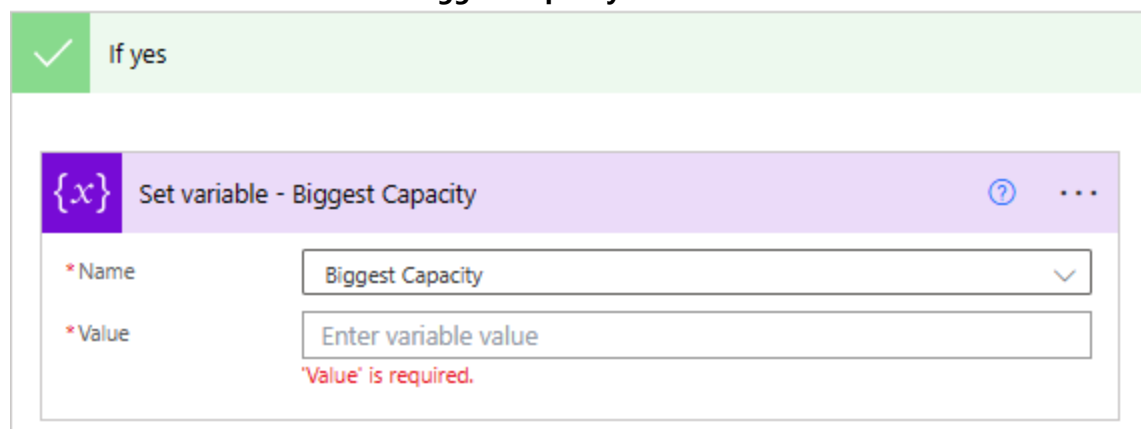


11. In the **Choose a value** textbox, select the **Dynamic value – Variables – Biggest Capacity** :

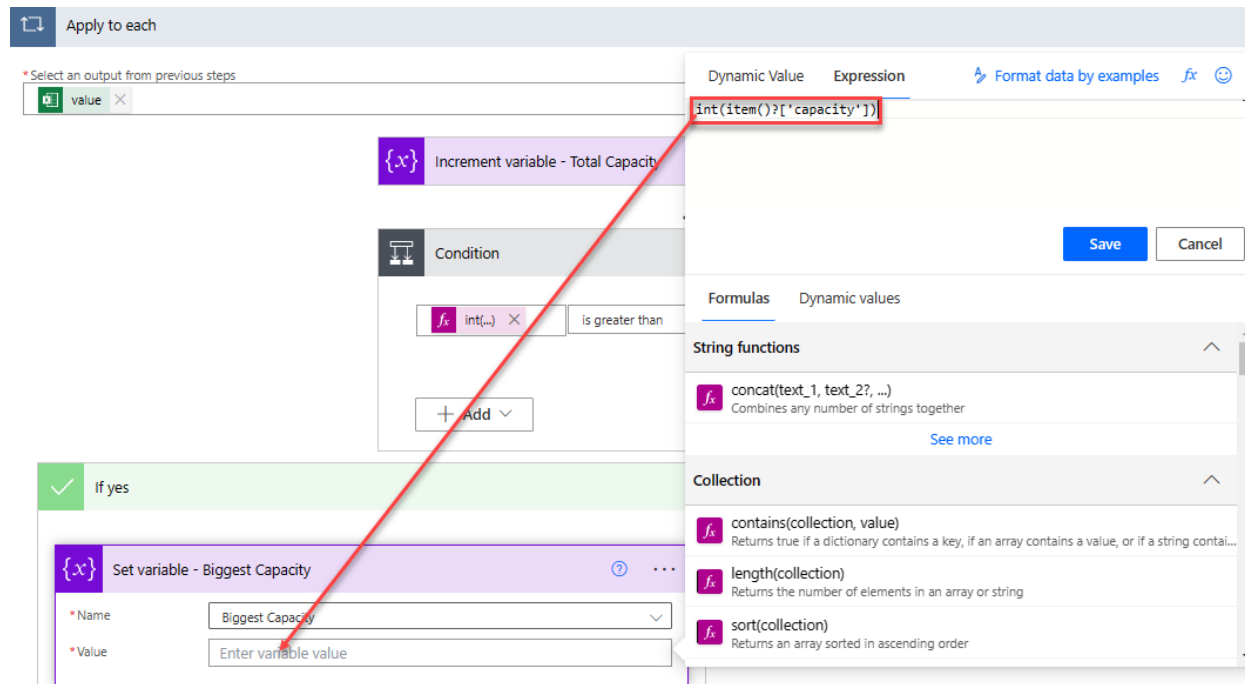


12. Now, in the left **If yes** branch, add a new action **Variables – Set variable** for our **MaxCapacity** variable.

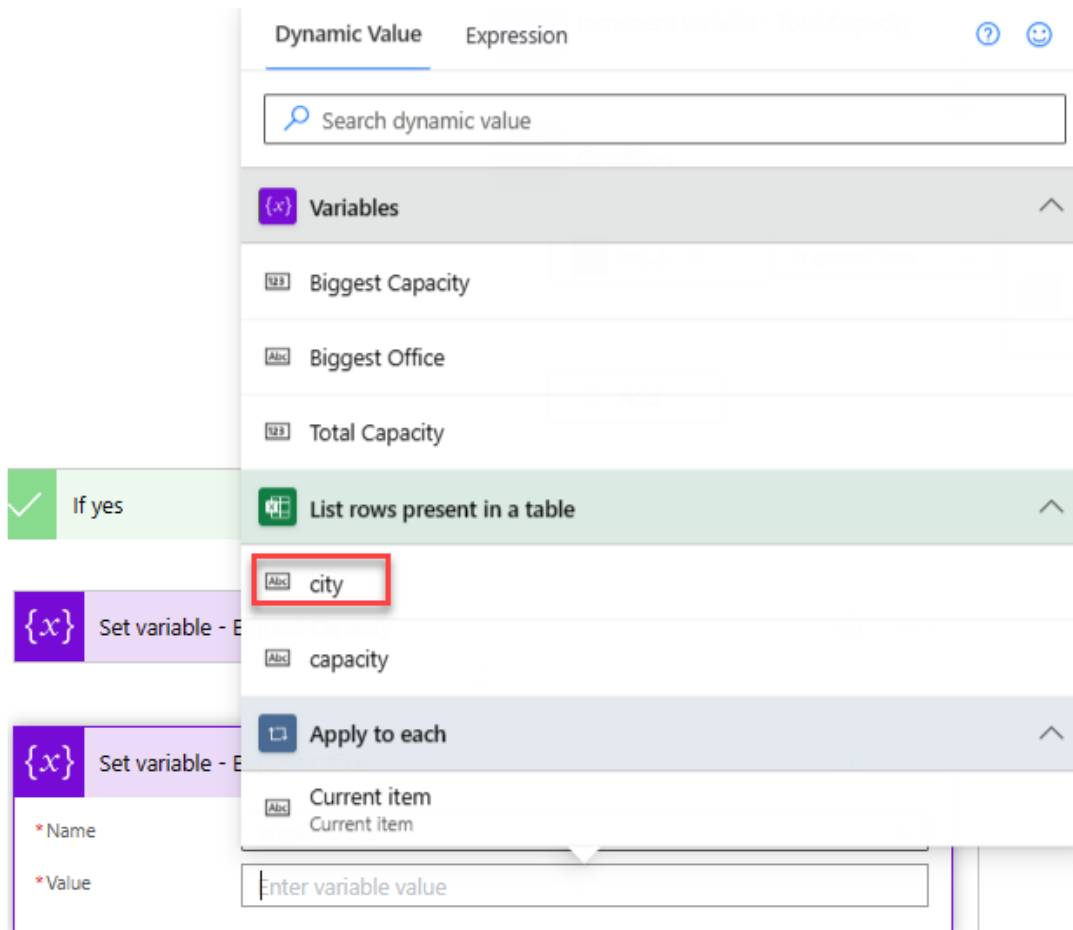
13. Rename the action **Set variable – Biggest Capacity**:



14. and in the **Expression** panel type of this variable **int(items()?['Capacity'])** as illustrated in the next picture.

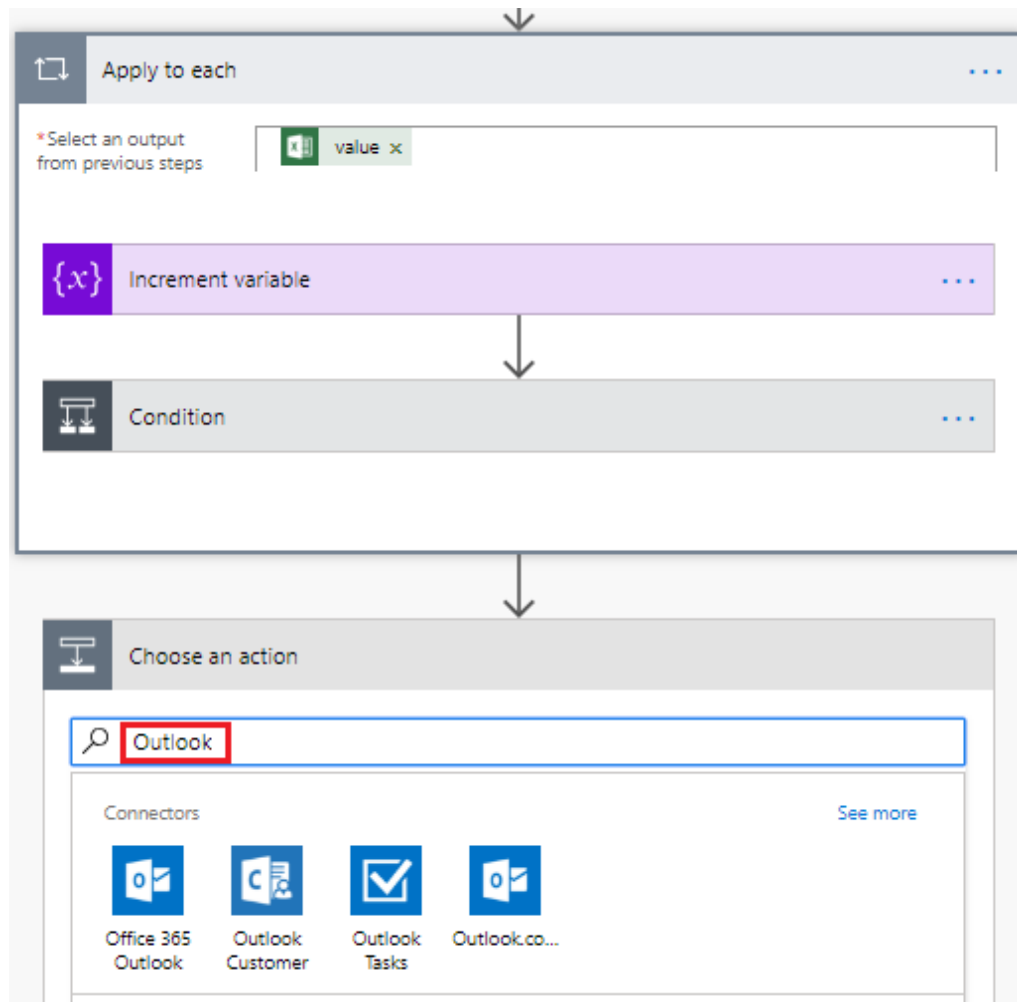


15. In the same left branch of the condition, add another **set variable action** and select the variable **Bigger Office** and assign it a value of **city**. Click on the Dynamic value button to retrieve city:

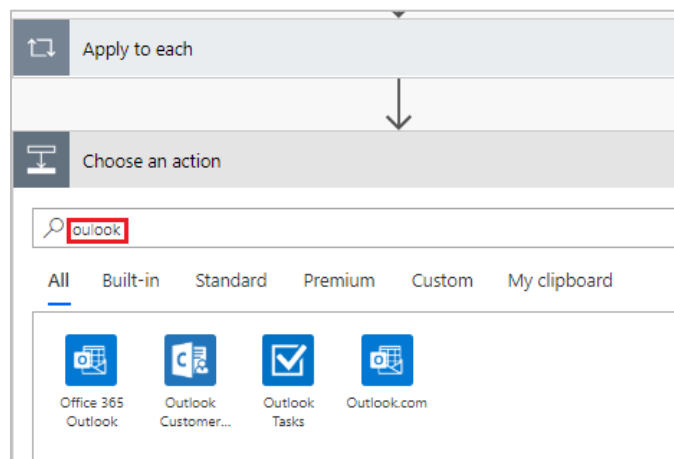


Or, as an alternative, you can create an expression with **item()?['city']**

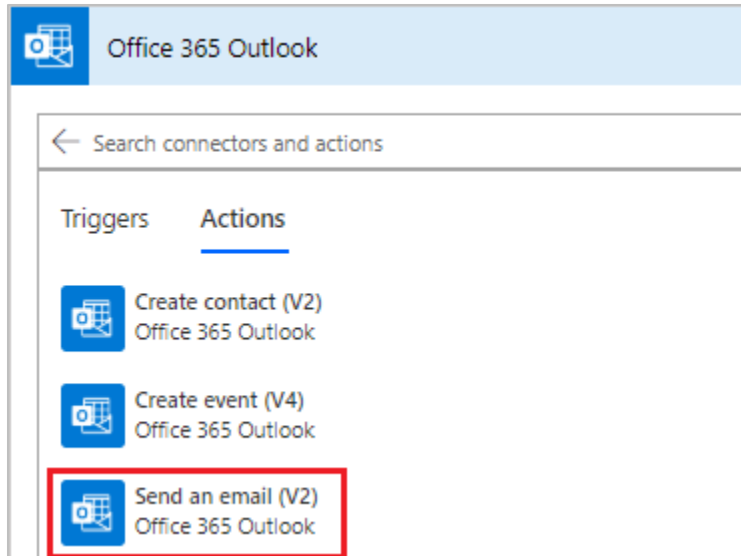
16. Save and test the flow to determine which city has the bigger capacity (Toronto in our case). You can debug the flow or add a notification (or send an e-mail to yourself).
17. Next, let's send an e-mail by adding an **Outlook 365 Outlook - Send an e-mail (v2)** action **after the Apply to each:**
 - a. Find the action by typing Outlook:



- b. In the Connectors list click **Office 365 Outlook**:



- c. Select the action **Office 365 Outlook – Send an e-mail (V2)**:



- d. Fill-in the Send an e-mail action with the following values
- In the **To field** provide your e-mail address
 - In the **Subject**, type "**Office Capacity Report.**"
 - In the **body** type the following text:

The screenshot shows the configuration form for the 'Send an email (V2)' action. The form has three main sections: 'To*', 'Subject*', and 'Body*'. The 'To*' field contains the text 'SL'. The 'Subject*' field contains the text 'Office Capacity Report'. The 'Body*' field contains the text: 'The biggest office is:', 'Its capacity is:', and 'The total capacity is:'. Below the body field, there is a 'Show advanced options' link with a downward arrow.

- We will now add the variable's value directly in the **body**
- Move the cursor just after the colon of *The biggest office is:*

Send an email (V2)

To*

Subject*

Office Capacity Report

Body*

Font 12 B I U

The biggest office is: |
 Its capacity is:
 The total capacity is:

Show advanced options

vi. Click Dynamic value and select the variable Biggest office:

Dynamic Value Expression

Search dynamic value

Variables

Biggest Office

Apply to each

Current item
Current item

List rows present in a table

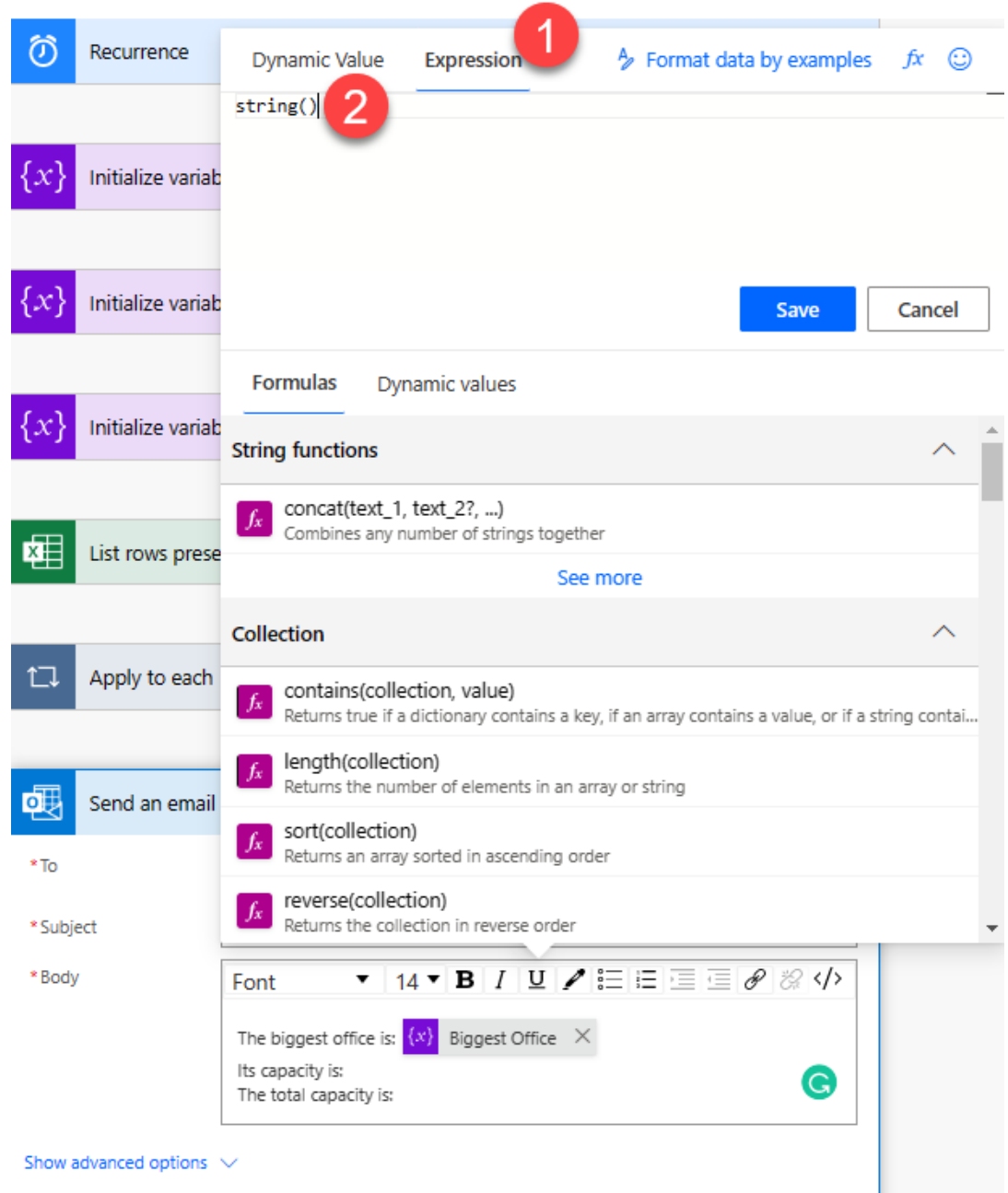
city

capacity

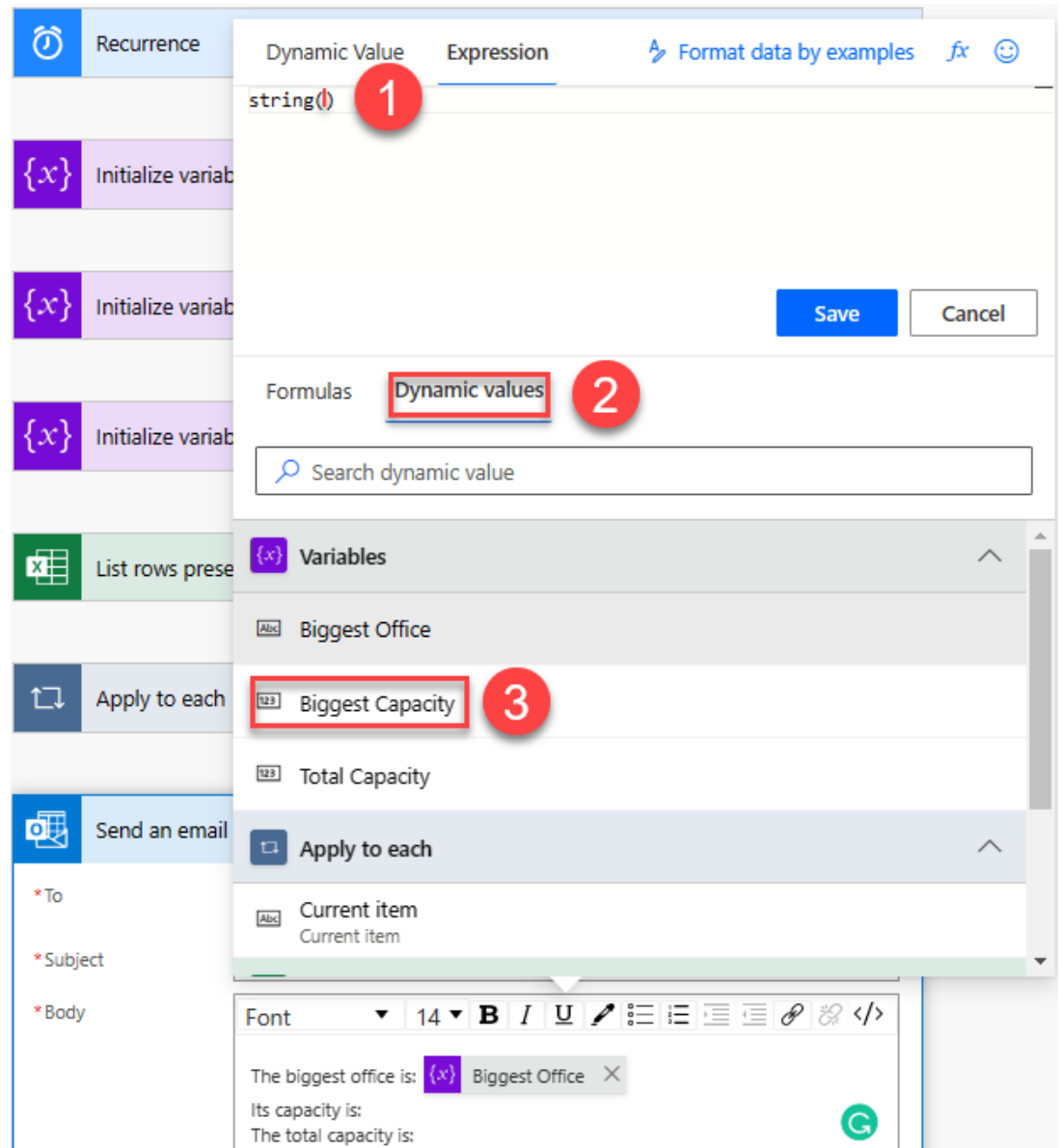
Font 14 B I U

The biggest office is: |
 Its capacity is:
 The total capacity is:

- vii. Unfortunately the other variables **Biggest Capacity** and **Total Capacity** are numbers, and the Send an e-mail designer expects strings; you will have to rely on a custom expression: move the cursor next to its capacity and click Expression, where you can type string



Move the cursor within the string () parentheses and click dynamic value to grab the variable Biggest Capacity (and click more if you don't find it in the list of variables):



viii. Eventually, the e-mail body should look like this:

Send an email (V2)

*To: [Redacted] X

*Subject: Office Capacity Report

*Body:

Font 14 B I U [Icons]

The biggest office is: {x} Biggest Office X

Its capacity is: f_x string(...) X

The total capacity is: f_x string(...) X

Show advanced options V

18. Save your flow and test it.

19. Check your e-mail; you should receive something like this:

Office Capacity Report



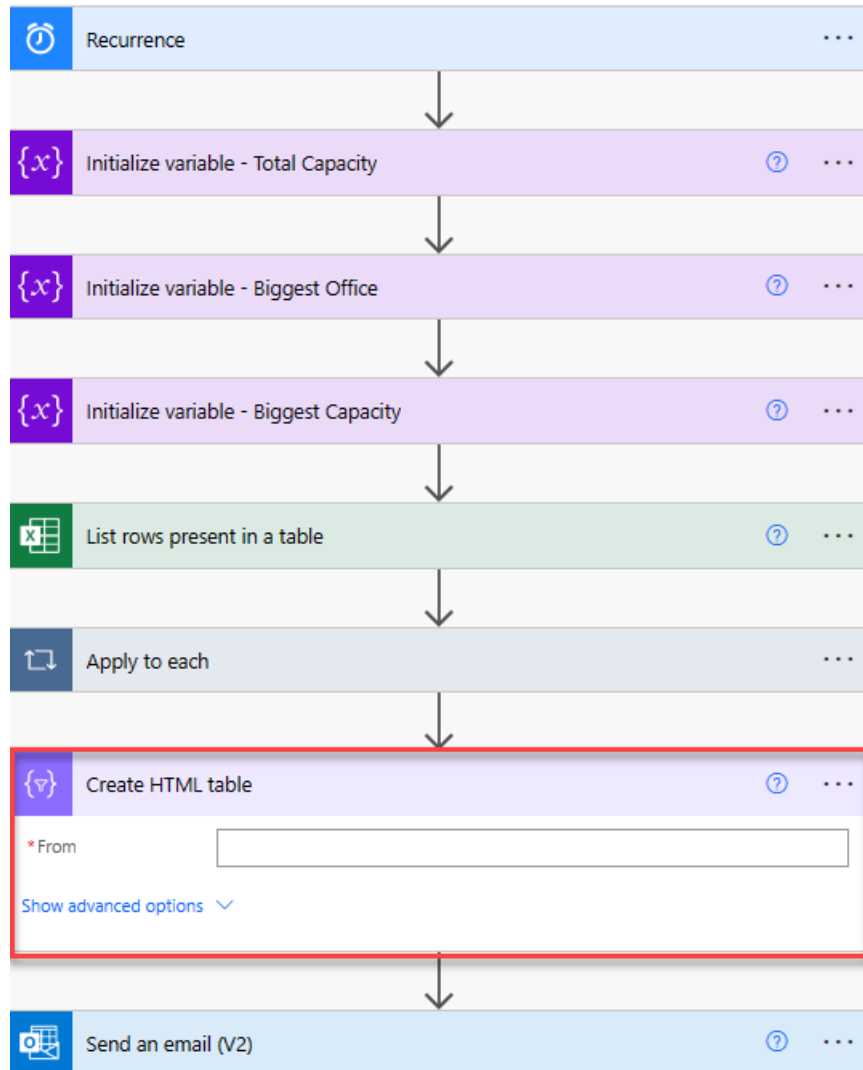
The biggest office is: Toronto
 Its capacity is:400
 The total capacity is:1607

Reply

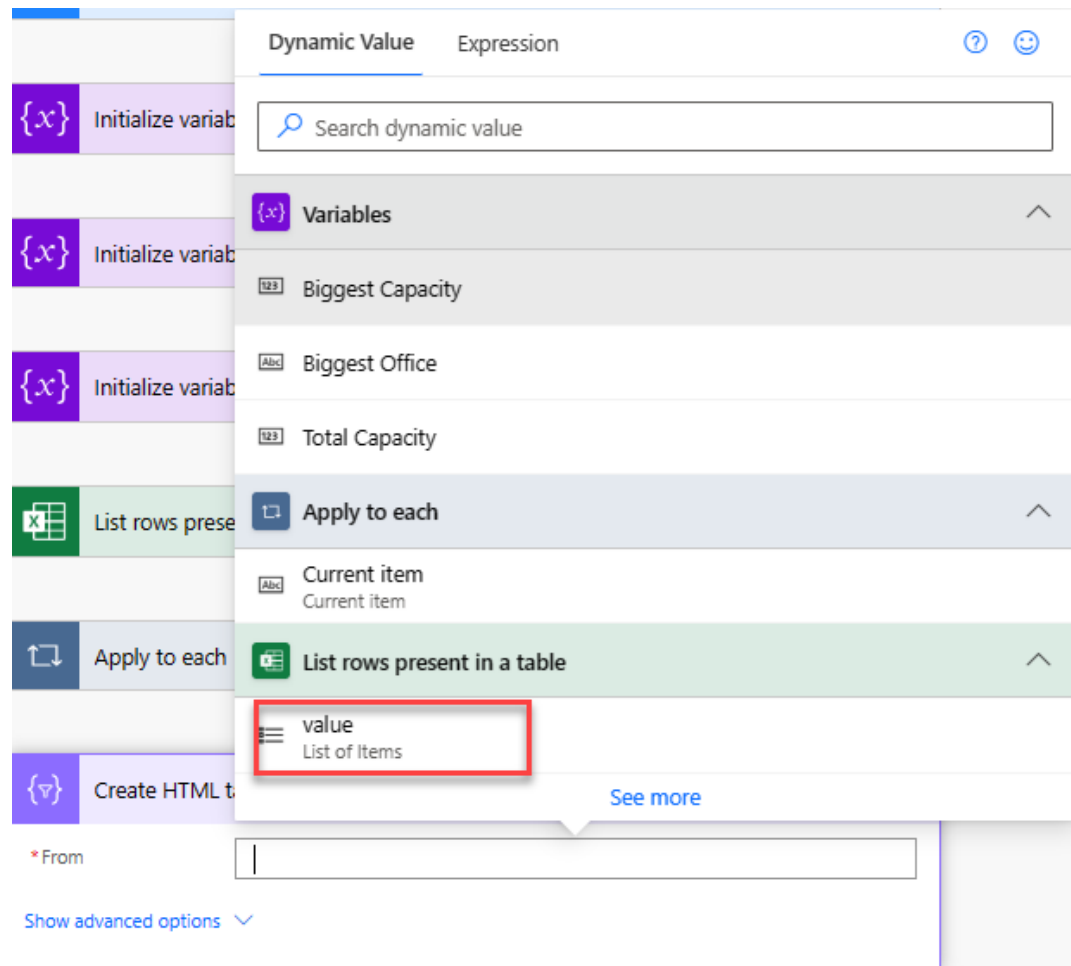
Forward

20. In the following steps, we will display the list of offices, so we will have to define a list formatting logic and create an HTML table based on this logic.

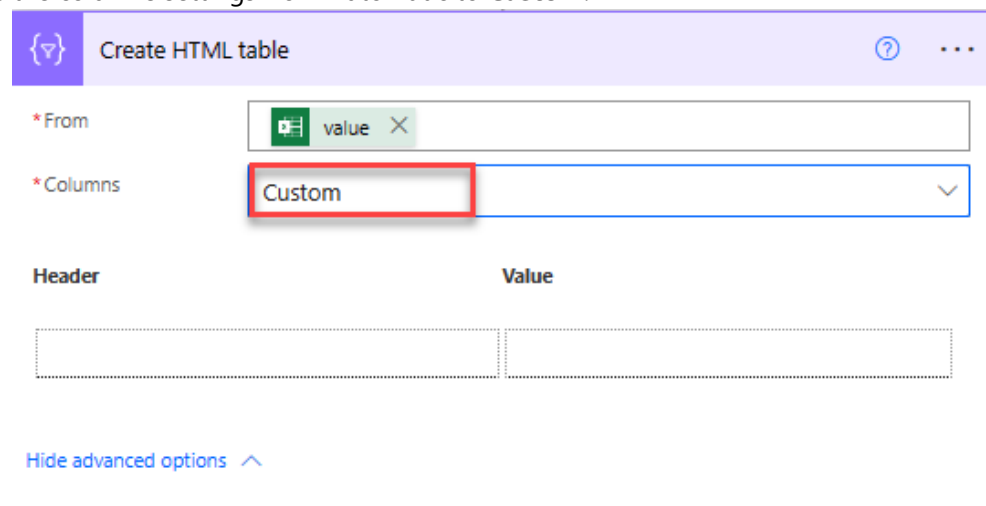
21. Let's define the list formatting logic. Before the **Send an e-mail action**, add a **Data Operations – Create HTML table** action:



22. Move the cursor in the From field and select the dynamic value associated with **the List rows present in the table** action:



23. Change the columns settings from Automatic to **Custom**:



24. You can now define the logic of your report : the header name (in **Header**) and the dynamic value or the expression associated with each header in **Value**.

25. In the **Header** field, type **City**, and in **Value** select **the dynamic value city** :

The screenshot shows the PowerApps interface during the 'Create HTML table' step. The 'Header' row is being edited, with 'City' entered in the first cell. The 'Dynamic Expression' pane is open, showing a list of variables. The 'city' variable is selected, highlighted with a red box and labeled with a red circle '3'. A red circle '1' is next to the 'City' text in the header cell, and a red circle '2' is next to the 'Dynamic Expression' title.

Dynamic Expression

Search dynamic value

Variables

- Biggest Capacity
- Biggest Office
- Total Capacity

Apply to each

- Current item

List rows present in a table

- value
- body
- body/value - Item
- city

Create HTML table

* From: value

* Columns: Custom

Header

City	

Hide advanced options ^

26. Follow the same procedure for Capacity:

{ }
Create HTML table
?
...

* From

value
X

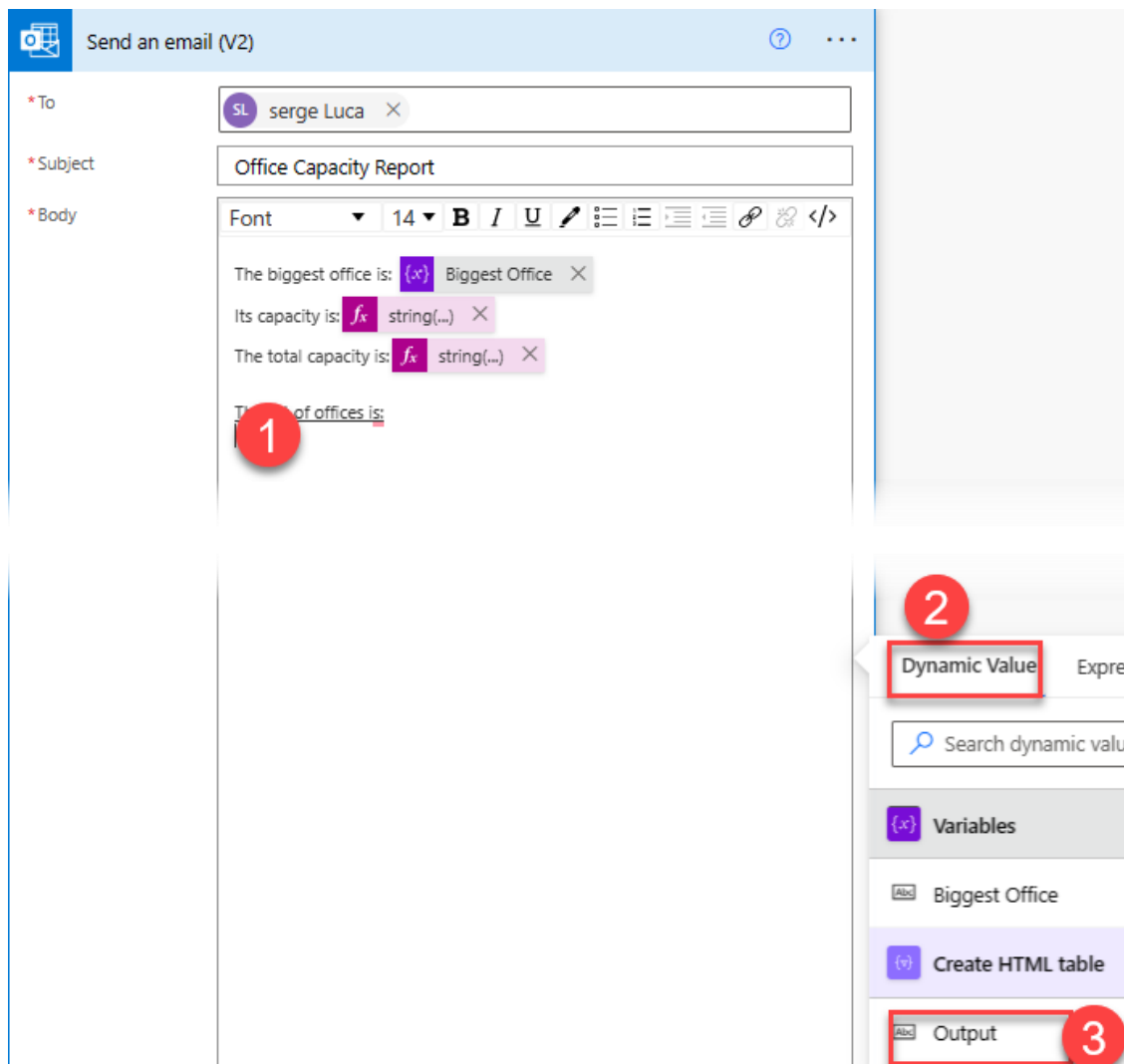
* Columns

Custom
v

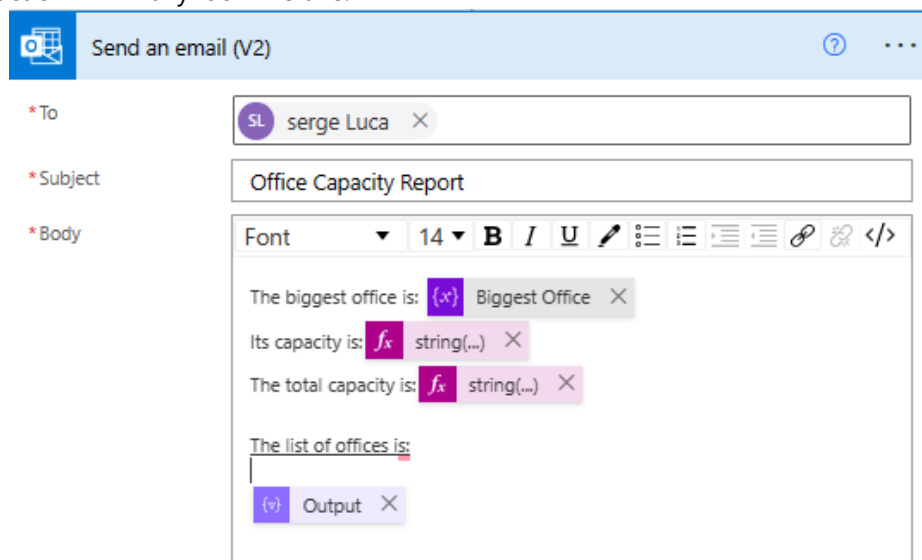
Header	Value
City	city X
Capacity	capacity X

Hide advanced options ^

27. Go back to the **Send an e-mail** action and update the **Body** text box to include the Create HTML Output value: just type the Total list of offices is : and add the output of The create HTML table action:



28. The action will finally look like this:



29. Test your flow and check your e-mail:

Office Capacity Report



[Redacted]

To: [Redacted]

The biggest office is: Toronto

Its capacity is:400

The total capacity is:1607

The list of offices is:

City	Capacity
London	100
Brussels	250
Seattle	80
Vancouver	200
Toronto	400
Antwerpen	15
Warsaw	300
Paris	54
Berlin	70
Amsterdam	60
Montreal	78

The rest of the exercise will be about customizing step by step the returned table, which will eventually look like this :

City	Capacity	Size
London	100	Small
Brussels	250	Big
Seattle	80	Small
Vancouver	200	Big
Toronto	400	Big
Antwerpen	15	Small
Warsaw	300	Big
Paris	54	Small
Berlin	70	Small
Amsterdam	60	Small
Montreal	78	Small

30. Let's extend our flow by adding a new column named Size in our report: the logic is that if the quantity is smaller than 100, we would like to see "Small", otherwise we will see "Big".

Answer; you will need to use the following functions **if()** and **greater()**. The expression is :

```
if ( greater(int(item()?['Capacity']), 100), 'Big', 'Small')
```

The screenshot displays a workflow editor with the following steps:

- Initialize variable - Biggest Office
- Initialize variable - Biggest Capacity
- List rows present in a table
- Apply to each
- Create HTML table

The 'Create HTML table' step is expanded, showing a table structure with the following headers and values:

Header	Value
City	city
Capacity	capacity
Size	if(...)

The 'Size' header is highlighted with a red box and a red circle labeled '1'. The 'if(...)' value is also highlighted with a red box and a red circle labeled '1'.

A 'Dynamic Value' dialog box is open, showing the 'Expression' tab. The expression is: `if ( greater(int(item()['Capacity']), 100), 'Big', 'Small')`. The expression is highlighted with a red box and a red circle labeled '3'. The 'Save' button is highlighted with a red circle labeled '4'.

The dialog box also shows a list of functions under the 'Collection' section:

- `contains(collection, value)`: Returns true if a dictionary contains a key, if an array contains a value, or if a string contains a substring.
- `length(collection)`: Returns the number of elements in an array or string.
- `sort(collection)`: Returns an array sorted in ascending order.
- `reverse(collection)`: Returns the collection in reverse order.

The generated mail will look like this:

The biggest office is: Toronto
Its capacity is:400
The total capacity is:1607

The list of offices is:

City	Capacity	Size
London	100	Small
Brussels	250	Big
Seattle	80	Small
Vancouver	200	Big
Toronto	400	Big
Antwerpen	15	Small
Warsaw	300	Big
Paris	54	Small
Berlin	70	Small
Amsterdam	60	Small
Montreal	78	Small

31. Now we would like to customize the generated html code by adding a table border.
- First, let analyze the html generated code by clicking on an existing flow run and then click the option "Show raw outputs" of the Create HTML Table action as illustrated below:

Send an email (V2)

* To: sergeluca@ShareQL.com

* Subject: Office Capacity Report

* Body:

Font 14 **B** *I* U [Link] [Image] [Code]

The biggest office is: {x} Biggest Office

Its capacity is: fx string(...)

The total capacity is: fx string(...)

The list of offices is:

Add the cursor where you want to display the html table and click **expression** where you can type the **replace()** function:

* Body:

Font 14 **B** *I* U [Link] [Image] [Code]

The biggest office is: {x} Biggest Office

Its capacity is: fx string(...)

The total capacity is: fx string(...)

The list of offices is:

Dynamic Value **Expression** [Format data by examples] [fx] [Smiley]

replace()

[Save] [Cancel]

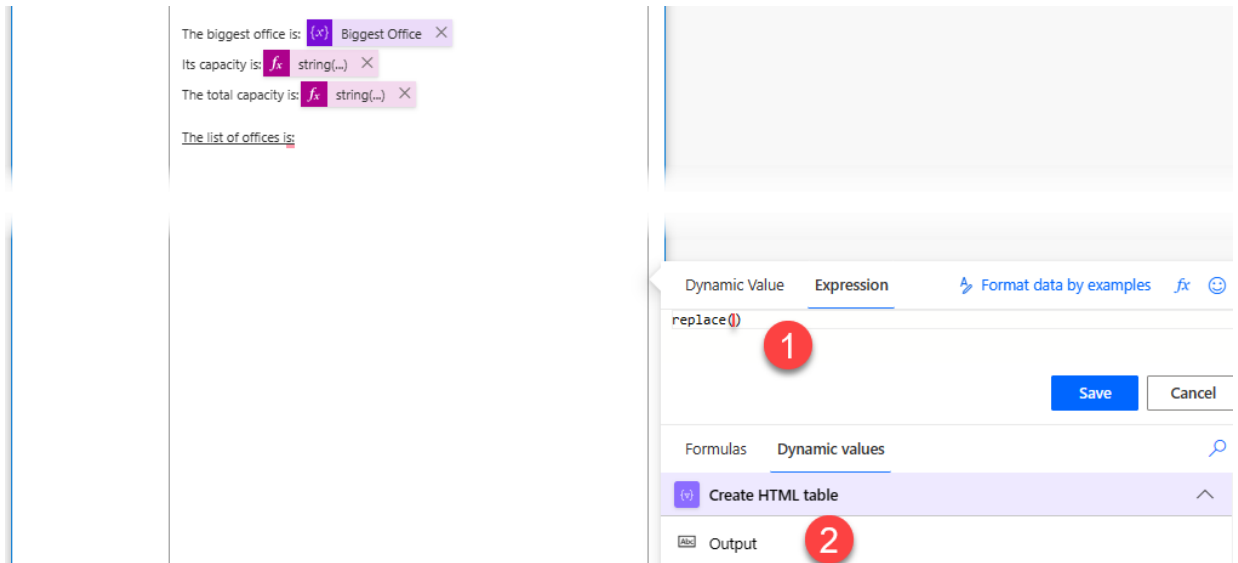
Formulas Dynamic values

String functions

concat(text_1, text_2?, ...)
Combines any number of strings together

[See more](#)

Move the cursor in the replace parentheses, and Click **Dynamic Value** (just down below, not the next one), to grab the Create HTML table – Output:



In replace, type `, '<table>', '<table border="1">' )` , the final code should be `replace(body('Create_HTML_table'), '<table>', '<table border="1">' )`, and click **Save**.

The mail will look like this:

The biggest office is: Toronto
Its capacity is:400
The total capacity is:1607

The list of offices is:

City	Capacity	Size
London	100	Small
Brussels	250	Big
Seattle	80	Small
Vancouver	200	Big
Toronto	400	Big
Antwerpen	15	Small
Warsaw	300	Big
Paris	54	Small
Berlin	70	Small
Amsterdam	60	Small
Montreal	78	Small

32. A more advanced exercise is to display the capacity number in red if it is greater than 200, and in green otherwise.

- Edit the Create HTML table action, and in the capacity value field, add the following expression just to the left of capacity: (and click **Save**>

The screenshot shows the 'Create HTML table' configuration window. The table structure is as follows:

Header	Value
City	city
Capacity	capacity
Size	if(...)

The 'Capacity' field is selected, and a dynamic value expression is being added to its left. The expression is:

```
if (greater(int(item()['capacity']),200),'<p style="color:red">','<div style="color:green">')
```

The 'Save' button is highlighted.

- b. Close the Div by adding `</div>` in plain text:

Create HTML table

*From

value

Header	Value
City	city
Capacity	if(...) capacity </div>
Size	if(...)

Show advanced options

The

- c. The generated e-mail will look like this:

The biggest office is: Toronto
 Its capacity is:400
 The total capacity is:1607

The list of offices is:

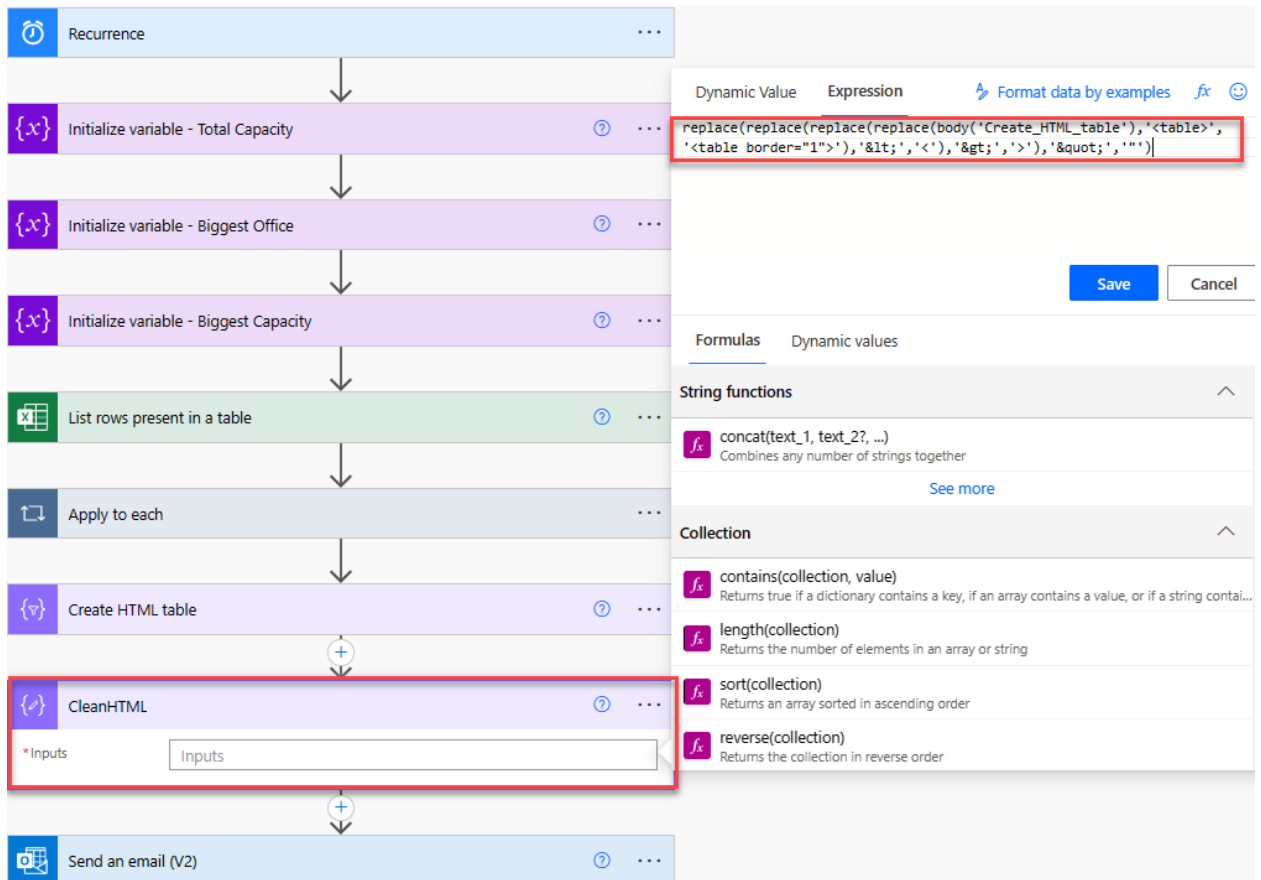
City	Capacity	Size
London	100	Small
Brussels	250	Big
Seattle	80	Small
Vancouver	200	Big
Toronto	400	Big
Antwerpen	15	Small
Warsaw	300	Big
Paris	54	Small
Berlin	70	Small
Amsterdam	60	Small
Montreal	78	Small

Actually the html mbols <, > and ' are automatically replaced with their encoded value as illustrated in the following table

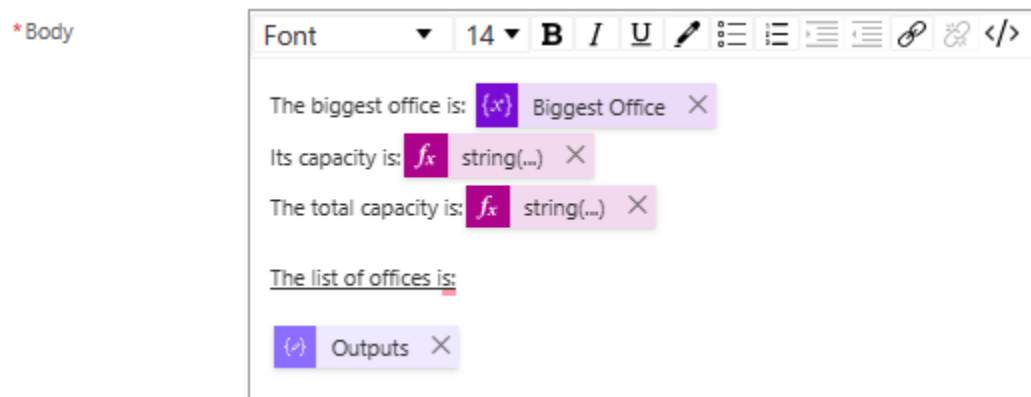
<	<
>	>
'	"

- d. After the create HTML Table action, add a **Compose** action (rename it **CleanHTML**) where you can add the following expression:

```
replace(replace(replace(replace(body('Create_HTML_table'),'<table>','<table border="1">'),'&lt;','<'),'&gt;','>'),'&quot;','"'))
```



e. Send the output of CleanHTM into the Send an e-mail action:



f. Test your flow; the received e-mail will look like this:

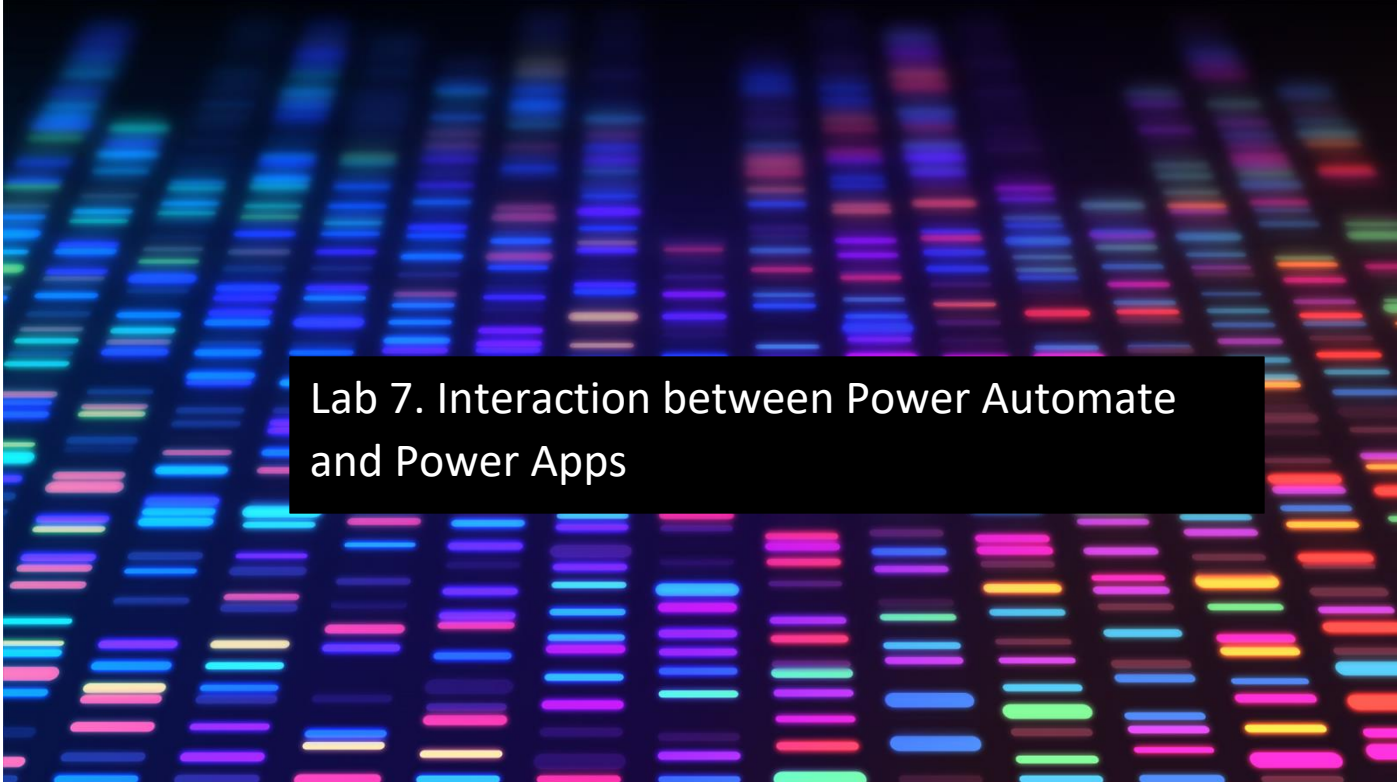
The biggest office is: Toronto
Its capacity is:400
The total capacity is:1607

The list of offices is:

City	Capacity	Size
London	100	Small
Brussels	250	Big
Seattle	80	Small
Vancouver	200	Big
Toronto	400	Big
Antwerpen	15	Small
Warsaw	300	Big
Paris	54	Small
Berlin	70	Small
Amsterdam	60	Small
Montreal	78	Small

We need your feedback

Do you want to report an issue or suggest something? We need your feedback:
<https://github.com/Power-Automate-in-a-day/Training-by-the-community/issues>



Lab 7. Interaction between Power Automate and Power Apps

Lab 7. Interaction between Power Automate and Power Apps

Author: Serge Luca, aka "Doctor Flow"

Learning objective: in many business scenarios, a Power Apps application must call a Power Automate flow because some actions are unavailable on Power Apps, some processes take some time to complete, or because we want a clean boundary between the user interface and the business logic.

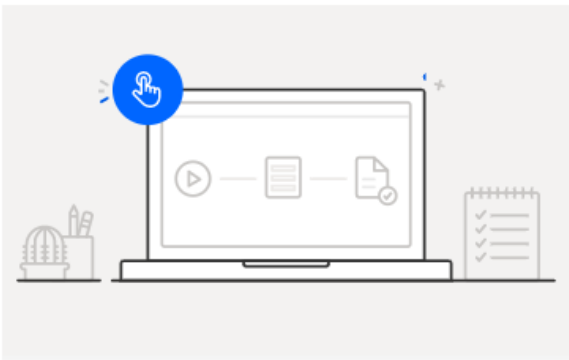
Duration: 30 minutes

Scenario: you will create a PowerApps form allowing any user to create a Team. The first part of the lab will call a dummy Flow that you will make. Then, if time permits, you will invoke the Team creation Flow you will create later.

Tasks

1. Create an instant Flow with From Power Apps as the trigger:

Build an instant cloud flow



Triggered manually from any device, easy-to-share instant flows automate tasks so you don't have to repeat yourself.

Examples:

- Get an automatic mobile alert whenever a VIP client emails you
- Save all your email attachments to a folder automatically

Flow name

Create a team

Choose how to trigger this flow *



Manually trigger a flow

Flow button for mobile

ⓘ

☒



PowerApps

PowerApps

ⓘ



When Power Virtual Agents calls a fl...

Power Virtual Agents

ⓘ



When a flow step is run from a busi...

Microsoft Dataverse

ⓘ



For a selected message (V2)

Microsoft Teams

ⓘ



From the compose box (V2)

Microsoft Teams

ⓘ



When someone responds to an ada...

...

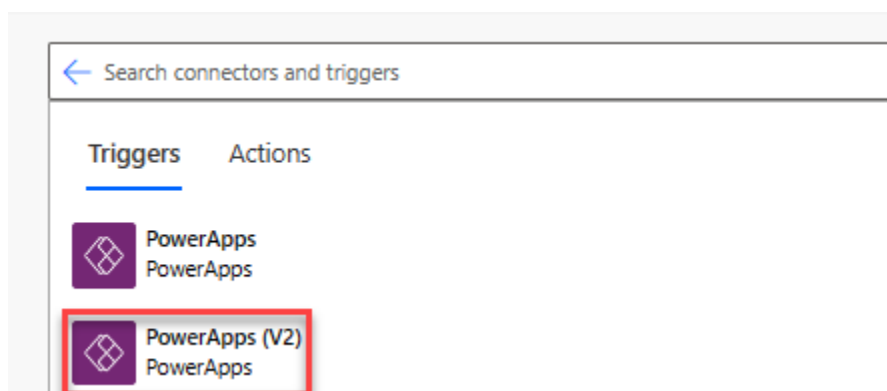
ⓘ

Skip

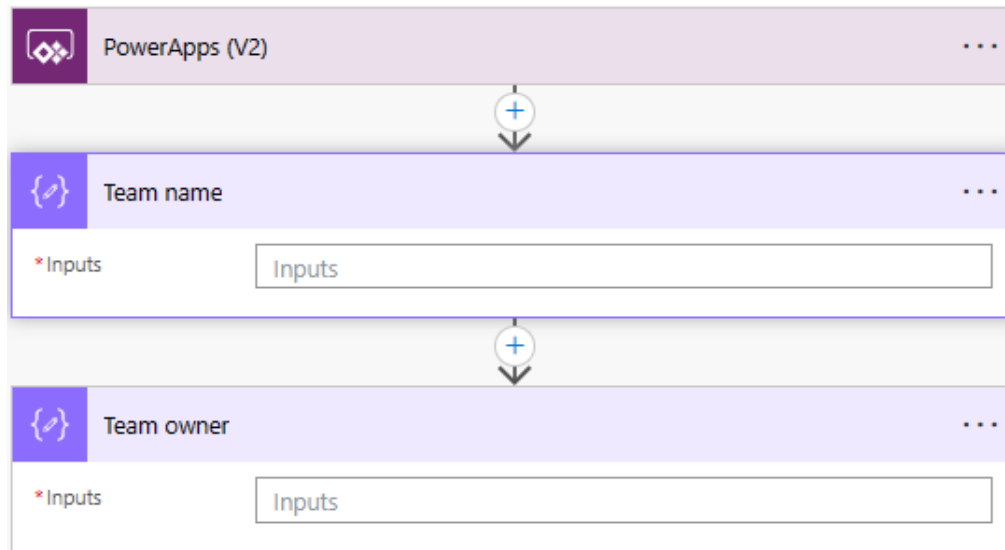
Create

Cancel

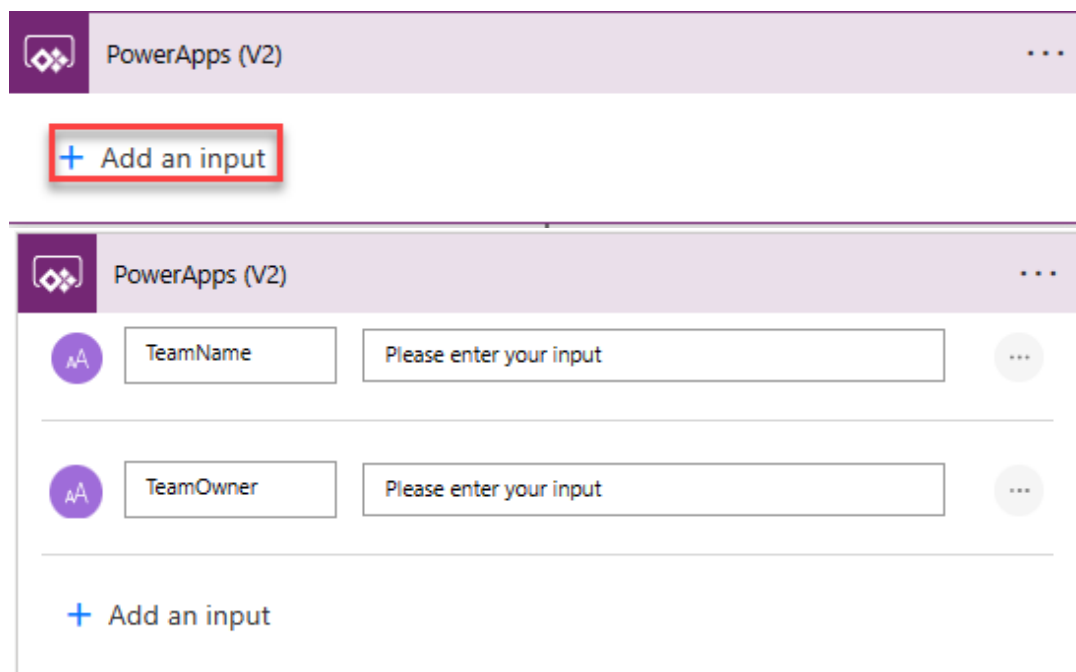
- Remove the Power Apps trigger to use a new version of the trigger.
- Replace the trigger with the Power Apps v2:



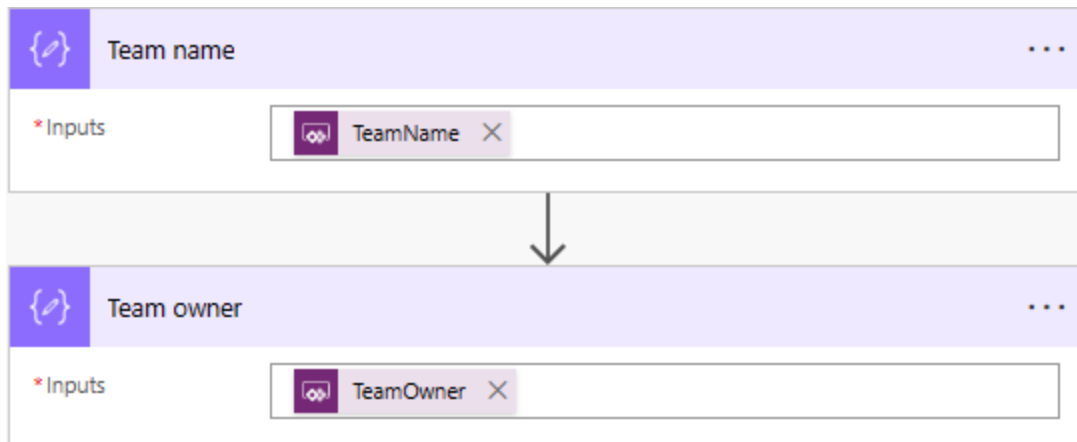
- In this flow, add 2 Compose actions and rename them **Team name** and **Team owner** :



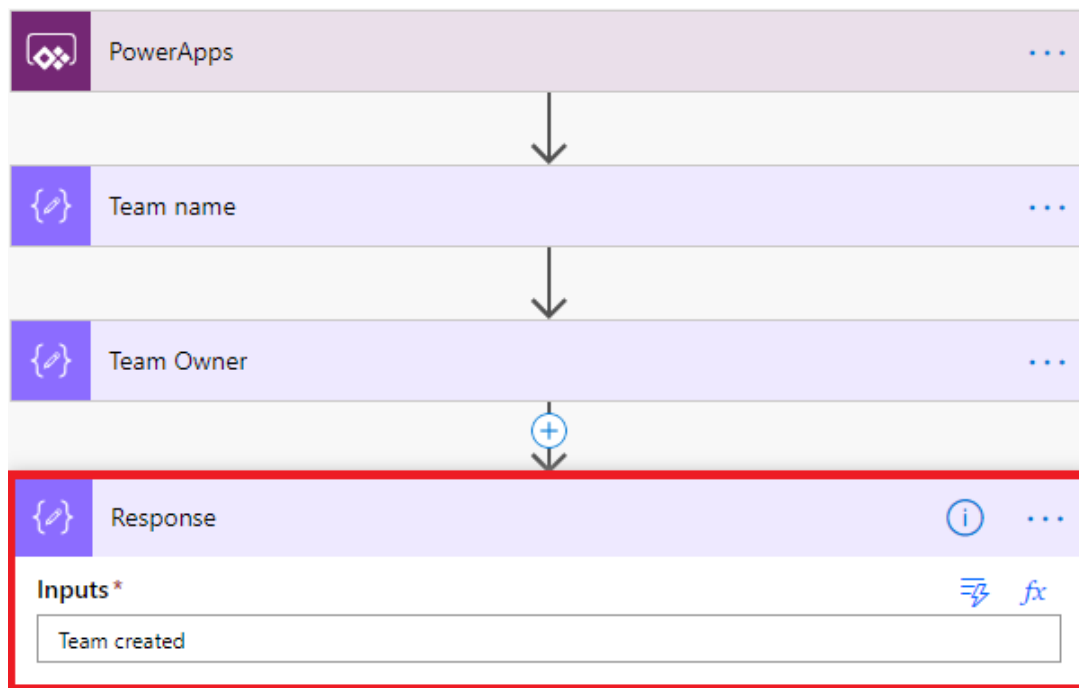
5. Click the trigger to define two parameters of type Text by clicking Add an input, and name them TeamName and TeamOwner



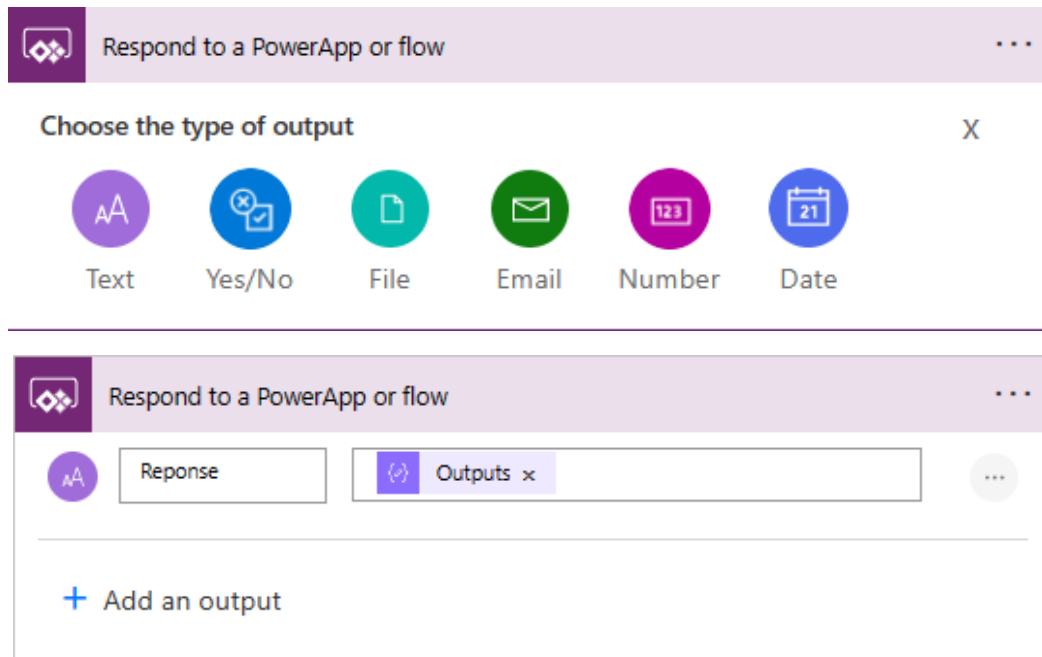
6. And store these parameters in the corresponding Compose actions:



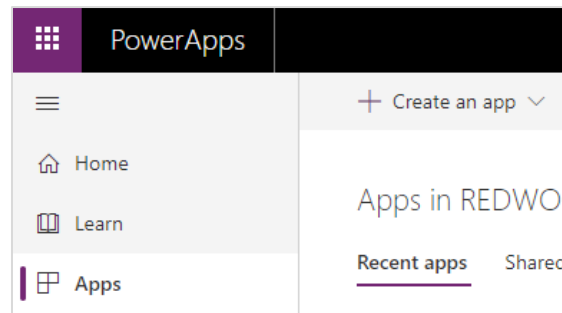
7. Add new **Compose** action, name it **response**, and set the Inputs to **Team created**:



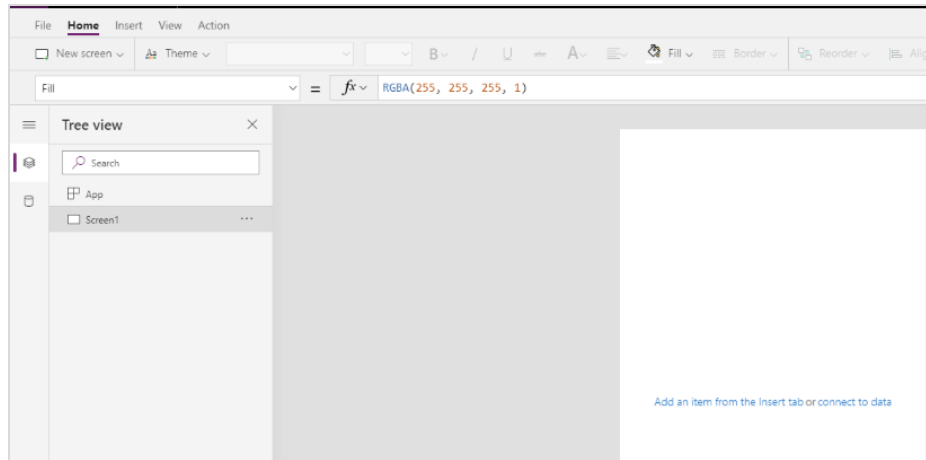
8. Add an action **Respond to a PowerApp or flow** that will return our response. Click Text as the output type:



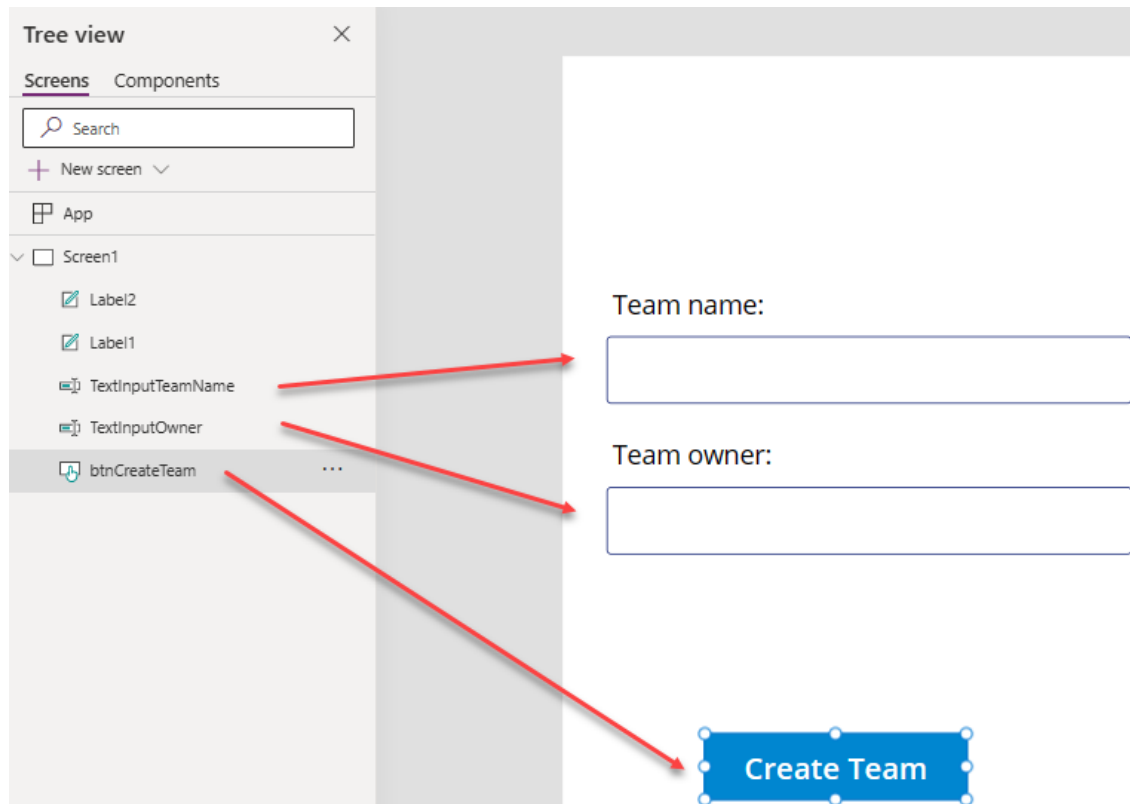
9. Save your Flow.
10. Now you will create a **Power Apps application** that will invoke our flow. Go to **PowerApps.com**. Click on **Apps** menu and create an App



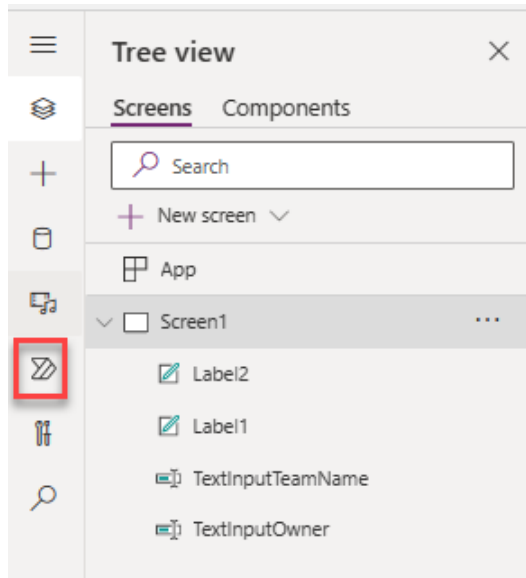
11. Select the **Canvas app**.
12. Click **Phone Layout**. The **PowerApps Designer** should show up:



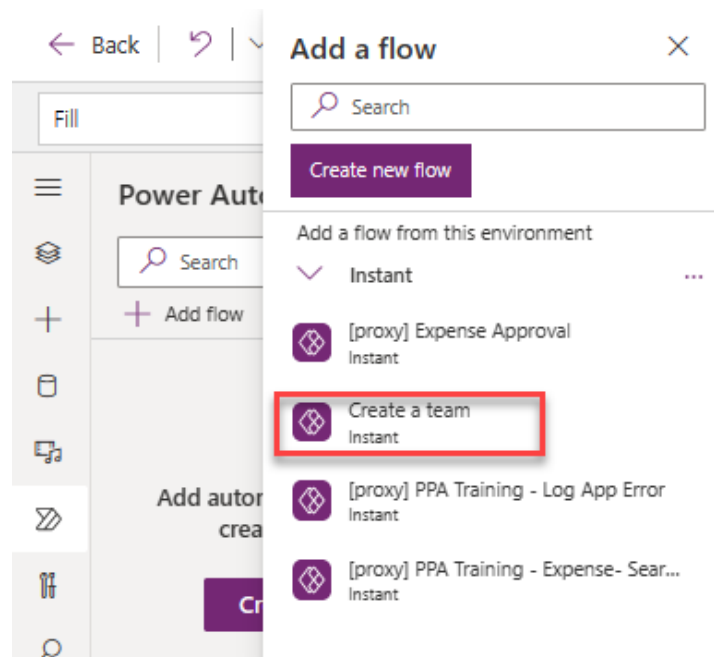
13. Add 2 Text labels, two text Input, (rename them **TextInputTeamName** and **TextInputOwner**), and one button (rename it **btnCreateTeam**):



14. In the menu on the left, click the **Power Automate** icon:



15. Click **Add flow** to bring the list of flows starting with a Power Apps trigger, and **select Create a team** :



16. In the OnSelect property of the button, call your flow:

Set (FlowStatus, Createateam.Run(TextInputTeamName.Text,TextInputOwner.Text))

If you are testing your code in a Dataverse for Teams environment, the code should be:

Set (FlowStatus, Createateam.Run(TextInputTeamName.Value,TextInputOwner.Value))

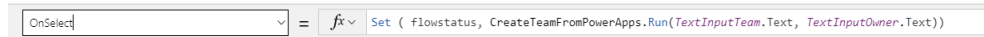
[Note:

If you are using a non US version of Power Apps, the comma in the formula should become a semicolon ; the code becomes **Set (Flowstatus;**

CreateTeamFromPowerApps.Run(TextInputTeam.Text; TextInputOwner.Text)) .

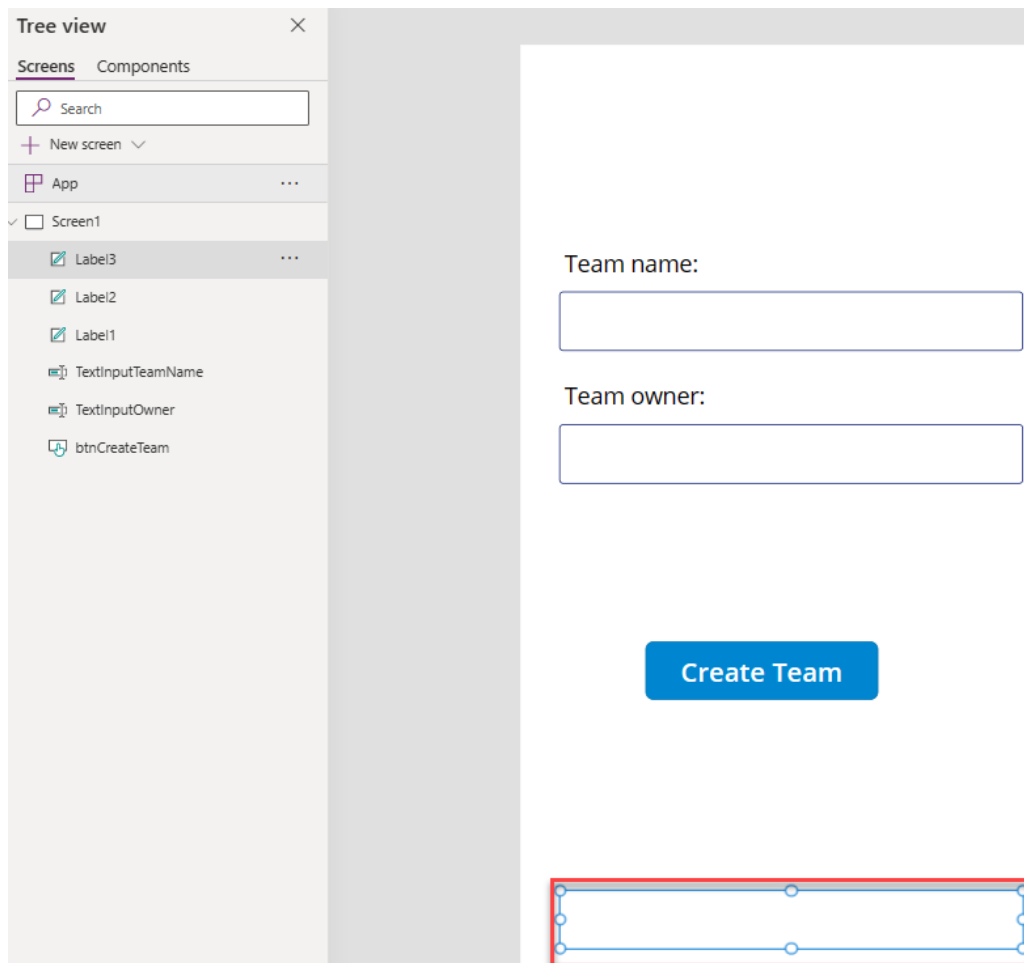
]

17. The value the flow returns will be stored in a variable called **FlowStatus**.



OnSelect = fx Set (flowstatus, CreateTeamFromPowerApps.Run(TextInputTeam.Text, TextInputOwner.Text))

18. We will display the status in another label; add another label to the form:



19. This new label Text property must be **FlowStatus.response**.

20. Run the Application, click the Submit button, and you should see the returned value:

Team name:

doctor

Team owner:

doctorflow@drflow.com

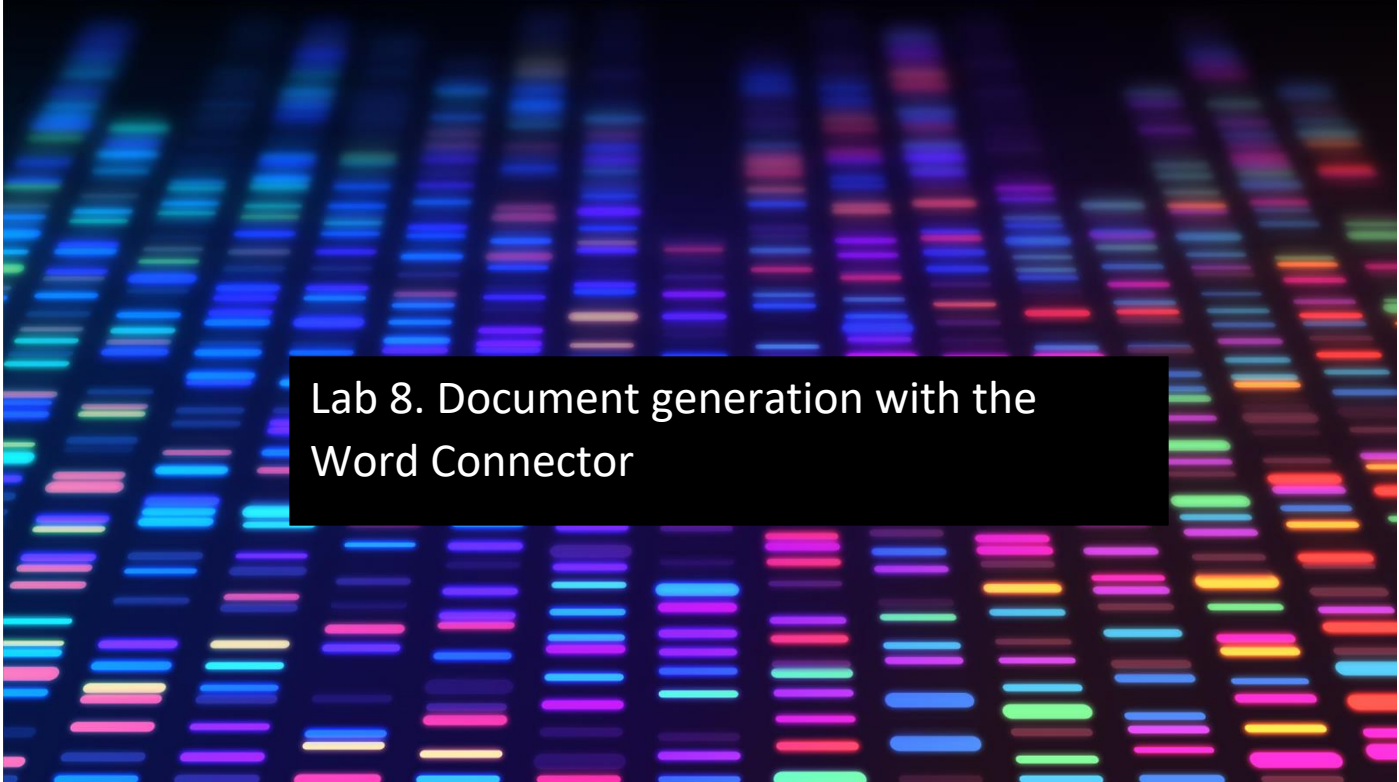
Create Team

Team created

We need your feedback

Do you want to report an issue or suggest something? We need your feedback:

<https://github.com/Power-Automate-in-a-day/Training-by-the-community/issues>



Lab 8. Document generation with the Word Connector

Lab 8. Document generation with the Word Connector

Author: Serge Luca, aka "Doctor Flow".

Learning objective: create a Flow that will generate an invoice (in PDF) based on a Word template and on data stored in an Excel document.

Duration: 30 minutes

Scenario: Excel files containing invoice data are stored in OneDrive for Business. A Flow will read these data, parse the data, and generate an invoice based on a word document template. The invoice will be transformed into a PDF file.

Prerequisites: The Word Connector is a Premium connector. You need to test this lab with a trial premium subscription.

When using the Word Connector, the designer will ask you to start a premium trial or you must have activated the developer license (see lab 0).

The files for the lab can be downloaded from the folder "Resources for the labs", download the labv9.zip file: the files are "**Invoice.xlsx**" and "**Invoice Template start.docx**"

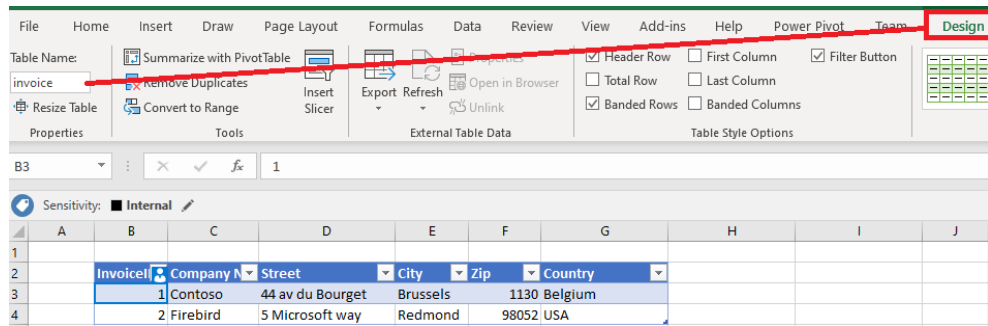
Tasks:

1. Upload the file **invoice.xlsx** provided by the trainer to the Documents folder of your OneDrive for Business
2. Look at this file: there are two tables:
 - a. The table **invoice** consists of the invoice headers
 - b. The table **invoicelines** consists of the invoice lines

InvoiceID	Company Name	Street	City	Zip	Country
1	Contoso	44 av du Bourget	Brussels	1130	Belgium
2	Firebird	5 Microsoft way	Redmond	98052	USA

Invoice	Product	Amount	Price	Total
1	Internal training PowerAp	5	1400	4200
1	Internal training Flow	4	1400	5600
2	Internal training PowerAp	3	1400	4200
2	Internal training PowerBI	2	1400	2800

3. Each table has a name that we will reference from our Flow; to find out the table name, open the file in Excel desktop, click the table, and select the menu Design as illustrated in the following picture:



- Each invoice can have several lines.
- Upload the file **Invoice Template start.docx** provided by the trainer into the Documents folder of your OneDrive for Business and take a look at this file:

(header)

Doctor Flow 133 av Dieweg 1180 Uccle Belgique		INVOICE	
BILL TO Name Company Name Street address City, Postal Code		INVOICE#	

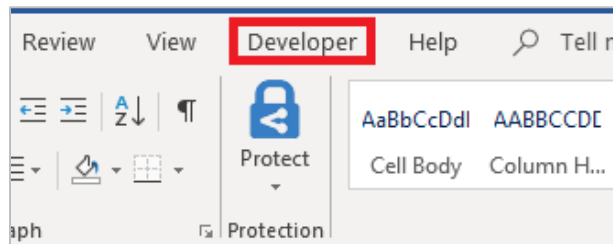
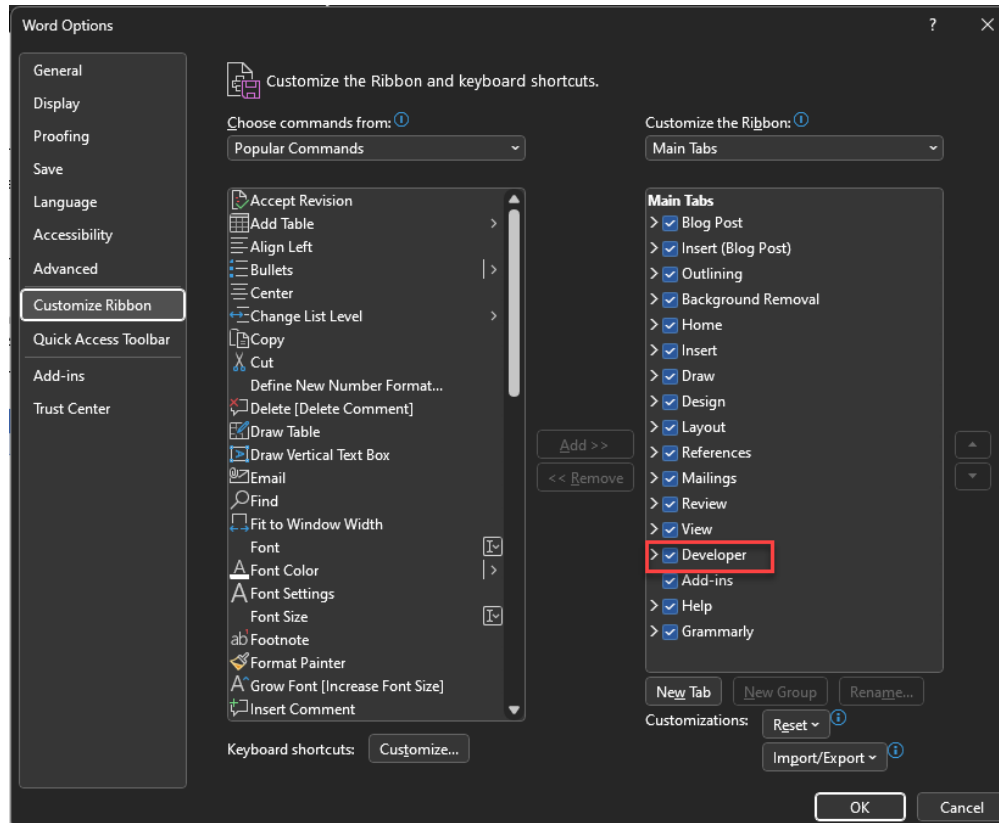
Description	Amount

(lines)

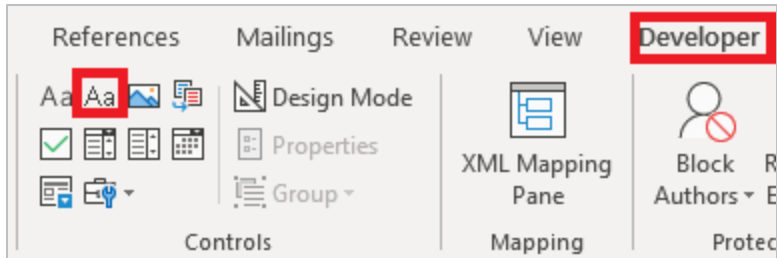
The section "header" and "lines" are two different tables. The lines section is a table with one row for the header and one row for the data.

- Go to the Word **Options**, select Customize Ribbon, select **Developer**, and click OK.

This will display the **Developer** tab in Word:



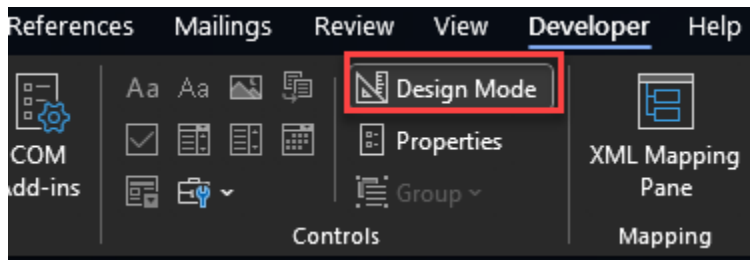
7. You will customize the word template/ next to each field of the template, you will add a **Word plain text content** control, and we will use the properties button to name them accordingly.



8. Add a control next to the following fields: **Company Name**, **Street address**, **City** (add 2 controls here, one for the city, one for the zip code) and one next to **invoice#**;

Doctor Flow 133 av Dieweg 1180 Uccle Belgique	INVOICE	
BILL TO Company Name: Click or tap here to enter text. Street address: Click or tap here to enter text. City: Click or tap here to enter text., Click or tap here to enter text.	INVOICE#	Click or tap here to enter text.

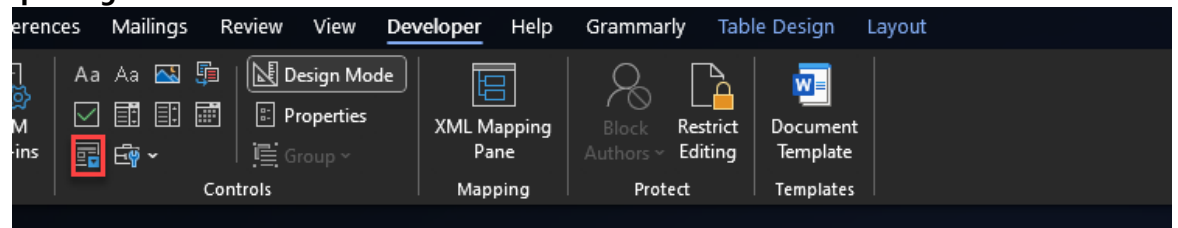
9. Click **Design Mode** to see where each control starts and ends:



10. With the **Properties** button, name (Title of Property) the controls **company**, **street**, **city**, **zip**, **invoiceid** as illustrated below:

Doctor Flow 133 av <u>Dieweg</u> 1180 Uccle <u>Belgique</u>	INVOICE	
BILL TO Company Name: company (Click or tap here to enter text.) Street address: street (Click or tap here to enter text.) City: city (Click or tap here to enter text.) , zip (Click or tap here to enter text.)	INVOICE#	invoiceid (Click or tap here to enter text.) invoiceid

11. You will focus now on the invoice lines; select the second row of the table (the one with the empty data, not the row with the header); select the whole row (including the two columns) and add a **Repeating Section Content Control**:



Doctor Flow 133 av Dieweg 1180 Uccle Belgique	INVOICE	
BILL TO Company Name: company (Click or tap here to enter text.) Street address: street (Click or tap here to enter text.) City: city (Click or tap here to enter text.) , zip (Click or tap here to enter text.)	INVOICE#	invoiceid (Click or tap here to enter text.) invoiceid

Description	Amount

12. Rename the repeating section **lineitems**.
13. In this repeating section, content control, add 2 Plain text content controls and name them **Product** and **Amount**:

Doctor Flow 133 av Dieweg 1180 Uccle Belgique		INVOICE	
BILL TO Company Name: company Click or tap here to enter text.		INVOICE#	
Street address: street Click or tap here to enter text.		invoiceid Click or tap here to enter text.	
City: city Click or tap here to enter text. ,			
zip Click or tap here to enter text.			

Description	Amount
lineitems Product Click or tap here to enter text.	Amount Click or tap here to enter text.

14. After the repeating section, adds a **Picture content control** that will display the manager signature. Name this control **managersignature**.
15. Eventually, the template should look like this:

Doctor Flow 133 av Dieweg 1180 Uccle Belgique		INVOICE	
BILL TO Company Name: company (Click or tap here to enter text.) Street address: street (Click or tap here to enter text.) City: city (Click or tap here to enter text.) , zip (Click or tap here to enter text.)		INVOICE#	invoiceid (Click or tap here to enter text.) invoiceid

Description	Amount
lineitems Product (Click or tap here to enter text.) Product	Amount (Click or tap here to enter text.) Amount lineitems



16. Rename the file to **invoice template.docx**
17. Let's create a Flow that starts from a button (Instant cloud flow > Manually trigger a flow).
18. Add an action **Excel Online (Business)- List rows present in a table**, and retrieve the **invoice table** of your **invoice.xlsx** documents:

Manually trigger a flow

List rows present in a table - Invoices

* Location: OneDrive for Business

* Document Library: OneDrive

* File: /Power Automate in a day - Labs/Invoice.xlsx

* Table: invoice

Show advanced options

19. Add an Apply to each action to get each invoice line, and rename it:

Manually trigger a flow

List rows present in a table - Invoices

Apply to each - Invoice

* Select an output from previous steps: value

Dynamic Value Expression

Search dynamic value

Environment Variables

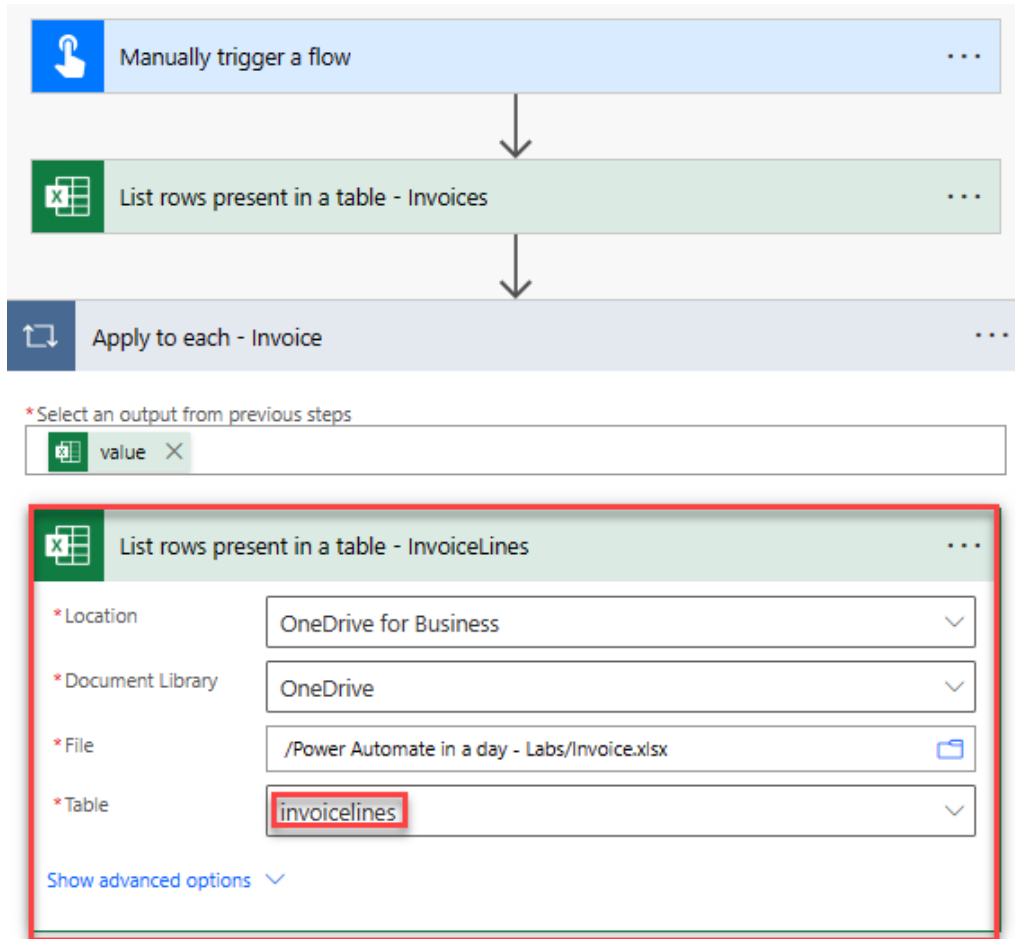
Should only leaf node selection be allowed (msdyn_AllowSelectLeafOnly)

Should the Peek Button Be Shown (msdyn_ShouldShowPeekButton)

List rows present in a table - Invoices

value
List of Items

20. For each invoice, you will fetch the corresponding invoice lines: Add another Excel action that will retrieve the **invoicelines** table List rows present



In the Excel for business action

21. You need to filter the retrieved lines. Click the **Show advanced options**, and in the **Filter Query** field, type **Invoice eq ''** (2 single quotes). The filter query is case-sensitive, be careful here.

↺

Apply to each Invoice

...

Select an output from previous steps *

value

×

Find InvoiceLines

i

...

Location *

OneDrive for Business

▼

Document Library *

OneDrive

▼

* File

/Flow Workshop/Invoice.xlsx

Table *

invoicelines

▼

Filter Query

Invoice eq "

⚙

fx

Order By

An ODATA orderBy query for specifying the order of entries.

Top Count

Total number of entries to retrieve (default = all).

Skip Count

The number of entries to skip (default = 0).

Select Query

Comma-separated list of columns to retrieve (first 500 by default).

Hide advanced options

^

22. Move the cursor between the ' ' and add the field "InvoiceID" under the quotes:

23.

The image shows a Power Automate flow configuration with the following steps:

- Manually trigger a flow**
- List rows present in a table - Invoices**
- Apply to each - Invoice**

Below the 'Apply to each' step, there is a section for selecting an output from previous steps, showing a 'value' output.

The main configuration area for the 'List rows present in a table - InvoiceLines' step is visible, with the following settings:

- Location:** OneDrive for Business
- Document Library:** OneDrive
- File:** /Power Automate in a day - Labs/Invoice.xlsx
- Table:** invoicelines
- Filter Query:** Invoice eq

On the right, a **Dynamic Value** pane is open, showing a search bar and a list of available fields. The **InvoiceID** field is highlighted with a red box.

Below the main configuration, an **Expression** builder window is open. It shows a search bar and a list of functions. The **Find Invoices** function is highlighted with a red box. Below it, the **InvoiceID** field is also highlighted with a red box. The expression builder shows the text 'Invoice eq' followed by a redacted area.

The expression builder also includes sections for **Order By**, **Top Count**, **Skip Count**, and **Select Query**.



List rows present in a table - InvoiceLines



* Location

OneDrive for Business



* Document Library

OneDrive



* File

/Power Automate in a day - Labs/Invoice.xlsx



* Table

invoicelines



Filter Query

Invoice eq '  InvoiceID ' 

24. Add an action **Populate a Microsoft Word template**, and you will notice that the Word content controls show up as fields in this connector; fill in the fields with the corresponding Excel values:

The screenshot shows a Power Automate workflow connector for the action "Populate a Microsoft Word template - Invoice Template". The connector is titled "Apply to each - Invoice" and has a dropdown menu for "Select an output from previous steps" with "value" selected. Below this, there is a step "List rows present in a table - Invoicelines". The connector itself has several fields: "Location" (OneDrive for Business), "Document Library" (OneDrive), "File" (/Power Automate in a day - Labs/Invoice Template.docx), and "invoiceid" (InvoiceID). There is a section for "1242361815 Product - 1" and "1242361815 Amount - 1" with text input fields. Below this is an "Add new item" button. At the bottom, there are fields for "company" (Company N...), "zip" (Zip), "street" (Street), "city" (City), and "managersignature".

Apply to each - Invoice

*Select an output from previous steps

value

List rows present in a table - Invoicelines

Populate a Microsoft Word template - Invoice Template

* Location OneDrive for Business

* Document Library OneDrive

* File /Power Automate in a day - Labs/Invoice Template.docx

invoiceid InvoiceID

1242361815 Product - 1

Click or tap here to enter text.

1242361815 Amount - 1

Click or tap here to enter text.

+ Add new item

company Company N...

zip Zip

street Street

city City

managersignature

25. You still have to generate a file in OneDrive for Business with the content of the previous action output content: add a **OneDrive for Business Create file** action and rename
26. Fill in the File Content from with the output of the previous action and dynamically generate the file name:

Create file - Invoice in Word

* Folder Path: /Power Automate in a day - Labs

* File Name: invoice_ InvoiceID .docx

* File Content: Microsoft W...

27. Add an action **Convert Word Document to PDF** and grab the path of the generated file
28. Add a **OneDrive for Business Create file** action to create the pdf file and pass the body of the previous action. The next two actions should look like this:

Convert Word Document to PDF

* Location: OneDrive for Business

* Document Library: OneDrive

* File: Path

[Add dynamic content](#)

[Add an action](#)

[+ New step](#) [Save](#)

Dynamic content

Search dynamic content

- Name without extension**
The name of the file or folder, without the file extension.
- Display name**
The display name of the file or folder.
- Path**
The path of the file or folder.

Convert Word Document to PDF

Create file - Invoice in PDF

* Folder Path: /Power Automate in a day - Labs

* File Name: invoice_ InvoiceID .pdf

* File Content: PDF docum...

29. Run the Flow and check the generated PDF files. The first file should look like this:

Doctor Flow 133 av Dieweg 1180 Uccle Belgique BILL TO Company Name: Contoso Street address: 44 av du Bourget City: Brussels , 1130	INVOICE	
	INVOICE#	1

Description	Amount
Click or tap here to enter text.	Click or tap here to enter text.

30. You will now update the Flow to implement the line items. First, edit the Flow and define an array variable called **invoiceLines** and initialize it as an empty array:

Manually trigger a flow
...

{x}
Initialize variable - InvoiceLines
...

* Name

InvoiceLines

* Type

Array v

Value

[]

31. Add a Set variable action at the start of Apply to each to clean up this variable again:

Apply to each - Invoice
...

* Select an output from previous steps

value
X

{x}
Set variable - InvoiceLines - Empty
...

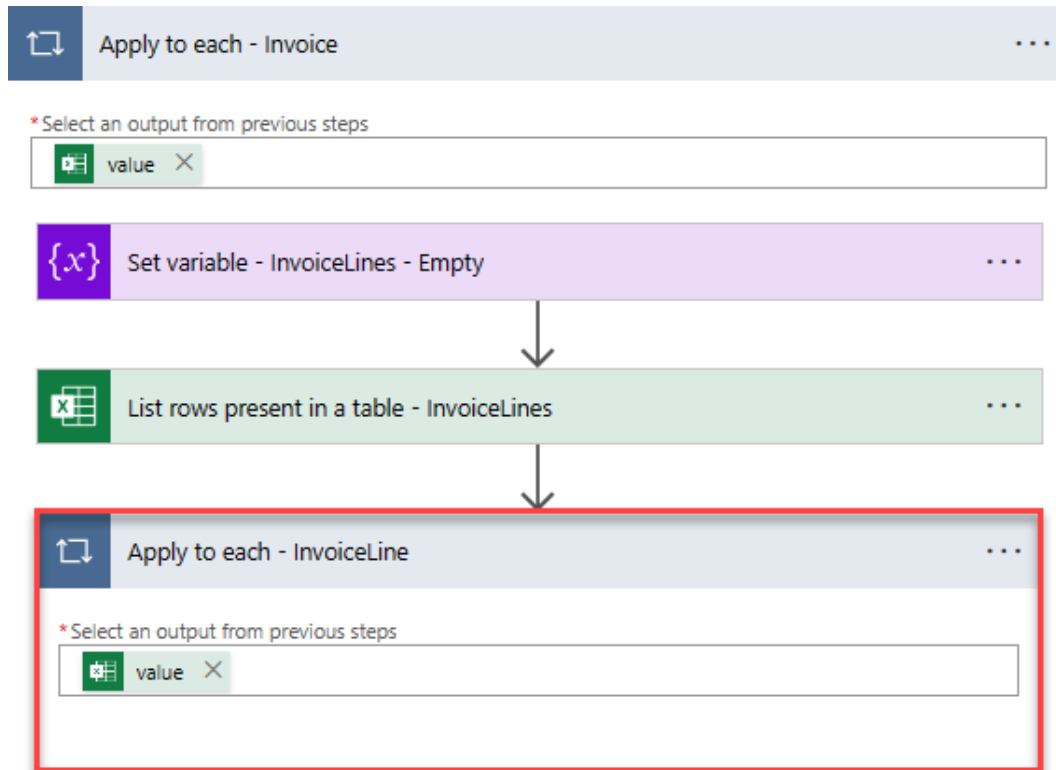
* Name

InvoiceLines v

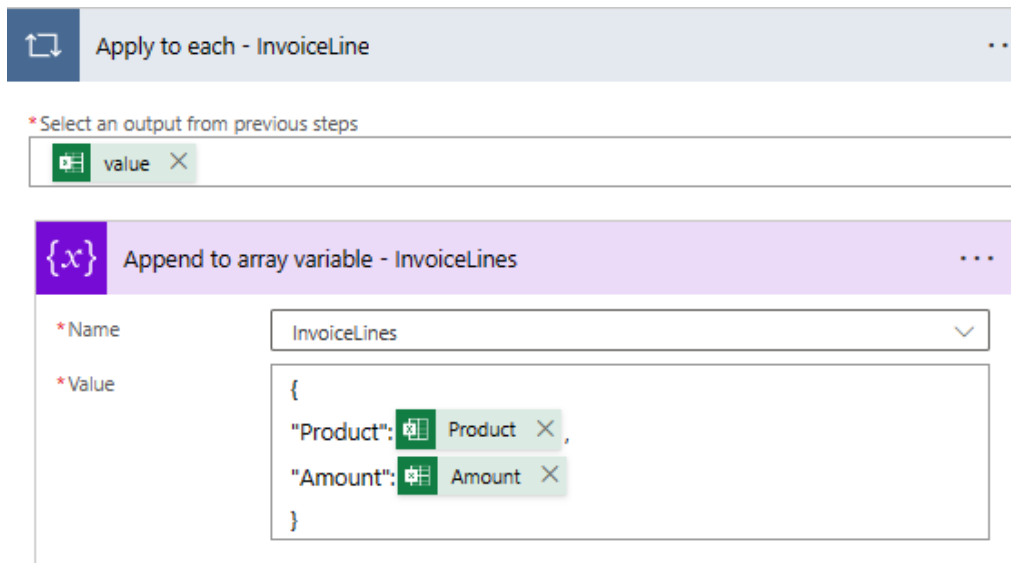
* Value

[]

32. After the **List rows present in a table - InvoicesLines** action, add an **Apply to each invoice** to go through every invoice lines:



33. In this Apply to each, you will create a JSON array containing the **Invoice Lines** information; add an **Append to array variable** action:



34. You will now update **the Populate a Microsoft Word template** action with this array. Select this action and click on the **Switch to input entire array** button:

Populate a Microsoft Word template - Invoice Template

* Location: OneDrive for Business

* Document Library: OneDrive

* File: /Power Automate in a day - Labs/Invoice Template.docx

invoiceid: InvoiceID

lineitems Product - 1: Click or tap here to enter text.

lineitems Amount - 1: Click or tap here to enter text.

+ Add new item

35. Store your array in this area:

Populate a Microsoft Word template - Invoice Template

* Location: OneDrive for Business

* Document Library: OneDrive

* File: /Power Automate in a day - Labs/Invoice Template.docx

invoiceid: InvoiceID

lineitems: InvoiceLines

company: Company N...

zip: Zip

street: Street

city: City

managersignature:

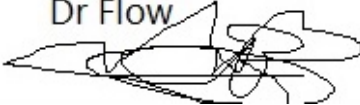
36. Run the Flow and check the generated documents. You should now see the invoice line items:

Doctor Flow 133 av Dieweg 1180 Uccle Belgique		INVOICE	
BILL TO Company Name: Contoso Street address: 44 av du Bourget City: Brussels , 1130		INVOICE#	1

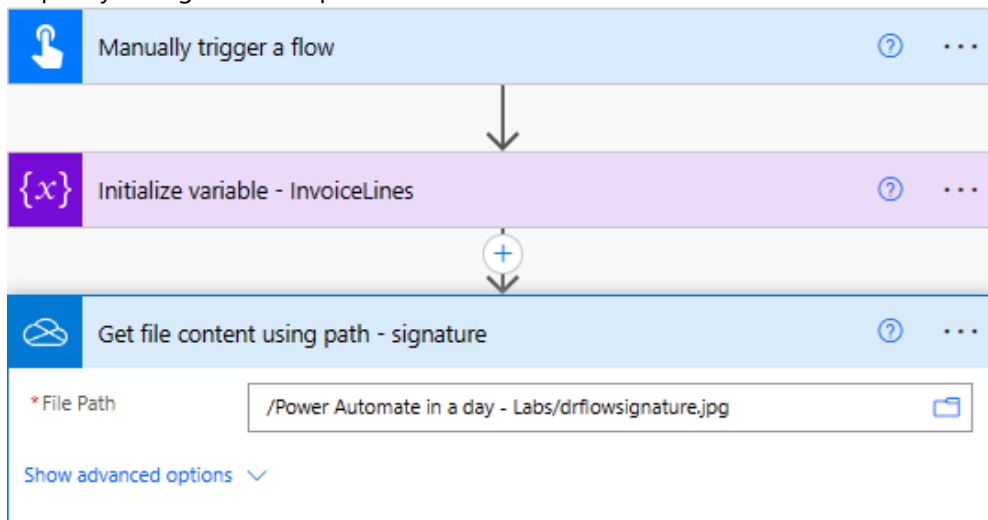
Description	Amount
Internal training PowerApps	5
Internal training Flow	4

37. We will now display the signature portion of the document. Create a jpeg file with your name and signature, and upload it to the Documents folder of your OneDrive for business. For instance, our signature looks like this:

Dr Flow



38. In the beginning section of the Flow, add a **OneDrive for business** action **Get file content using path** and pass your signature file path:



```

graph TD
    A[Manually trigger a flow] --> B[Initialize variable - InvoiceLines]
    B --> C[Get file content using path - signature]
    C --> D[Show advanced options]
  
```

* File Path:

Show advanced options ▾

39. Edit the existing action **Populate a Microsoft Word template** action and in the **managersignature** field, pass your signature content:

Populate a Microsoft Word template - Invoice Template

*Location: OneDrive for Business

*Document Library: OneDrive

*File: /Power Automate in a day - Labs/Invoice Template.docx

invoiceid: InvoiceID

lineitems Product - 1: Click or tap here to enter text.

lineitems Amount - 1: Click or tap here to enter text.

+ Add new item

company: Company N...

zip: Zip

street: Street

city: City

managersignature: [Red Box]

Workflow Steps:

- Create file - Invoice in Word
- Convert Word Document to PDF
- Create file - Invoice in PDF

Dynamic Value Pane:

- Search dynamic value
- Amount
- Price
- Total
- Get file content using path - signature
- File content: The content of the file.

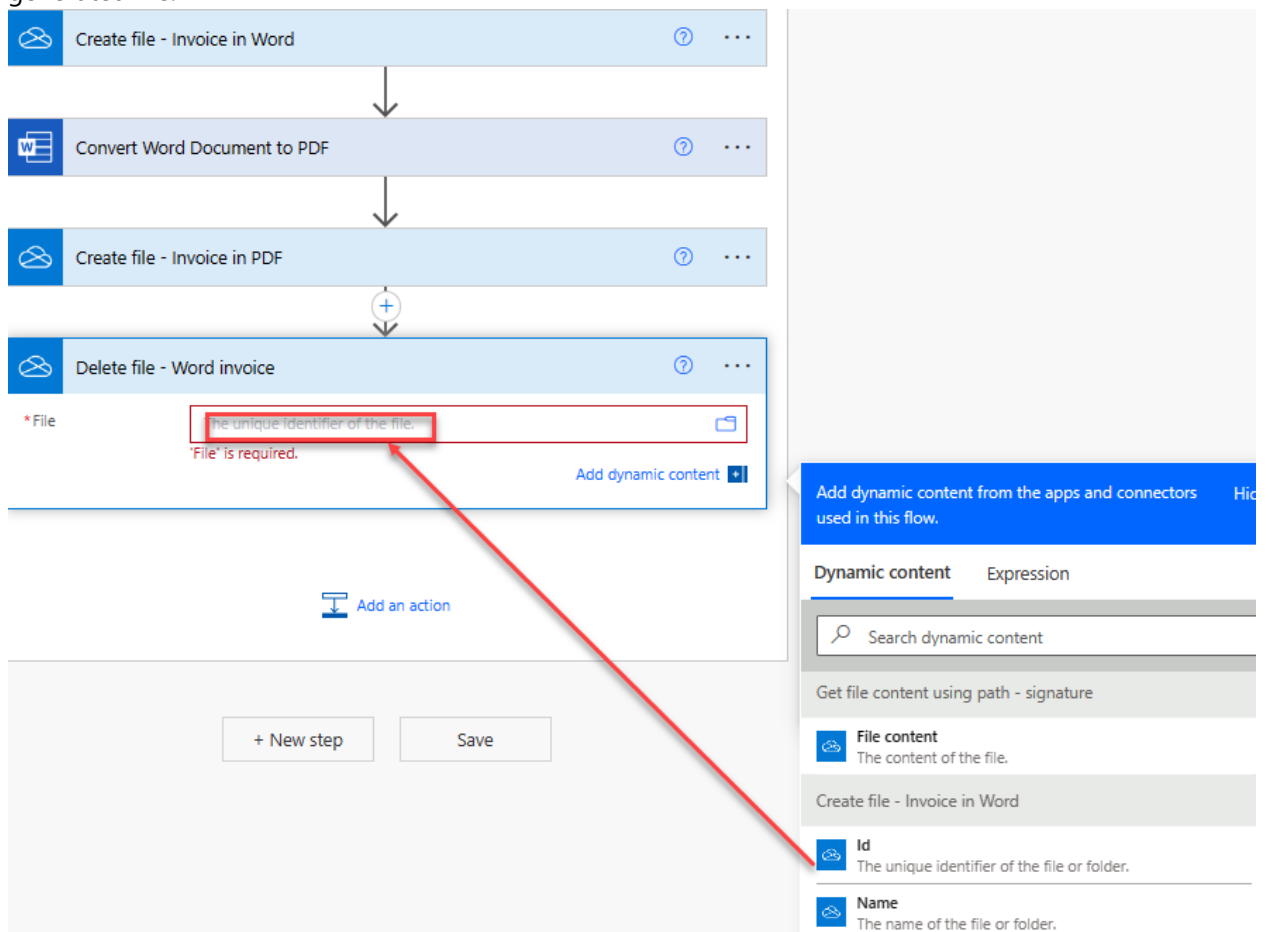
40. Test your Flow and check one of the generated documents; the signature should now be visible:

Doctor Flow 133 av Dieweg 1180 Uccle Belgique		INVOICE	
BILL TO Company Name: Contoso Street address: 44 av du Bourget City: Brussels , 1130		INVOICE#	1

Description	Amount
Internal training PowerApps	5
Internal training Flow	4



41. After the **Create file – Invoice in PDF action**, add a Delete file action to remove the Word generated file:



The screenshot shows a Power Automate flow with the following steps:

- Create file - Invoice in Word
- Convert Word Document to PDF
- Create file - Invoice in PDF
- Delete file - Word invoice

In the 'Delete file - Word invoice' step, the 'File' field is highlighted with a red box and contains the placeholder text 'The unique identifier of the file.'. A red arrow points from this field to the 'Id' dynamic content option in the right-hand pane.

The right-hand pane shows the 'Dynamic content' section with the following options:

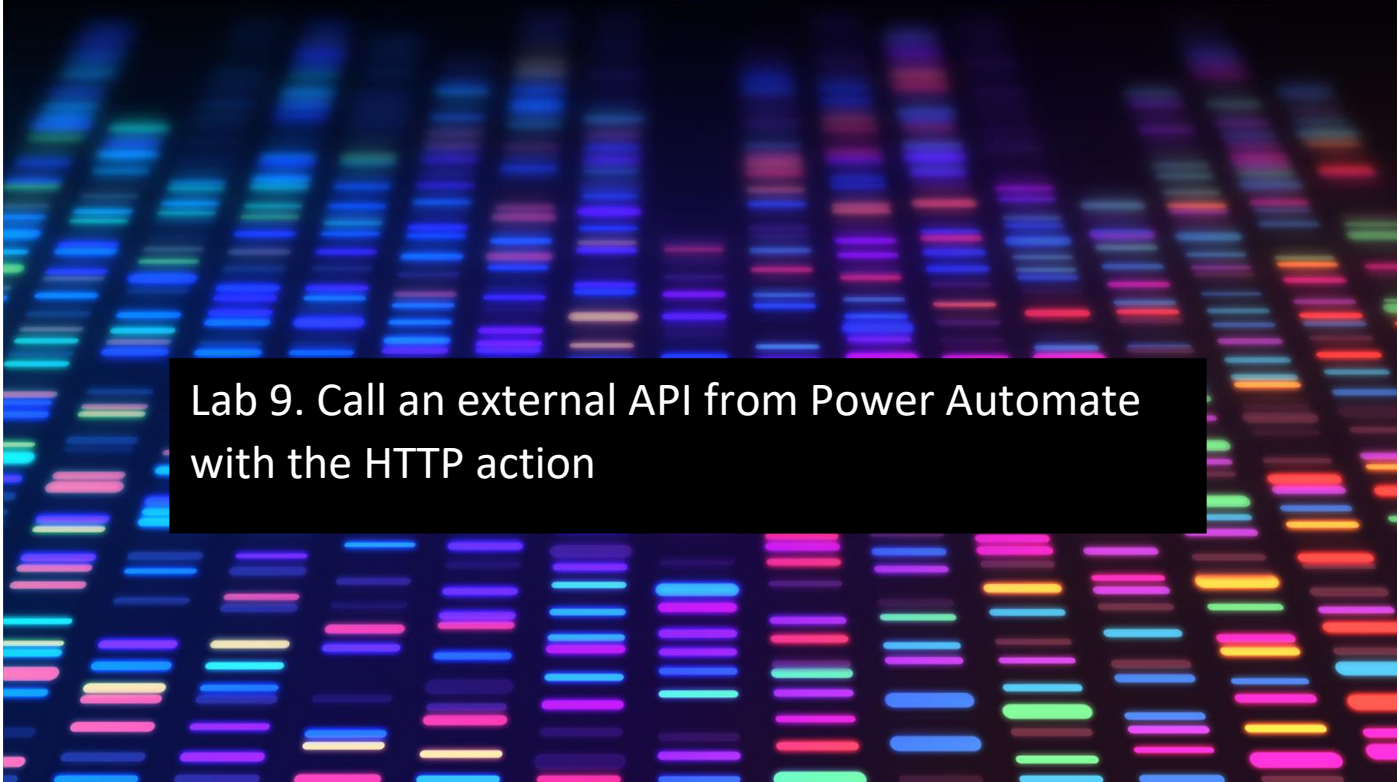
- File content: The content of the file.
- Create file - Invoice in Word
- Id**: The unique identifier of the file or folder.
- Name: The name of the file or folder.

42.

We need your feedback

Do you want to report an issue or suggest something? We need your feedback:

<https://github.com/Power-Automate-in-a-day/Training-by-the-community/issues>



Lab 9. Call an external API from Power Automate
with the HTTP action

Lab 9. Call an external API from Power Automate with the HTTP action

Author: Serge Luca, aka "Doctor Flow"

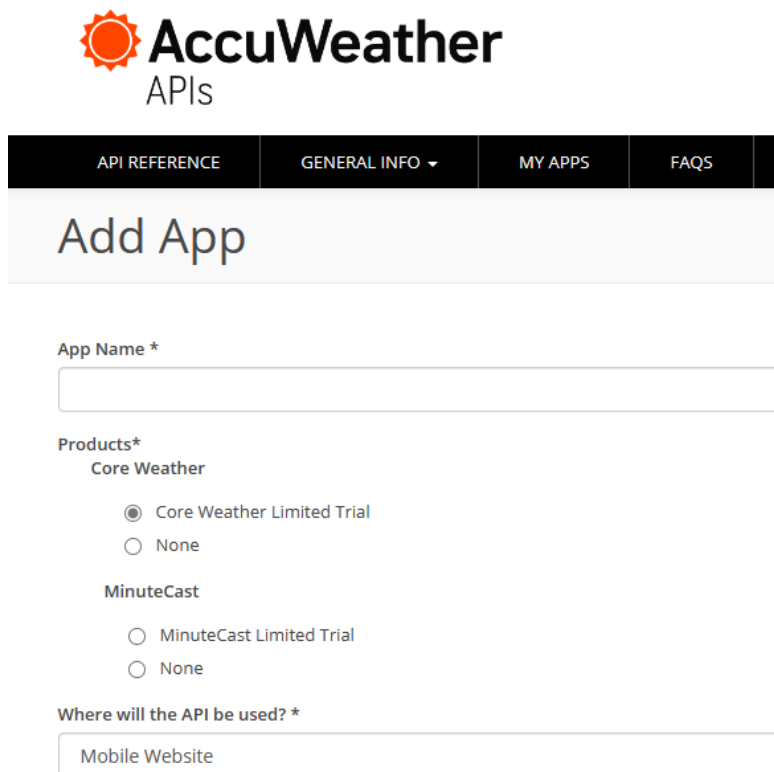
Learning objective: call an external REST API from Flow

Duration: 15 minutes

Prerequisites: This exercise aims to call the AccuWeather API to get the weather in a specific city.

Tasks:

1. Go to the site <https://developer.accuweather.com/> and register for a free account; log in with your account and create a new App – select the **Core Weather Limited Trial** product:



The screenshot shows the 'Add App' page on the AccuWeather APIs developer portal. At the top is the AccuWeather APIs logo. Below it is a navigation bar with links: API REFERENCE, GENERAL INFO (with a dropdown arrow), MY APPS, and FAQs. The main heading is 'Add App'. The form contains the following fields:

- App Name ***: A text input field.
- Products***: A section with two groups of radio buttons.
 - Core Weather**:
 - ☒ Core Weather Limited Trial
 - ☐ None
 - MinuteCast**:
 - ☐ MinuteCast Limited Trial
 - ☐ None
- Where will the API be used? ***: A dropdown menu with 'Mobile Website' selected.

An API key will be generated and associated with your flow:

These are your apps! Explore!

new **Power Automate in a day**

Keys Products Details Analytics Edit "Power Automate in a day" Delete "Power Auto

Power Automate in a day's Keys

The key below can be used to access the API products associated with this application.

API Key	
Key Issued	Sun, 02/19/2023 - 08:22

Each location (like a city) can be identified by a specific code called **location key**; let's find the location key of the city "Brussels".

- In your browser, type

<http://dataservice.accuweather.com/locations/v1/search?q=brussels&apikey=<yourapikey>>

where <yourapikey> is the key provided above.

Several locations in the world are called Brussels. Select the one in **Belgium**, and copy the key:

```
[{"Version":1,"Key":"27581","Type":"City","Rank":40,"LocalizedName":"Brussels","EnglishName":"Brussels","PrimaryPostalCode":"","ID":"BE","LocalizedName":"Belgium","EnglishName":"Belgium"},"AdministrativeArea":{"ID":"BRU","LocalizedName":"Brussels","EnglishName":"Brussels"},"Code":"CET","Name":"Europe/Brussels","GmtOffset":1.0,"IsDaylightSaving":false,"NextOffsetChange":"2023-03-26T01:00:00Z"},"G":{"Value":114.0,"Unit":"ft","UnitType":0},"IsAlias":false,"SupplementalAdminAreas":[],"DataSets":["AirQualityCurrentConditions","AirQualityForecasts","Alerts","ForecastConfidence","FutureRadar","MinuteCast","Radar"]},"Ver":{"ID":"CA","LocalizedName":"Canada","EnglishName":"Canada"},"AdministrativeArea":{"ID":"ON","LocalizedName":"Ontario","EnglishName":"Ontario"},"Code":"EST","Name":"America/Toronto","GmtOffset":-5.0,"IsDaylightSaving":false,"NextOffsetChange":"2023-03-12T07:00:00Z"},"Value":1115.0,"Unit":"ft","UnitType":0},"IsAlias":false,"SupplementalAdminAreas":[{"Level":2,"LocalizedName":"Huron","EnglishName":"Huron"}],"AirQualityCurrentConditions","AirQualityForecasts","Alerts","ForecastConfidence","FutureRadar","MinuteCast","Radar"]},"Ver":{"ID":"NAM","LocalizedName":"North America","EnglishName":"North America"},"Country":{"ID":"US","LocalizedName":"United States","EnglishName":"United States"},"ID":"IL","LocalizedName":"Illinois","EnglishName":"Illinois","Level":1,"LocalizedType":"State","EnglishType":"State","Count":12708,"GeoPosition":{"Latitude":38.95,"Longitude":-90.589,"Elevation":{"Metric":{"Value":151.0,"Unit":"m","UnitType":0},"Imperial":{"Value":495.7,"Unit":"ft","UnitType":0}}},"Level":2,"LocalizedName":"Calhoun","EnglishName":"Calhoun"},"DataSets":["AirQualityCurrentConditions","AirQualityForecast"]}]
```

- Create a **Power Automate instant flow**;
- add the **HTTP** action and provide the following values:

Manually trigger a flow

HTTP

* Method: GET

* URI: <http://dataservice.accuweather.com/currentconditions/v1/27581>

Headers:

test	555
Enter key	Enter value

Queries:

apikey	YrbpfOiWYasjqREjxSooqbS6kWSnRcgE
details	false
Enter key	Enter value

Body: Enter request content

Cookie: Enter HTTP cookie

Show advanced options

- The **method** must be **GET**
- The **URI**: <http://dataservice.accuweather.com/currentconditions/v1/27581>
- In Queries:

apikey : <your apikey>

details: false

5. Save and run the flow. The result value (in the body of the HTTP action looks like this:

```
[
{
  "LocalObservationDateTime": "2023-02-19T10:22:00+01:00",
  "EpochTime": 1676798520,
  "WeatherText": "Mostly cloudy",
  "WeatherIcon": 6,
  "HasPrecipitation": false,
  "PrecipitationType": null,
  "IsDayTime": true,
  "Temperature": {
    "Metric": {
      "Value": 9.4,
      "Unit": "C",
      "UnitType": 17
    }
  }
}
```

```

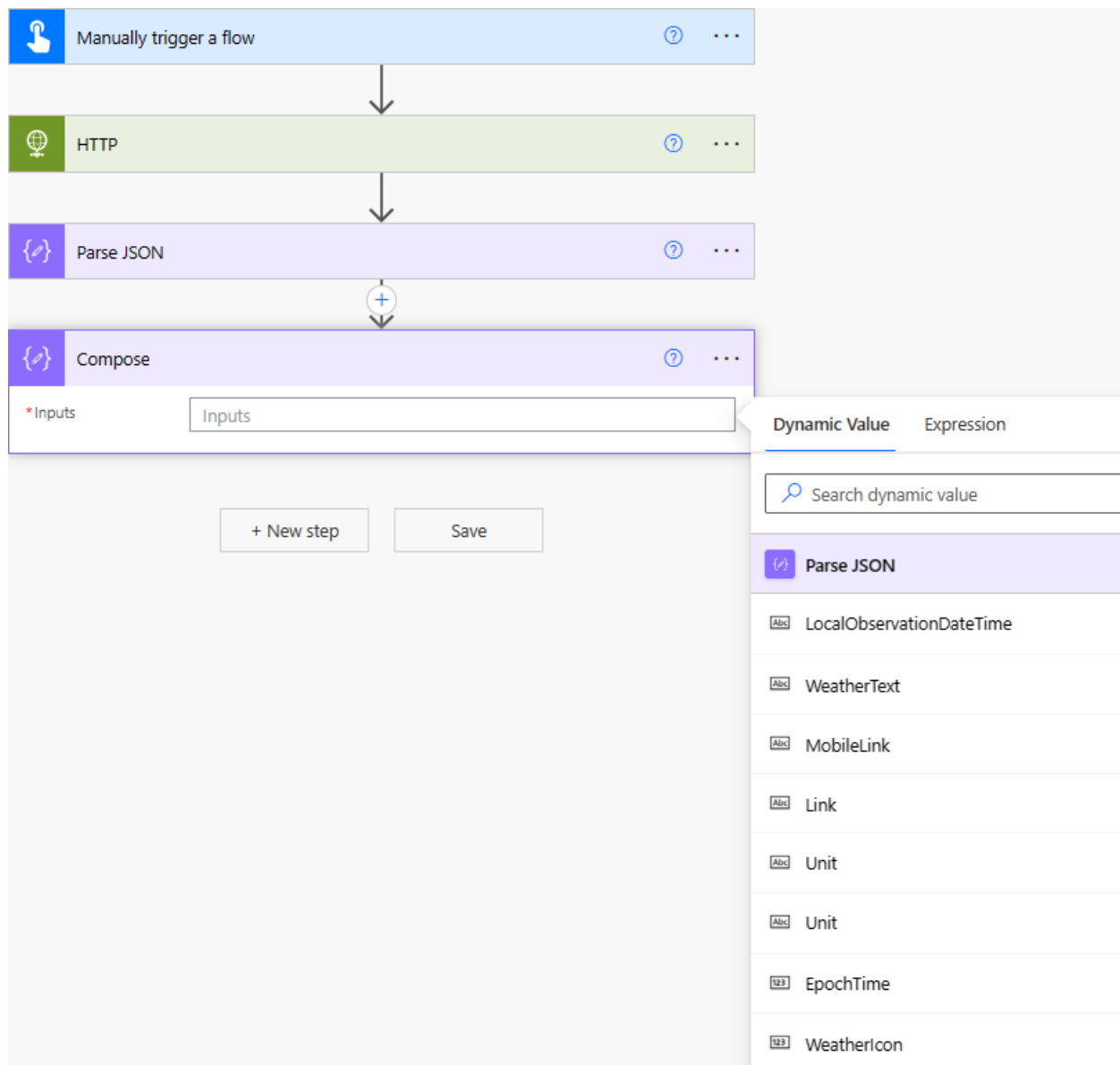
    },
    "Imperial": {
      "Value": 49,
      "Unit": "F",
      "UnitType": 18
    }
  },
  "MobileLink": "http://www.accuweather.com/en/be/brussels/27581/current-weather/27581?lang=en-us",
  "Link": "http://www.accuweather.com/en/be/brussels/27581/current-weather/27581?lang=en-us"
}
]

```

6. Copy the return value of the HTTP action into your clipboard.
7. After the HTTP action, add a **Parse JSON** action and connect the Content property to the body of the HTTP action:



8. Click **Generate from sample**, paste the content from your clipboard and click **Done**.
9. Add a compose and bring it the Dynamic value of Parse JSON:



This is a straightforward way to retrieve value from a complex JSON.

However, using the HTTP complex can still be error-prone. Therefore, we will encapsulate the HTTP call into a custom connector in the next lab.